

17/10/2026



Molecular network MetGem 1.5.2

David Touboul

Introduction



- Laboratory created in 2014
 - Ecole Polytechnique
 - 12 researchers and teachers
 - 7 engineers
-
- 3 scientific themes:
 - Catalysis and Activations
 - Complex mixtures
 - f Orbitals and Reactivity





- Mass Spectrometry
 - FT-ICR Solarix XR 9.4 Tesla (Bruker), 2012 (Infranalytics) (1.5 M€)
 - GC-MS 7000D GC TQ (Agilent), 2017
 - LC-HR-MS Tims-TOF (Bruker), 2018
 - AP-MALDI and LC/SFC Orbitrap (Excedlon Pro – Biopharma Option), Sep 2025





Dr. Edith Nicol



Dr. Sophie Bourcier

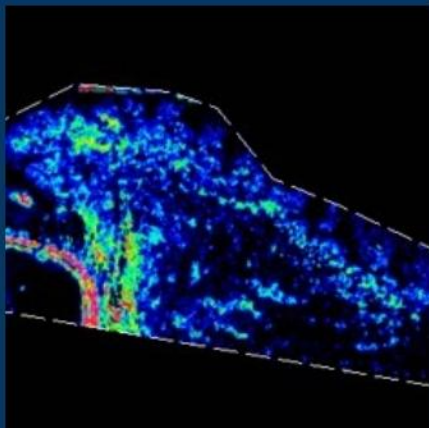


Paul Renquet



Dr. Covadonga Lucas-Torres

- NMR (ResoMag platform)
 - 300 MHz liquid Avance III TBO ^{31}P (Bruker)
 - 400 MHz liquid Avance I QNP (Bruker)
 - 300 MHz liquid Avance I BBFO (Bruker)
- In 2025 (1.2 M€, funded by IdF/Ecole Polytechnique)
 - **600 MHz liquid/solid + UV irradiation (Bruker)**
 - New consol for 400 MHz
 - New consol for 300 MHz



Mass Spectrometry Imaging

Spatial molecular analysis of tissues, organisms, plants and environmental samples.



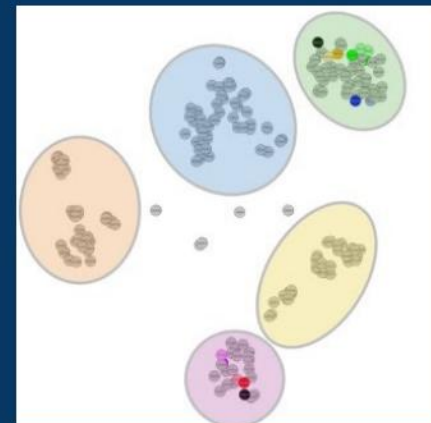
Natural Products Chemistry & Chemotaxonomy

LC-HRMS and MS/MS workflows for biodiversity, herbarium specimens and plant classification.



Metabolomics & Epitranscriptomics

Targeted and untargeted profiling of lipids, metabolites, specialized metabolites and modified nucleosides.



Computational Mass Spectrometry

Molecular networking, MS/MS annotation, AI-assisted curation and reproducible workflows.

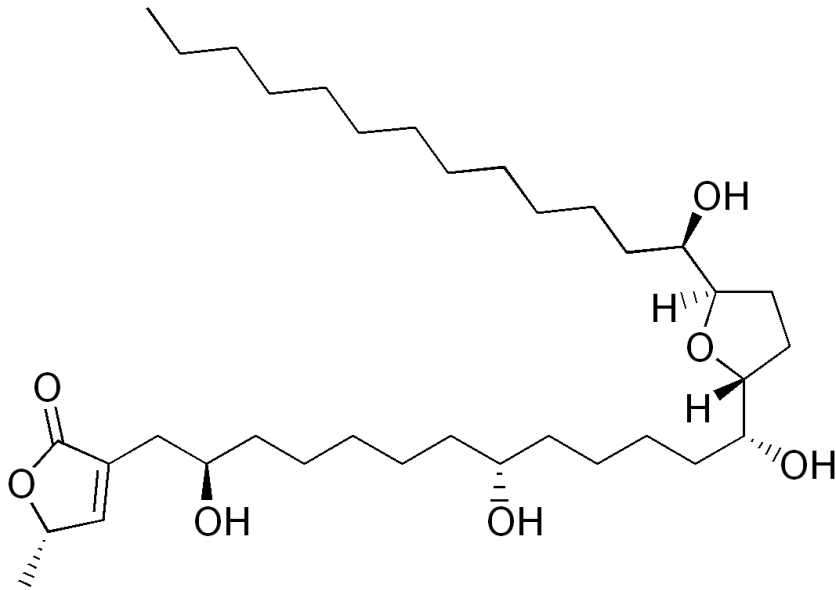
<https://sites.google.com/view/david-touboul-research/home>

Mon principal défi: le temps !

Métabolite totem:

Mon principal défi: le temps !

Métabolite totem:



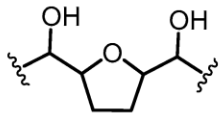
Annonacine



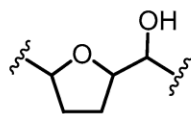
Mon principal défi: le temps !

Métabolite totem:

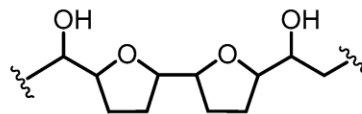
Types of tetrahydrofuran (THF) systems



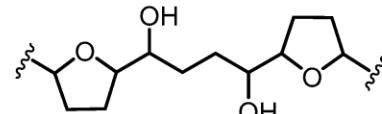
T-A



T-B

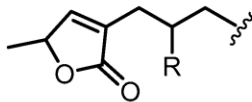


T-C

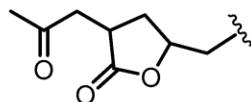
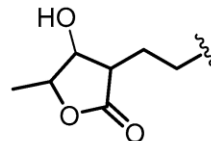


T-D

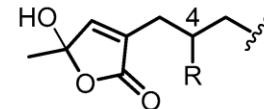
Types of γ -lactone rings



L-A

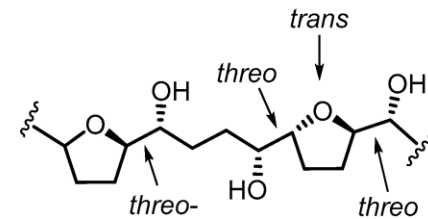
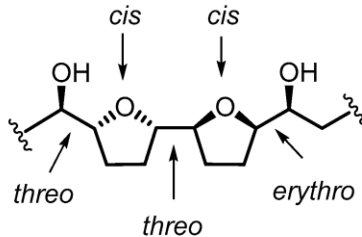
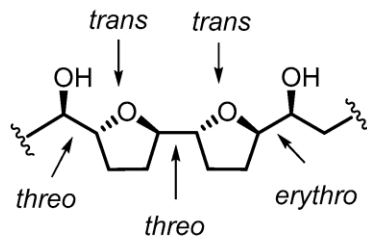
L-B₁

L-C

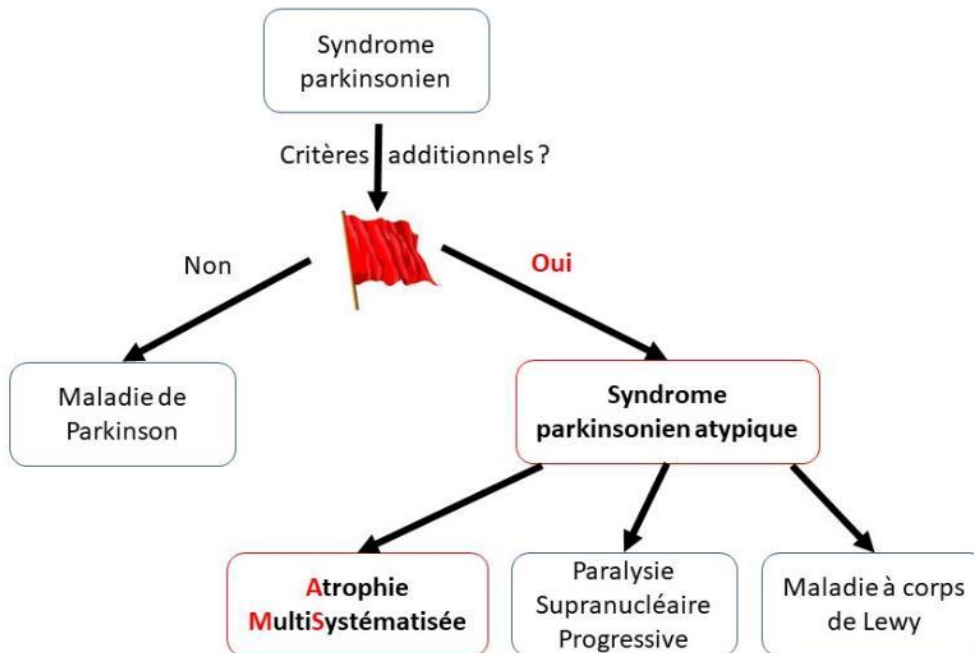


L-D R = H or OH

Relative configuration



Les syndromes parkinsoniens atypiques

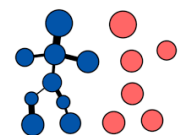


Parkinson «caribéen»

PSP-like / PDC ; peu sensible à la L-Dopa;
tauopathie ; association épidémiologique
avec exposition aux Annonaceae

1. How would you rate your expertise in mass spectrometry ?
2. How familiar are you with fragmentation in mass spectrometry ?
3. How familiar are you with molecular network ?
4. How comfortable are you with Python programming ?

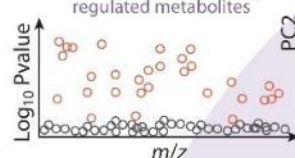
****Expert — ***Proficient — **Basic — *None



Statistical analysis

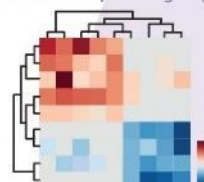
Univariate analysis

Identifying differentially regulated metabolites



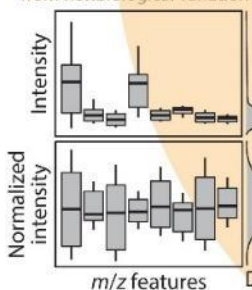
Clustering

Mass feature pattern grouping



Feature normalization

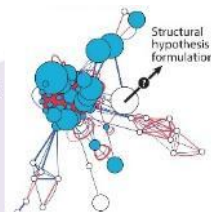
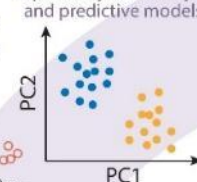
Separating biological variation from nonbiological variation



Normalization

Chemometrics

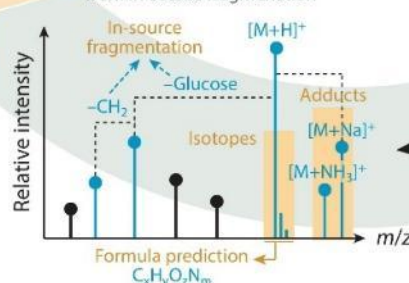
Exploratory data analysis and predictive models



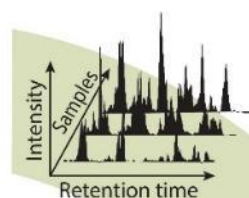
Scientific question

Spectral deconvolution

Resolving mass feature derived from in-source fragmentation



Peak shape/cross-sample intensity-based correlation



Raw data acquisition and preprocessing

Raw data



Standardized file formats



Feature detection and alignment

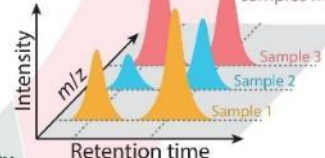
Mass feature detection

Detecting mass signal(s) for a molecule from mass spectrometry data



Mass feature alignment

Aligning mass features across multiple samples from a data set



MS spectral deconvolution

MZmine 3



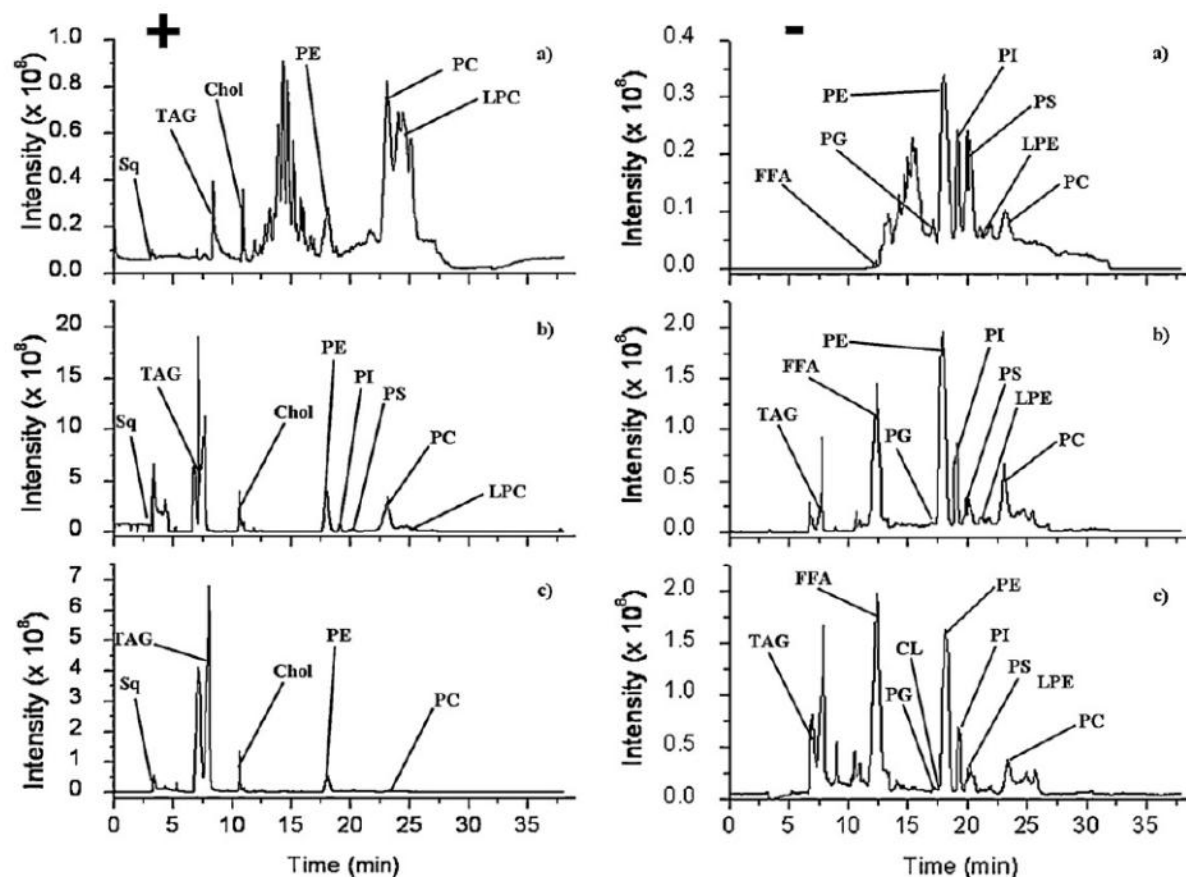
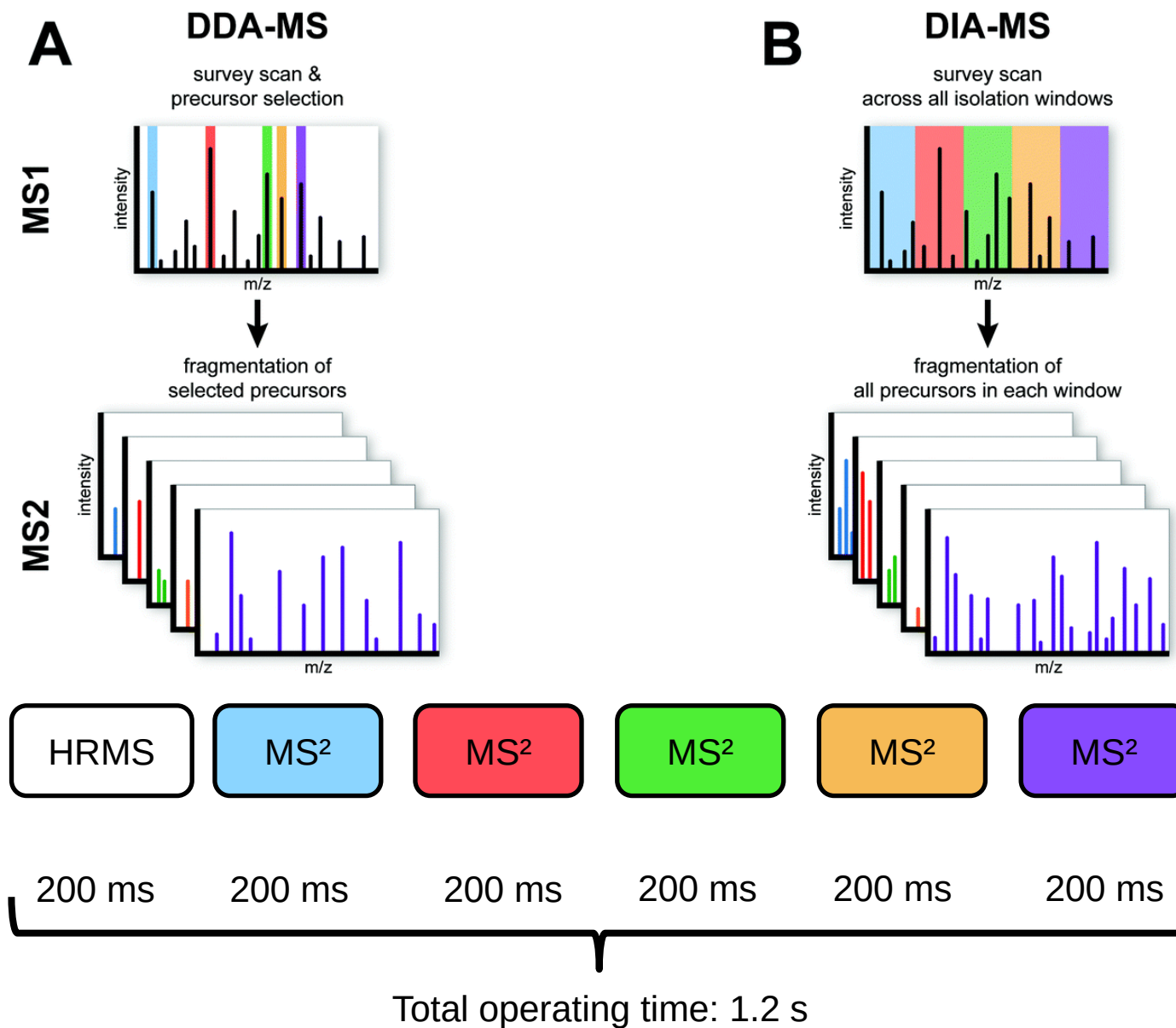


Fig. 4. Chromatograms obtained with ESI (a), APCI (b) and APPI (c) in positive (left) and negative (right) ion mode on a *Leishmania donovani* HePC-RT sample.

Phase A: n-heptane/ethyl acetate (99.8:0.2, v/v),
 Phase B : acetone/ethyl acetate (2:1; v/v) +0.02% acetic acid,
 Phase C : propan-2-ol/water (85:15, v/v) + 0.05% acetic acid and ethanolamine.

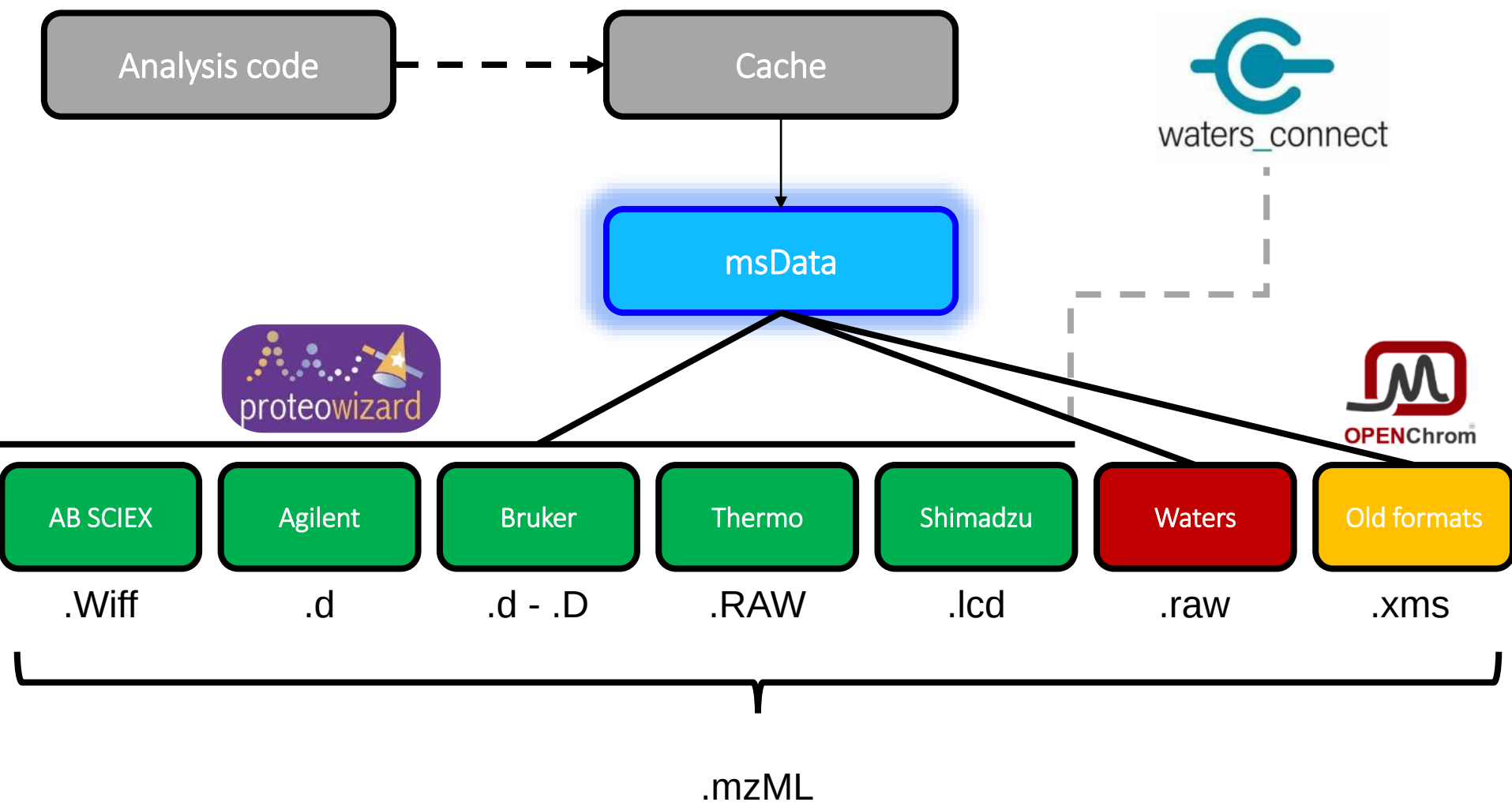


ProteoWizard → MSConvert

<https://proteowizard.sourceforge.io/download.html>

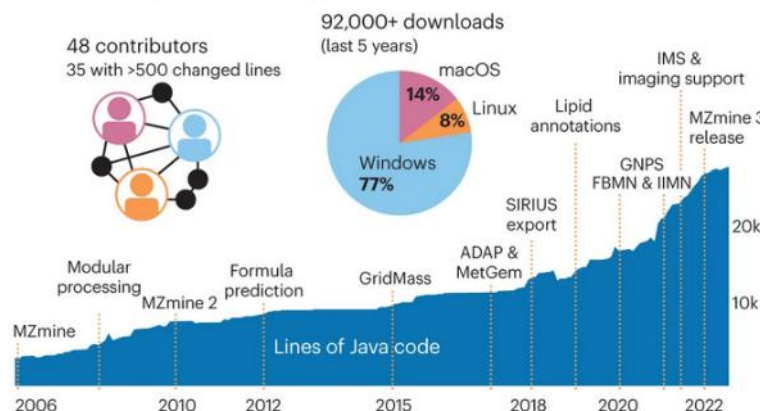
OpenChrom

<https://www.openchrom.net/>

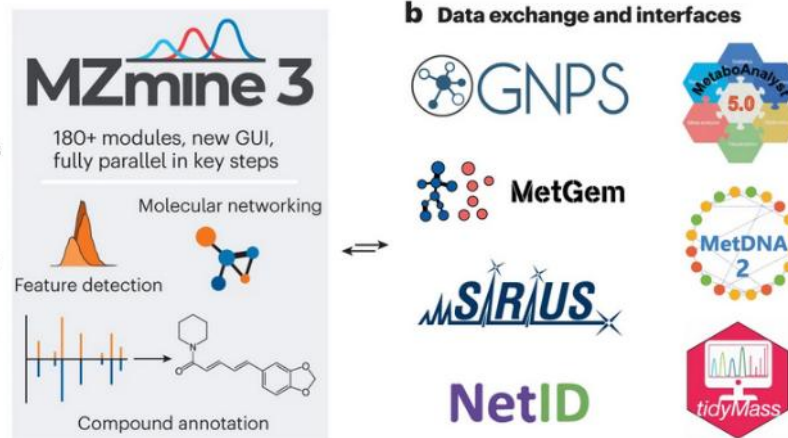


MZmine <https://mzio.io/#mzmine> Documentation: https://mzmine.github.io/mzmine_documentation/

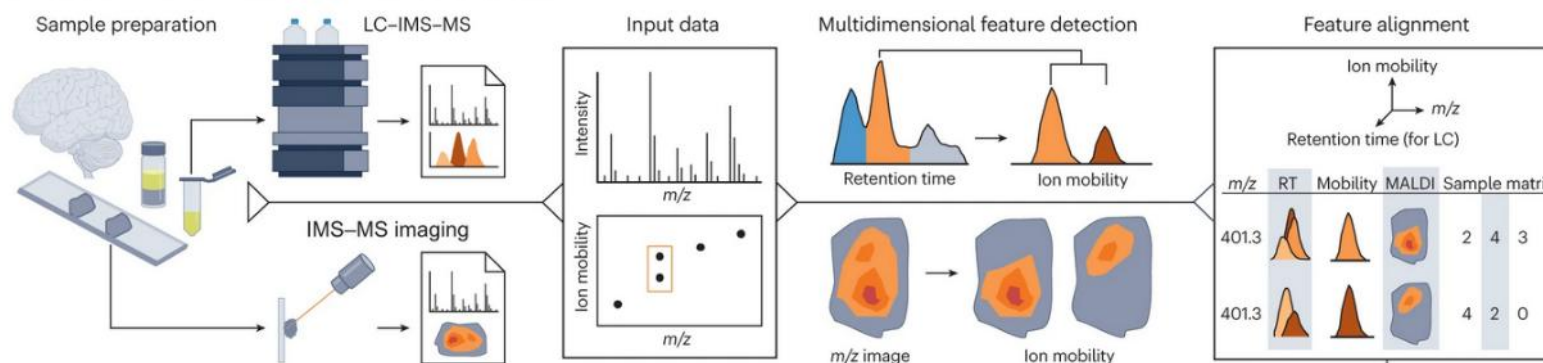
a Community-driven development



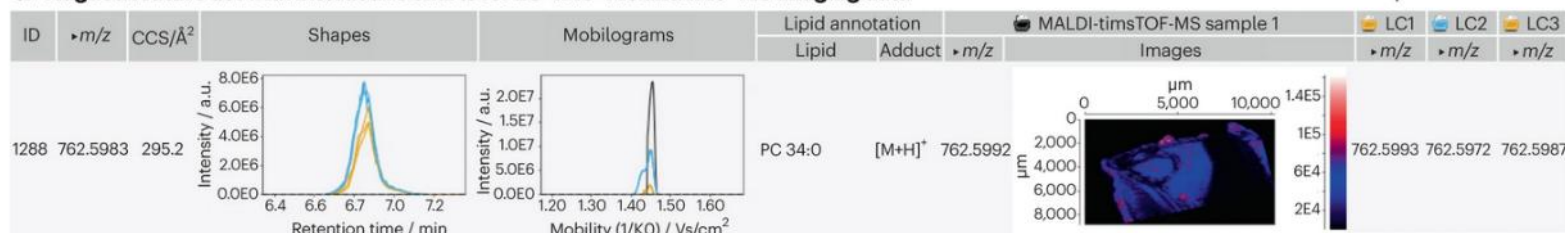
b Data exchange and interfaces

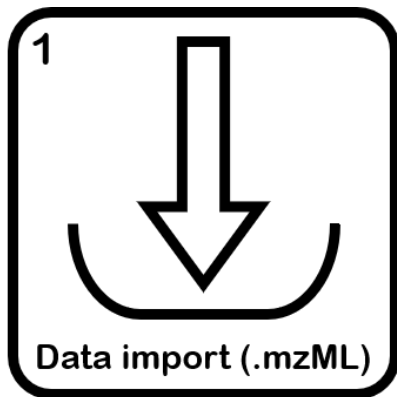


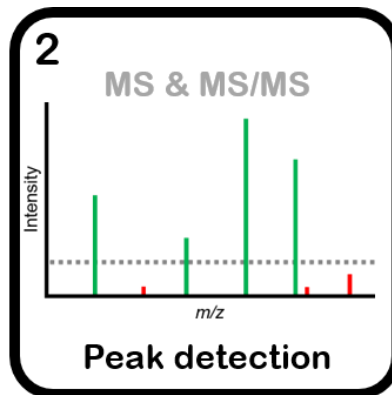
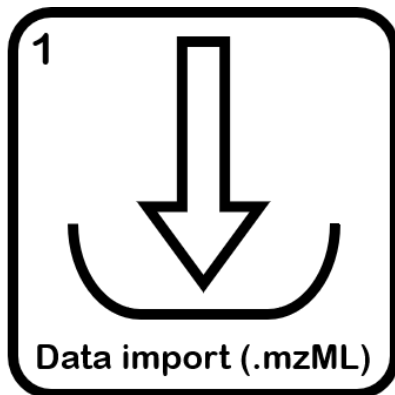
c Integrative LC-IMS-MS and IMS-MS imaging workflow

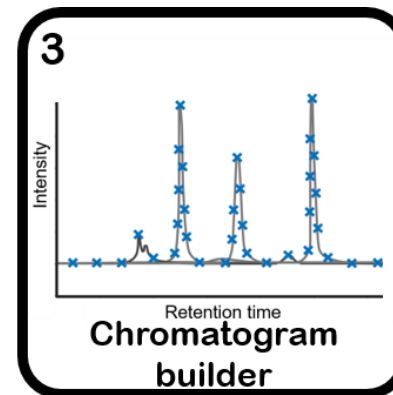
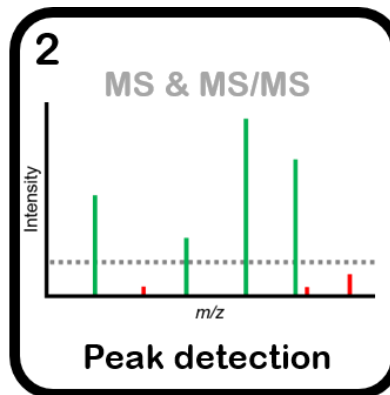
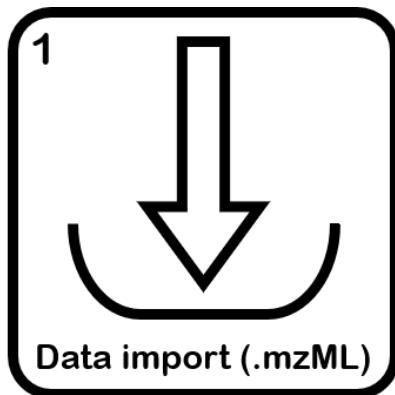


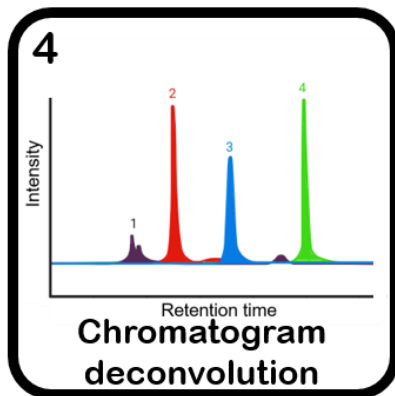
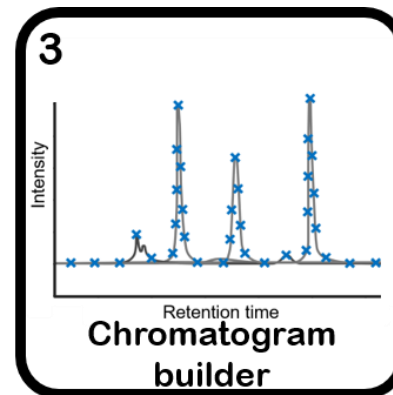
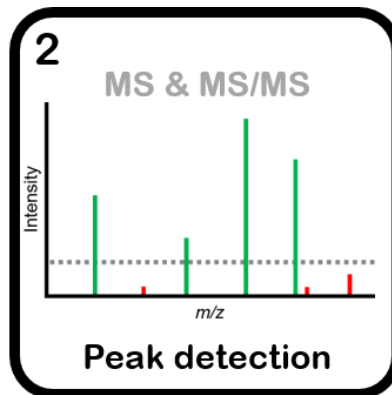
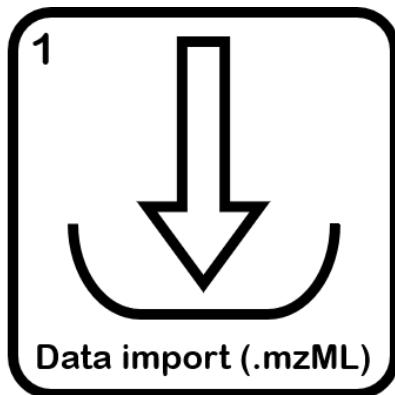
d Aligned feature list with interactive charts for LC-IMS-MS and IMS-MS imaging data

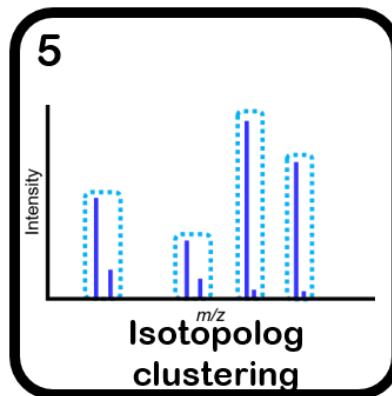
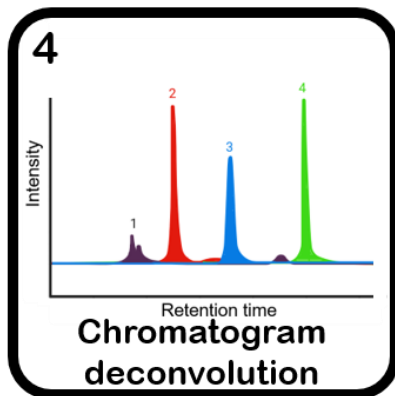
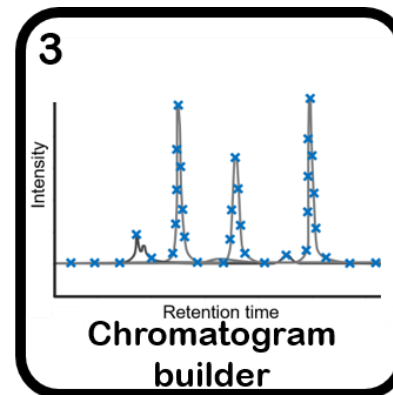
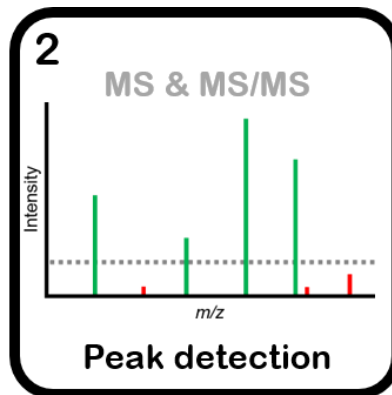
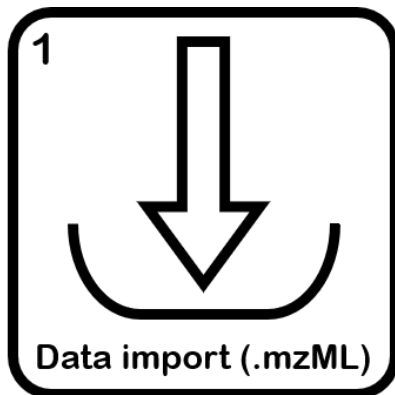


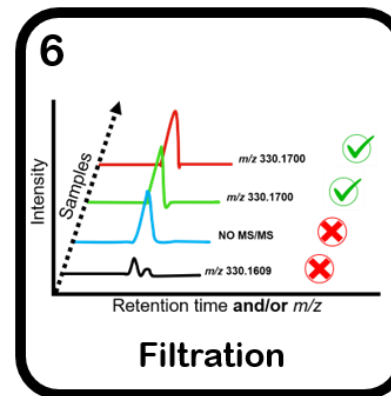
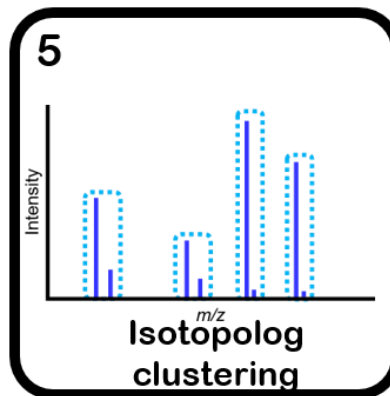
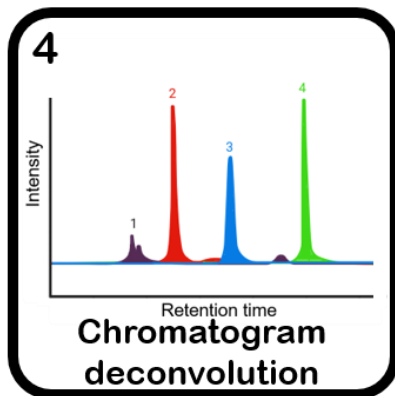
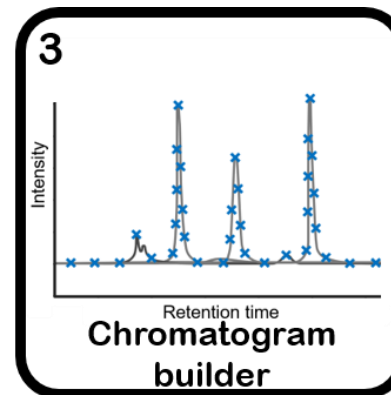
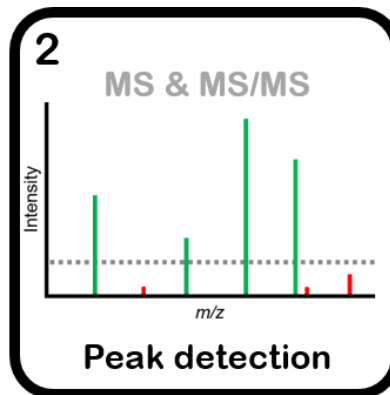
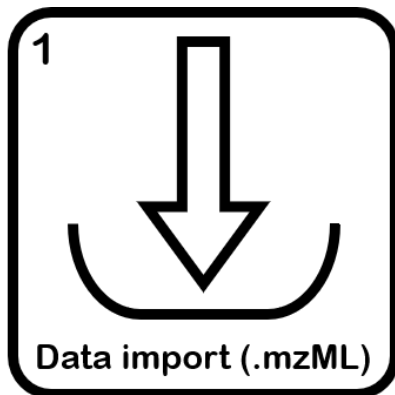


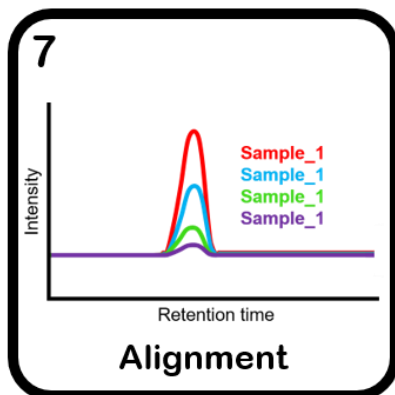
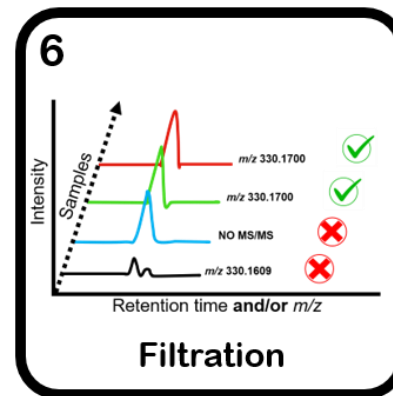
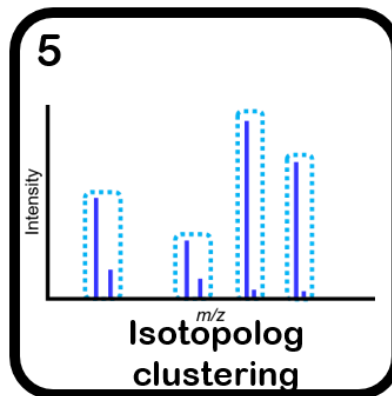
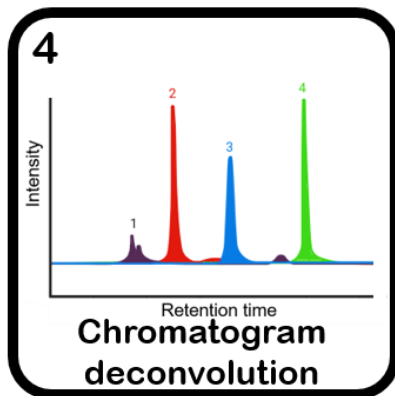
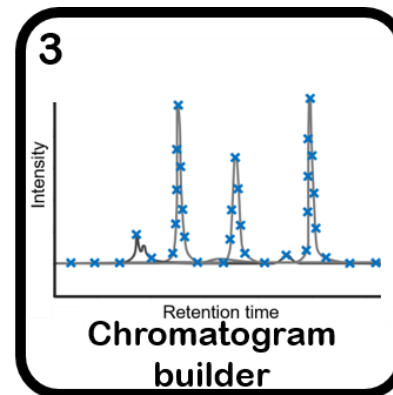
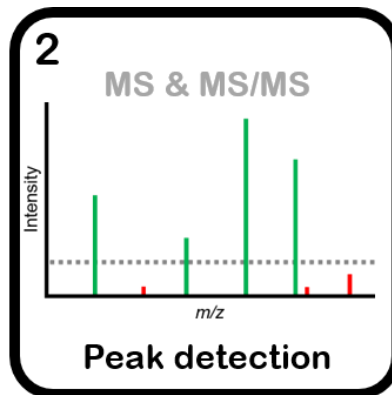
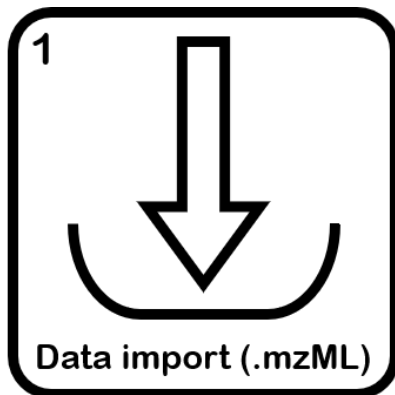


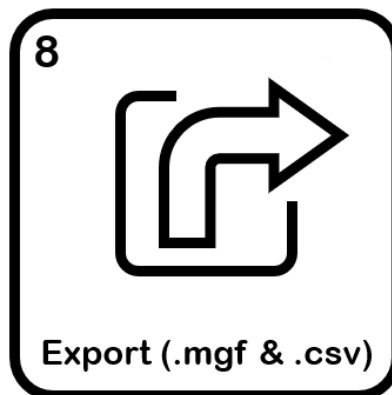
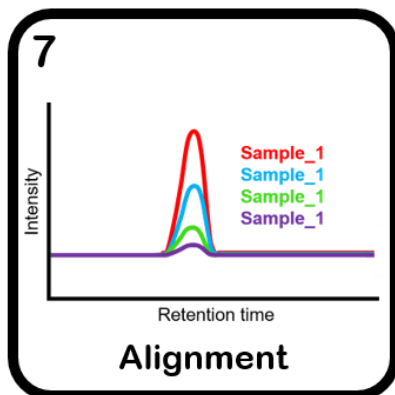
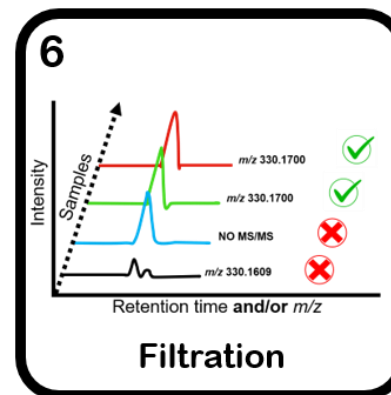
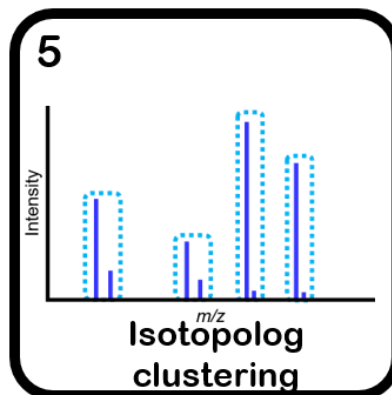
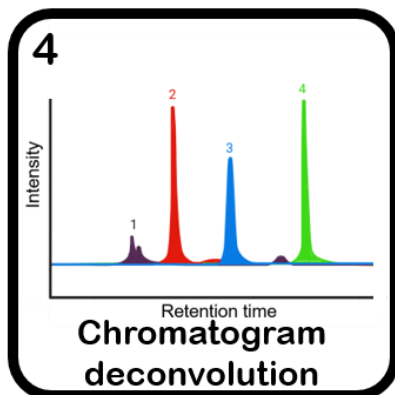
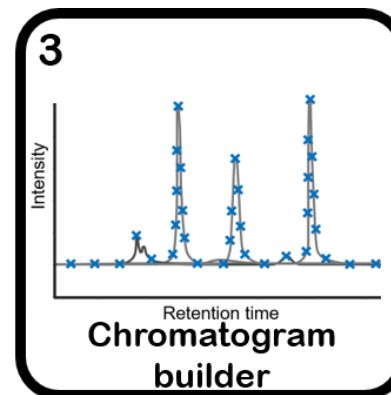
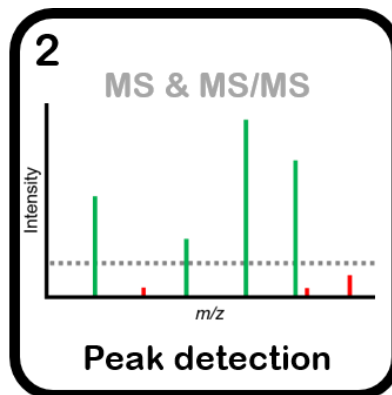
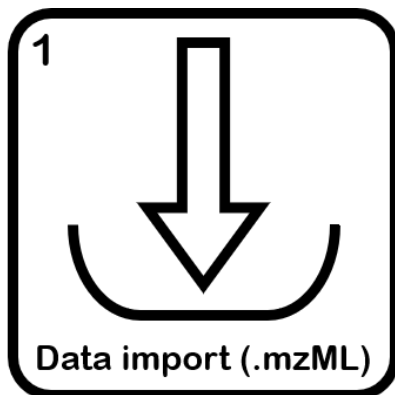


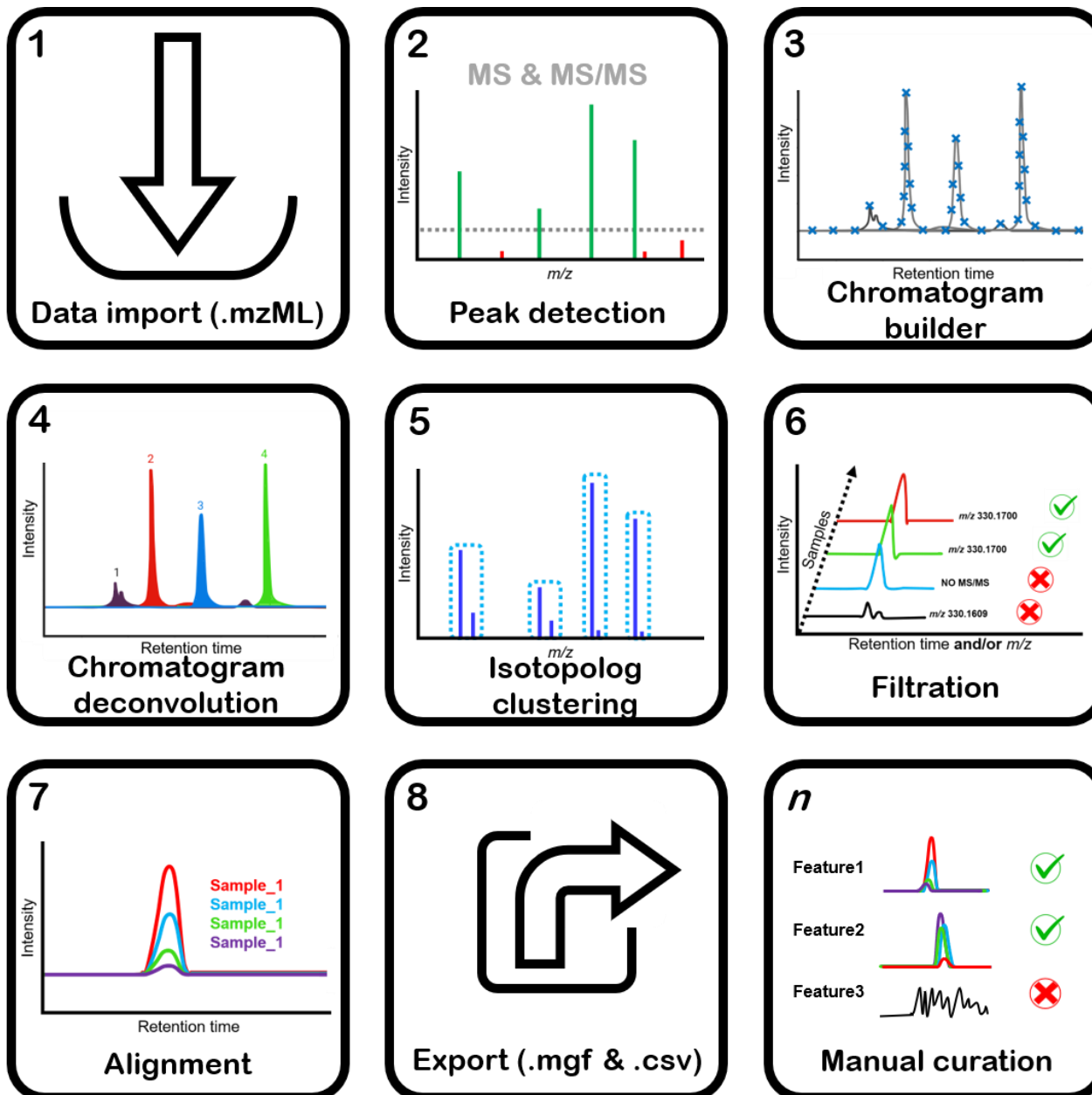












Optimization and Testing of Mass Spectral Library Search Algorithms for Compound Identification

Stephen E. Stein

NIST Mass Spectrometry Data Center, National Institute of Standards and Technology, Gaithersburg, Maryland, USA

Donald R. Scott

Atmospheric Research and Exposure Assessment Laboratory, U. S. Environmental Protection Agency, Research Triangle Park, North Carolina, USA

J Am Soc Mass Spectrom 1994, 5, 859-866

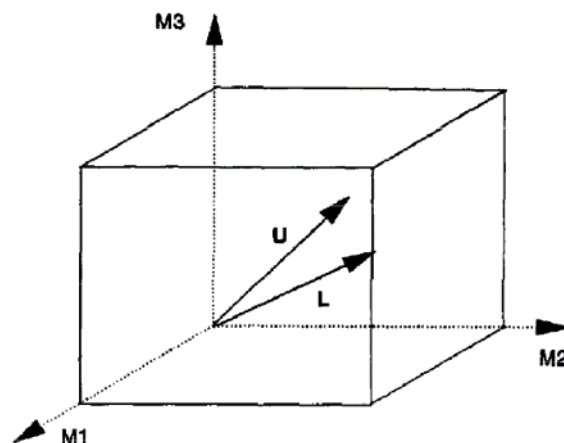


Figure 1. Vector representation of a hypothetical three-peak unknown (U) and library (L) mass spectrum in three-dimensional space (peaks have mass M1, M2, and M3).

Euclidean distance

$$\left(1 + \frac{\sum (w_L - w_U)^2}{\sum w_U^2}\right)^{-1}$$

Absolute Value Distance

$$\left(1 + \frac{\sum |w_L - w_U|}{\sum w_U}\right)^{-1}$$

Hertz et al. [9]

average of weighted peak intensity ratios
[1 + fraction of unmatched intensities]

Dot-product (cosine), F_D

$$\frac{(\sum w_L w_U)^2}{\sum w_L^2 \sum w_U^2}$$

Probability-based matching (PBM) [5b, 5d, 10c]

Uses probability that, by chance, peaks match within an abundance window (W value) by using uniqueness values for mass (U value) and abundance (A) along with a variety of rules and correlation tables.

Composite:

$$\frac{N_U F_D + N_{L\&U} F_R}{N_U + N_{L\&U}}$$

F_D = Dot-Product Term Above

F_R = Ratio of Peak Pairs (below)

W = [Peak Intensity]ⁿ [Mass]^m = Weighted Intensity

N = Number of peaks

$$F_R = \frac{1}{N_{L\&U}} \sum_i^{L\&U} \left(\frac{w_{L,i}}{w_{L,i-1}} \frac{w_{U,i-1}}{w_{U,i}} \right)^n$$

where $n = 1$ or -1 when the term in parentheses is less than or greater than unity, respectively

^a Subscripts: L = library, U = unknown, L & U = peak in both library and unknown spectrum.

Algorithm	% Correct at Rank		
	1	1-2	1-3
Dot-product	72.9	85.9	90.8
	73.2	86.3	91.0
	72.8	85.9	90.8
Optimized	74.9	86.9	91.7
Euclidean distance	65.8	79.3	84.9
	69.9	82.9	88.2
Optimized	71.9	83.9	88.9

Comparing Similar Spectra: From Similarity Index to Spectral Contrast Angle

Katty X. Wan, Ilan Vidavsky, and Michael L. Gross
Department of Chemistry, Washington University, St. Louis, Missouri, USA

J Am Soc Mass Spectrom 2002, 13, 85–88

$$\cos\theta = \frac{\sum_i a_i b_i}{\sqrt{\sum_i a_i^2 \sum_i b_i^2}}$$

Alternatively, a simpler approach using a spreadsheet can be utilized in cases where the number of comparisons to make is small. The approach should be useful for high-throughput applications such as identification of metabolites and characterization of combinatorial libraries.

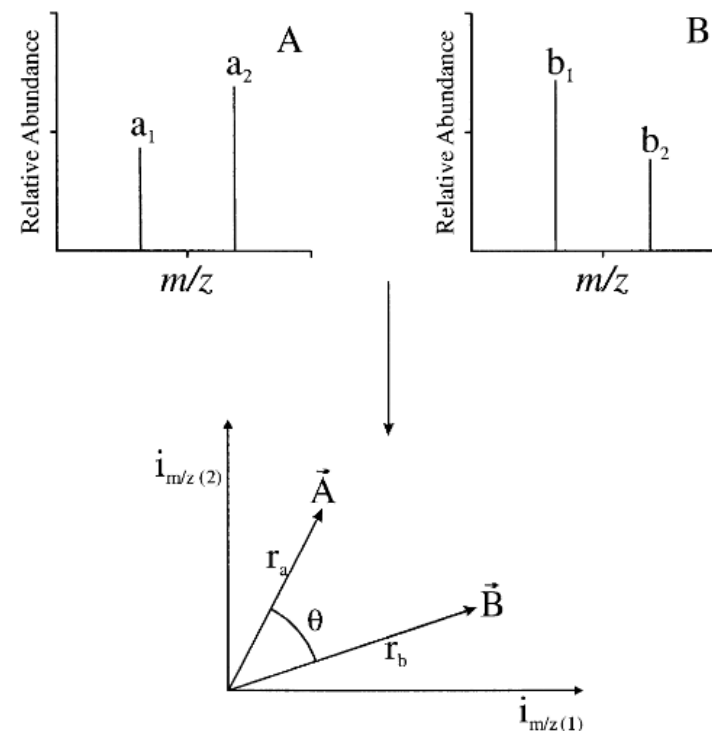
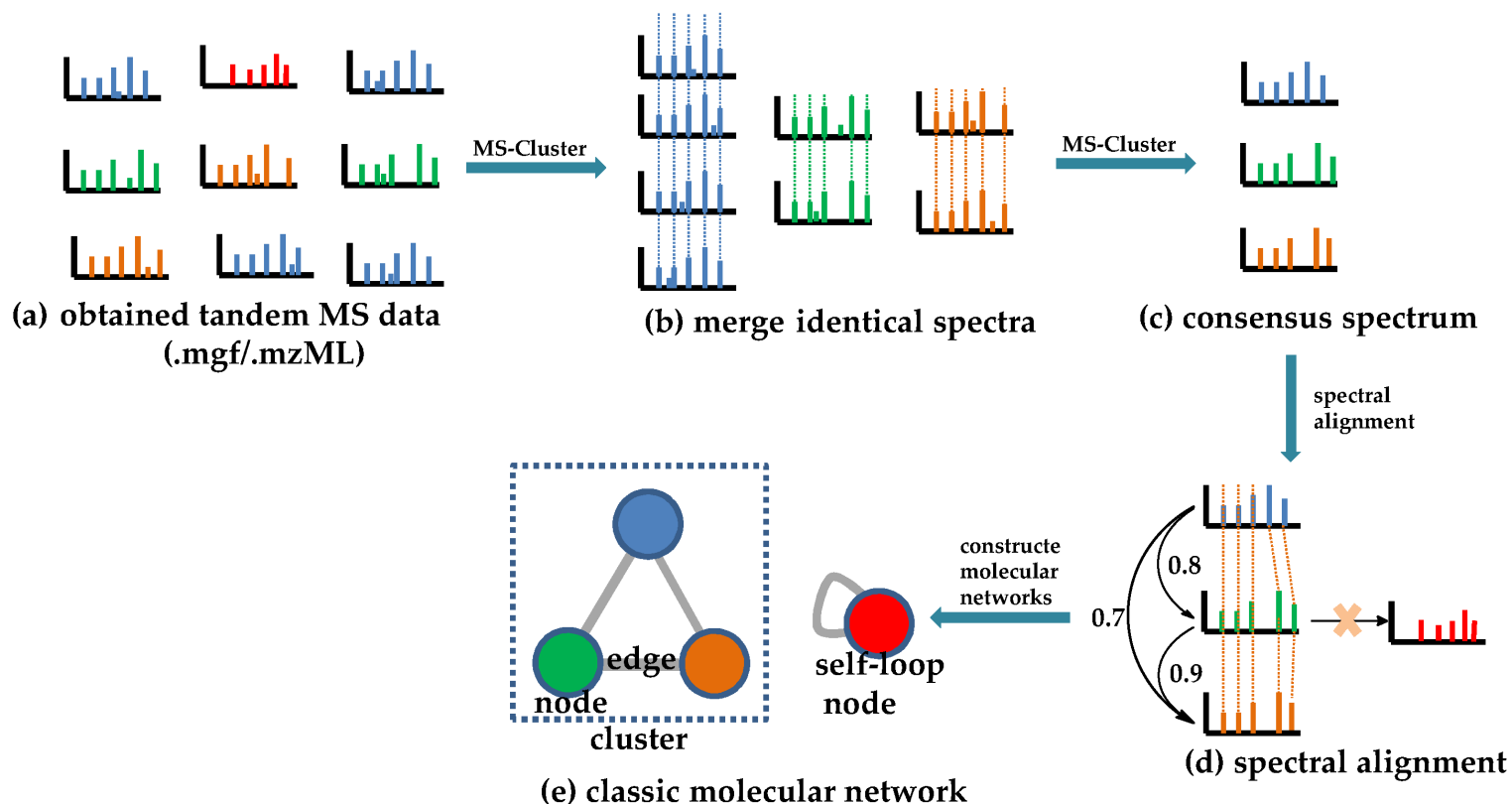


Figure 1. Schematic vector representations of the Spectra A and B.



Proc Natl Acad Sci U S A. 2012 Jun 26;109(26):E1743-52.
doi:10.1073/pnas.1203689109. Epub 2012 May 14.

<https://doi.org/10.3390/molecules28010157>

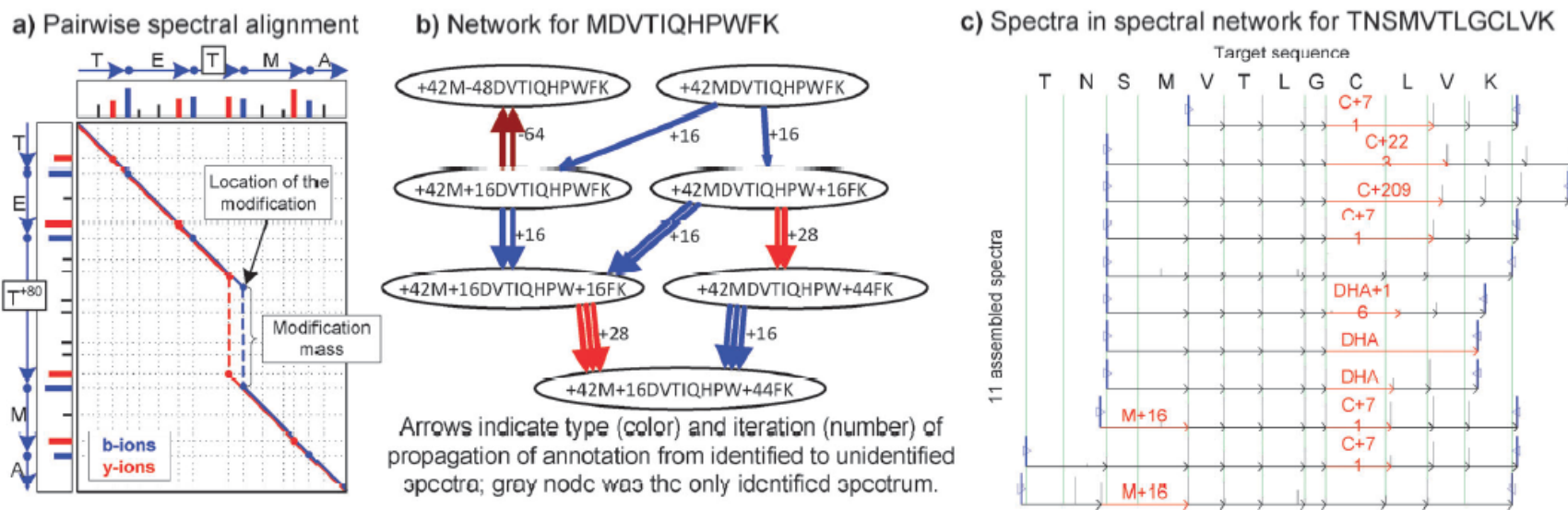


Fig. 1 Discovery and identification of post-translational modifications through spectral networks; (a) Spectral alignment between modified and unmodified variants of the peptide TETMA (*b*-ions shown in blue, *y*-ions in red, blue/red lines track consecutively matched *b*/*y*-ions); (b) Grouped modification states of the peptide MDVTIQHPWFK from a sample of cataractous lenses. Nodes in the spectral network represent individual MS² spectra and edges between nodes represent significant spectral alignments such as that shown in part (a); (c) Spectra assembled in the spectral network for TNSMVTLGCLVK with diverse Cysteine modifications on a monoclonal antibody. Each arrow corresponds to the propagation of a sequence and/or PTM from an identified spectrum to an unidentified spectrum (repeated arrows are iterative propagations). Arrow colors correspond to types of modifications transferred.

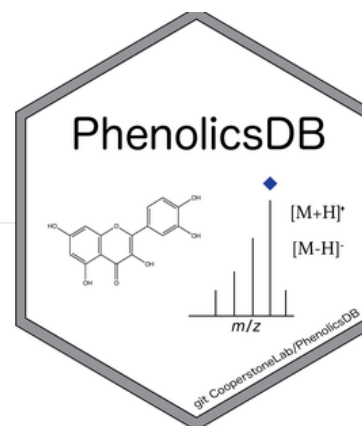
Mol. BioSyst., 2012, **8**, 2535–2544

and even thousands of spectra from the same peptide. Instead of repeating the identification process for each spectrum, it can be beneficial to perform the search only once using a representative consensus spectrum per peptide and later apply the results to all similar spectra.^{41–43} By analyzing only representative spectra (one per cluster of spectra from each precursor mass), our MS-Cluster algorithm¹¹ scaled this approach for the analysis of tens of millions of spectra and resulted in a significant speed-up of MS² database searches (up to 10 fold)

1. Are suffering of connexion problems to GNPS / GNPS2 ?
2. Are you allowed to deposit all your datasets on a public database in the USA ?
3. Do you need to redo all the GNPS pipeline when changing the cosinescore threshold of your molecular network ?
4. Do you want to learn Cytoscape after mastering GNPS website to generate figures for your article ?

PhenolicsDB

PhenolicsDB is a data package that contains raw MS/MS data of authentic analytical standards. **PhenolicsDB** raw data files and data frames are aimed to exemplify the use of [MS2extract](#) to create in-house metabolite MS/MS libraries. Between the list of metabolites we include phenolic acids, flavonoids, dihydrochalcones, carbohydrates, and others. Tandem mass spectrometry data was collected using a LC coupled to a Q-TOF mass analyzer.



Download PhenolicsDB

You can download PhenolicsDB in these links:

- Negative polarity [\(link\)](#)
- Positive polarity [\(link\)](#)

You can also access the .mzML files used for this library in the following [link](#)

Links

[Browse source code](#)

[Report a bug](#)

License

[Full license](#)

[CC BY 4.0](#)

Citation

[Citing PhenolicsDB](#)

Developers

Cristian Quiroz-Moreno

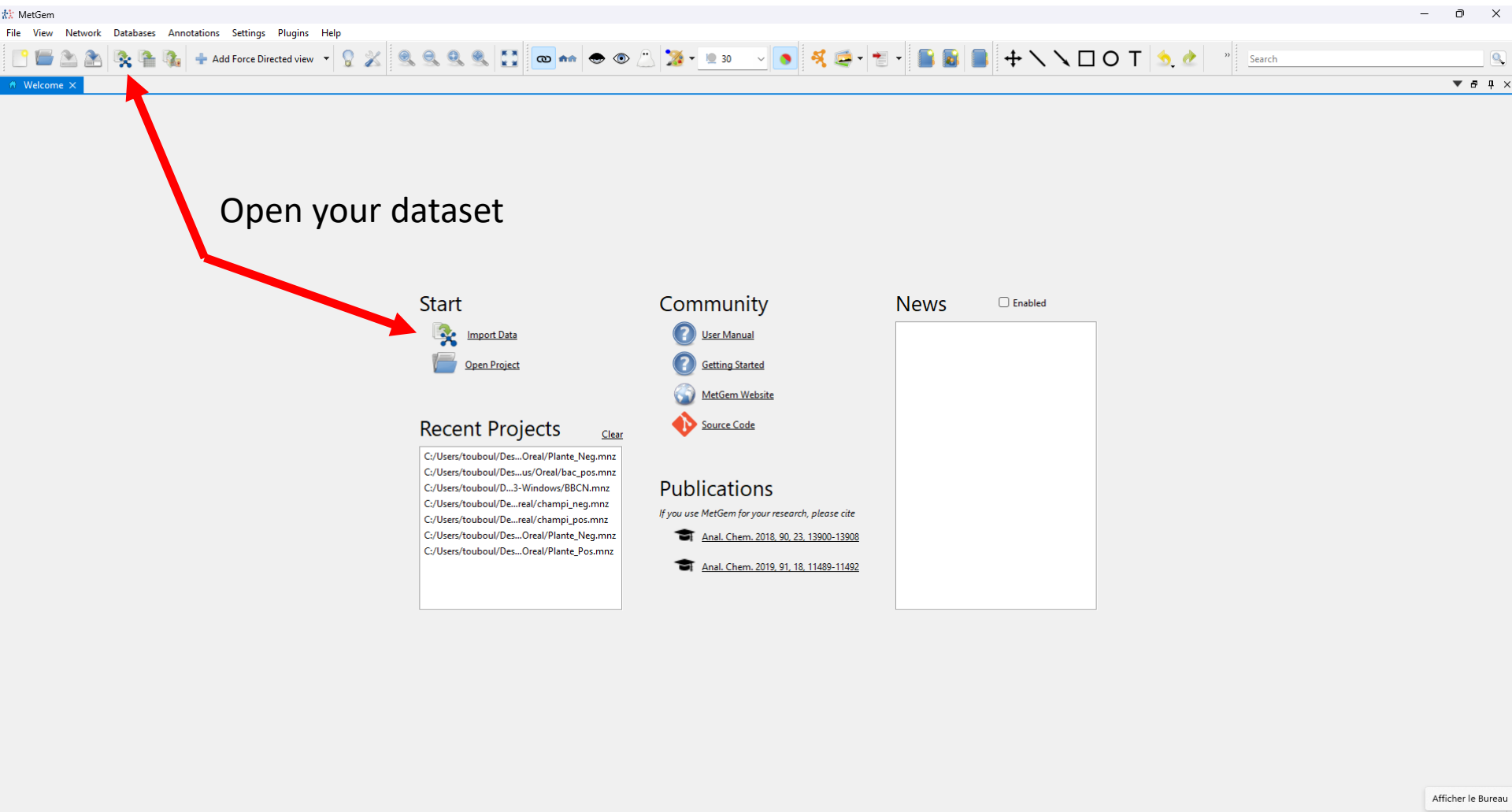
Author, maintainer 

Jessica Cooperstone

Author 

Dev status

[DOI](#)



The screenshot shows the MetGem software interface. At the top is a menu bar with 'File', 'View', 'Network', 'Databases', 'Annotations', 'Settings', 'Plugins', and 'Help'. Below the menu bar is a toolbar with various icons. A red arrow points from the text 'Open your dataset' to the 'Open Project' button in the 'Start' section. The 'Start' section also includes an 'Import Data' button. Below 'Start' is a 'Recent Projects' list with a 'Clear' link. To the right of 'Start' is a 'Community' section with links to 'User Manual', 'Getting Started', 'MetGem Website', and 'Source Code'. Below 'Community' is a 'Publications' section with two citations. To the right of 'Community' is a 'News' section with an 'Enabled' checkbox and a large empty box. The bottom right corner has a button labeled 'Afficher le Bureau'.

Open your dataset

Start

- Import Data
- Open Project

Recent Projects [Clear](#)

- C:/Users/touboul/Des...Oreal/Plante_Neg.mnz
- C:/Users/touboul/Des...us/Oreal/bac_pos.mnz
- C:/Users/touboul/D...3-Windows/BBCN.mnz
- C:/Users/touboul/De...real/champi_neg.mnz
- C:/Users/touboul/De...real/champi_pos.mnz
- C:/Users/touboul/Des...Oreal/Plante_Neg.mnz
- C:/Users/touboul/Des...Oreal/Plante_Pos.mnz

Community

- [User Manual](#)
- [Getting Started](#)
- [MetGem Website](#)
- [Source Code](#)

Publications

If you use MetGem for your research, please cite

- [Anal. Chem. 2018, 90, 23, 13900-13908](#)
- [Anal. Chem. 2019, 91, 18, 11489-11492](#)

News ☐ Enabled

Afficher le Bureau

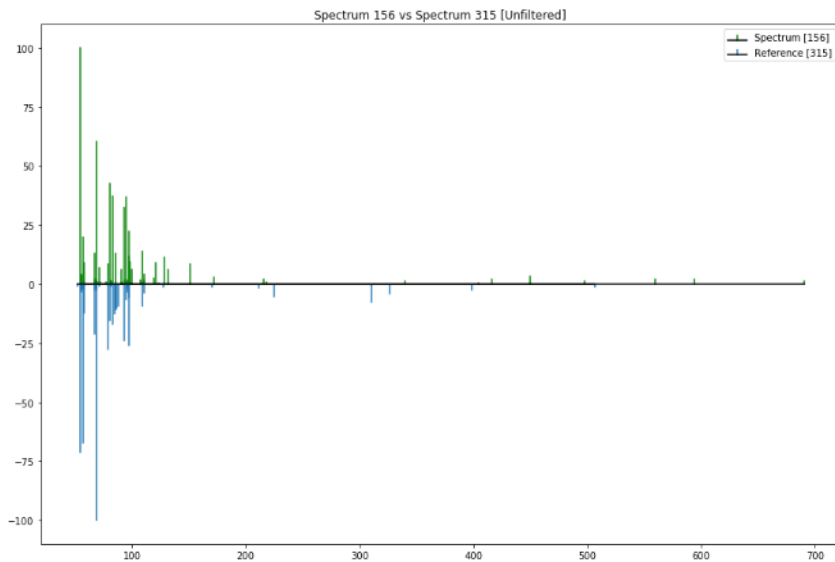
The screenshot shows the 'Process File' dialog box in the MetGem software. The dialog is divided into several sections: 'File to process', 'Metadata', 'Cosine Score Computing', and 'Filtering'. Red arrows point from external text labels to specific settings in the dialog:

- .mgf**: Points to the 'Choose a data file to import' field.
- .CSV**: Points to the 'Choose a metadata file' field.
- Only for GC-MS**: Points to the 'Load spectra as MS1 Data' checkbox.
- All filters**: Points to the 'Filtering' section, which includes:
 - ☒ Filtering
 - ☒ Keep peaks higher than 50 Th
 - ☒ Keep peaks outside of the ± 17 Th window
 - ☐ Keep peaks above 0% of maximum
 - ☒ Keep each peak in top 6 in the ± 50 Th window
- Important to fix !**: Points to the 'm/z Tolerance' field, which is set to 0.0200.

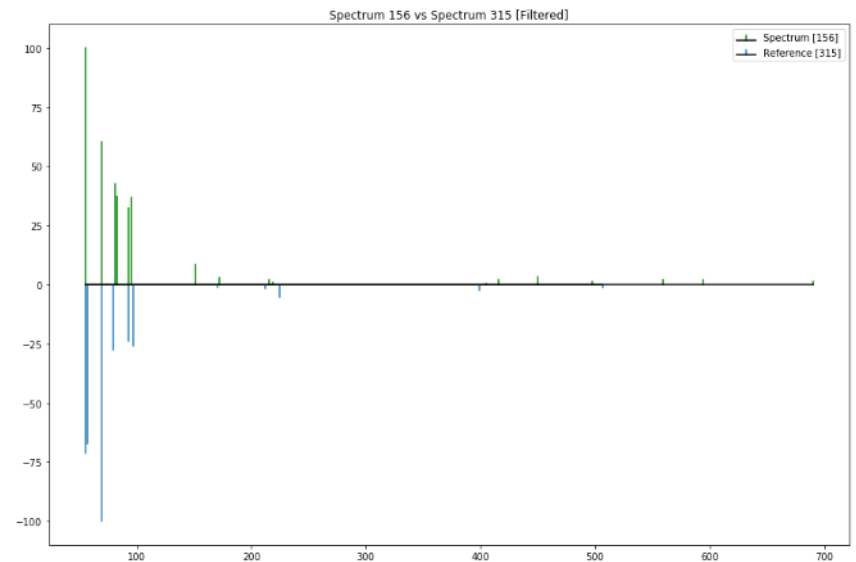
Other settings visible in the dialog include 'Minimum Matched Peaks' set to 4, and 'Use sparse similarity matrix' which is unchecked. The 'Add Views' panel on the right is empty. At the bottom are 'OK' and 'Cancel' buttons.

Filter noise (2)

Unfiltered



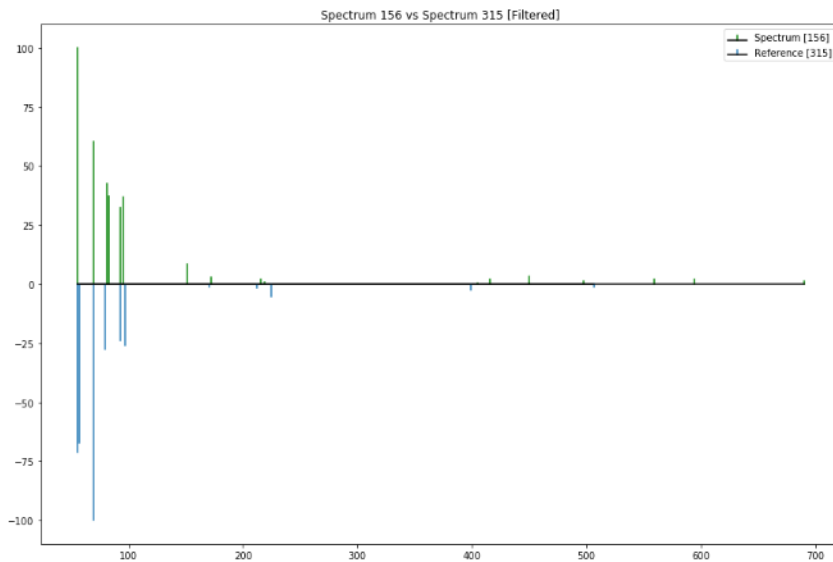
Filtered



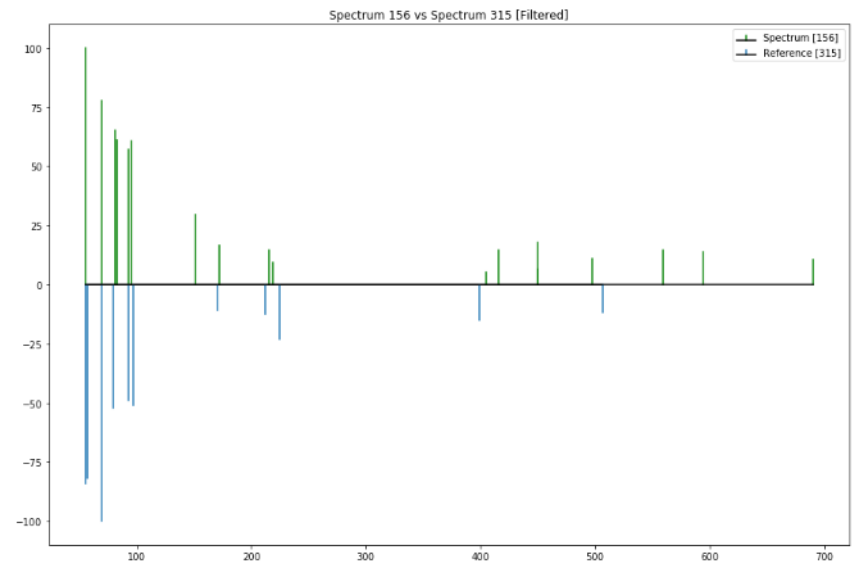
Use square root of intensities

(to minimize/maximize effects of high/low intensity peaks)

Before



After



Calculate cosine score similarity

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|_2 \|\mathbf{B}\|_2} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Find matching peaks...

Spectrum

precursor: m/z 336.3256

id	m/z	Normalized intensity
1	55.0542	100
2	69.0698	77.7
3	81.0697	65.3
4	83.0845	61.1
5	93.0692	57.1
6	95.0851	60.6
7	151.1463	29.4
8	172.0928	16.6
9	215.3468	14.5
10	218.4671	9.2
11	404.6468	5.5
12	416.0892	14.4
13	449.8349	17.7
14	449.9614	6.5
15	497.1872	10.9
16	559.1857	14.4
17	594.035	13.9
18	690.107	10.4

Reference

precursor: m/z 310.3105

id	m/z	Normalized intensity
1	55.0539	84.3
2	57.0695	81.9
3	69.0693	100
4	79.0539	52.3
5	93.07	48.9
6	97.065	50.8
7	170.1688	10.9
8	211.7954	12.5
9	225.028	23.2
10	398.7962	15.1
11	506.6284	11.6

...for fragments...

Spectrum

precursor: *m/z* 336.3256

id	<i>m/z</i>	Normalized intensity
1	55.0542	100
2	69.0698	77.7
3	81.0697	65.3
4	83.0845	61.1
5	93.0692	57.1
6	95.0851	60.6
7	151.1463	29.4
8	172.0928	16.6
9	215.3468	14.5
10	218.4671	9.2
11	404.6468	5.5
12	416.0892	14.4
13	449.8349	17.7
14	449.9614	6.5
15	497.1872	10.9
16	559.1857	14.4
17	594.035	13.9
18	690.107	10.4

Reference

precursor: *m/z* 310.3105

id	<i>m/z</i>	Normalized intensity
1	55.0539	84.3
2	57.0695	81.9
3	69.0693	100
4	79.0539	52.3
5	93.07	48.9
6	97.065	50.8
7	170.1688	10.9
8	211.7954	12.5
9	225.028	23.2
10	398.7962	15.1
11	506.6284	11.6

...and for neutral losses

Spectrum

precursor: m/z 336.3256

id	m/z	Normalized intensity
1	281.2714	100
2	267.2558	77.7
3	255.2559	65.3
4	253.2412	61.1
5	243.2564	57.1
6	241.2406	60.6
7	185.1794	29.4
8	164.2328	16.6
9	120.9788	14.5
10	117.8585	9.2
11	-68.3211	5.5
12	-79.7635	14.4
13	-113.5093	17.7
14	-113.6358	6.5
15	-160.8616	10.9
16	-222.86	14.4
17	-257.7094	13.9
18	-353.7814	10.4

Reference

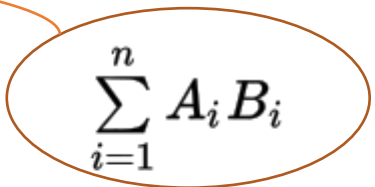
precursor: m/z 310.3105

id	m/z	Normalized intensity
1	255.2566	84.3
2	253.241	81.9
3	241.2412	100
4	231.2566	52.3
5	217.2406	48.9
6	213.2455	50.8
7	140.1418	10.9
8	98.5152	12.5
9	85.2825	23.2
10	-88.4857	15.1
11	-196.3178	11.6

Produce a list of matches

Peak id spectrum	Peak id reference	score
1	1	84.3
2	3	77.7
5	5	27.9
3	1	55.1
4	2	50.1
6	3	60.6

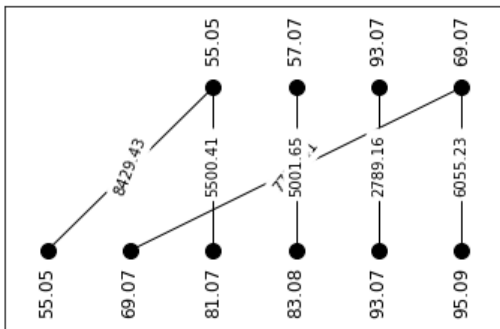
Score = product of intensities

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|_2 \|\mathbf{B}\|_2} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$


Headaches are coming!

Peak id spectrum	Peak id reference	score
1	1	84.3
2	3	77.7
5	5	27.9
3	1	55.1
4	2	50.1
6	3	60.6

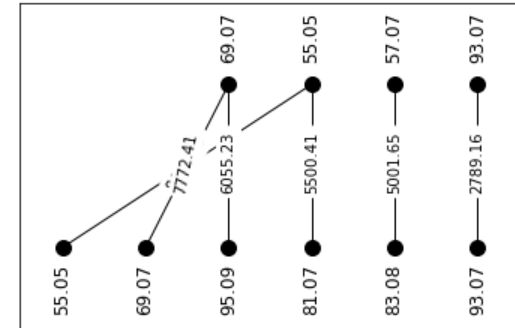
One peak in a spectrum can match with more than one peak in the other one. Which one do we need to choose?



Problem can be seen as maximum weight matching in bipartite weighted graph

Solutions

- In our case, a simple greedy algorithm will lead to the solution :
 - 1. Sort the table by decreasing score



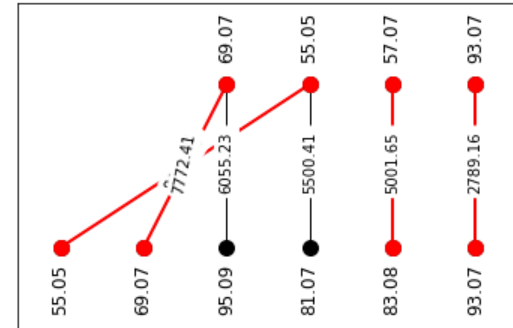
Peak id spectrum	Peak id reference	score
1	1	84.3
2	3	77.7
6	3	60.6
3	1	55.1
4	2	50.1
5	5	27.9



Solutions

- In our case, a simple greedy algorithm will lead to the solution :

- 1. Sort the table by decreasing score
- 2. For each line, keep it if both peaks in the two spectra were not used before



Peak id spectrum	Peak id reference	score
1	1	84.3
2	3	77.7
6	3	60.6
3	1	55.1
4	2	50.1
5	5	27.9



Solutions

- In our case, a simple greedy algorithm will lead to the solution :
 - 1. Sort the table by decreasing score
 - 2. For each line, keep it if both peaks in the two spectra were not used before
 - 3. Sum scores of all remaining peaks

Peak id spectrum	Peak id reference	score
1	1	84.3
2	3	77.7
6	3	60.6
3	1	55.1
4	2	50.1
5	5	27.9
		Σ 240.0



Solutions

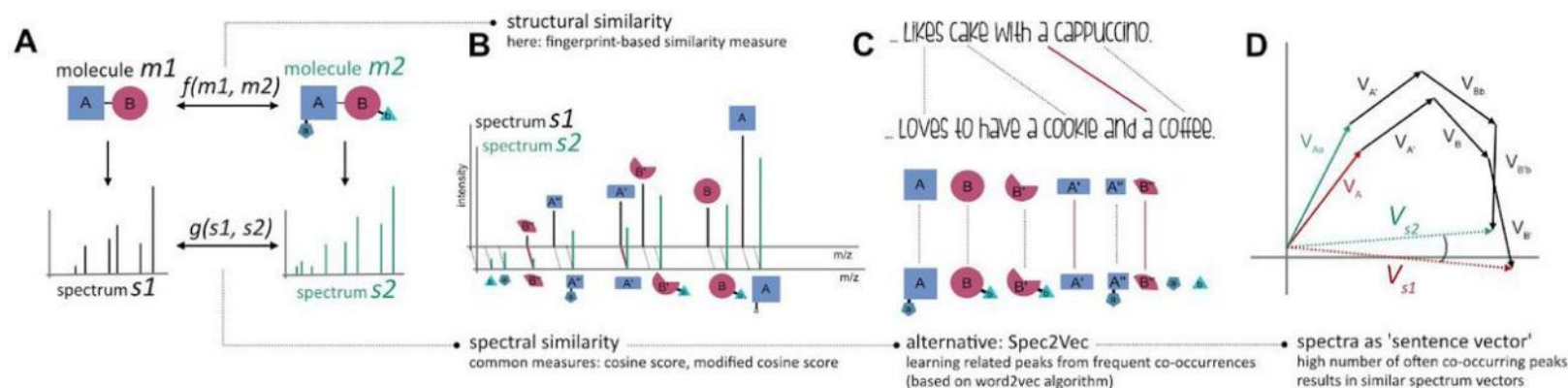
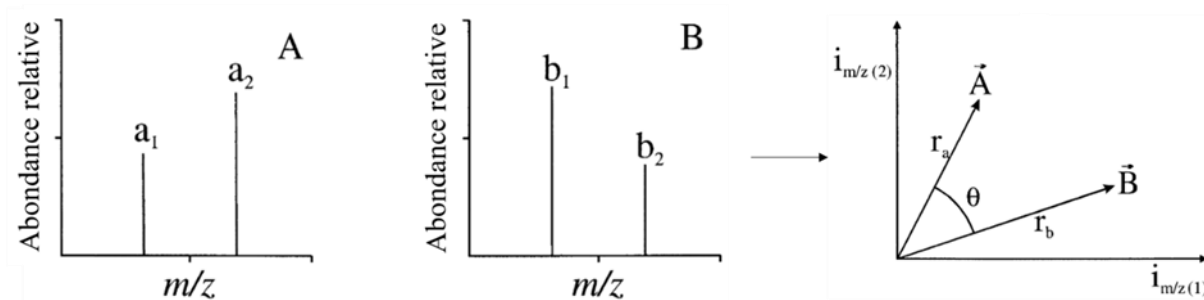
- In our case, a simple greedy algorithm will lead to the solution :
 - 1. Sort the table by decreasing score
 - 2. For each line, keep it if both peaks in the two spectra were not used before
 - 3. Sum scores of all remaining peaks
 - 4. Divide by the product of the norms of the two vectors

Peak id spectrum	Peak id reference	score
1	1	84.3
2	3	77.7
6	3	60.6
3	1	55.1
4	2	50.1
5	5	27.9
		Σ 240.0

$$240.0 / (183.3 * 180.8) \approx$$

0.72

$$\cos(\theta) = \frac{\sum_i a_i b_i}{\sqrt{\sum_i a_i^2 \sum_i b_i^2}}$$



2021

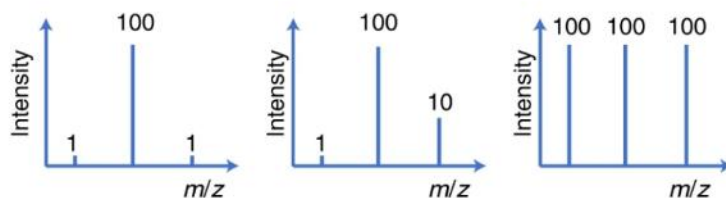
J. van der Hooft

Modified cosine score

Spectral entropy

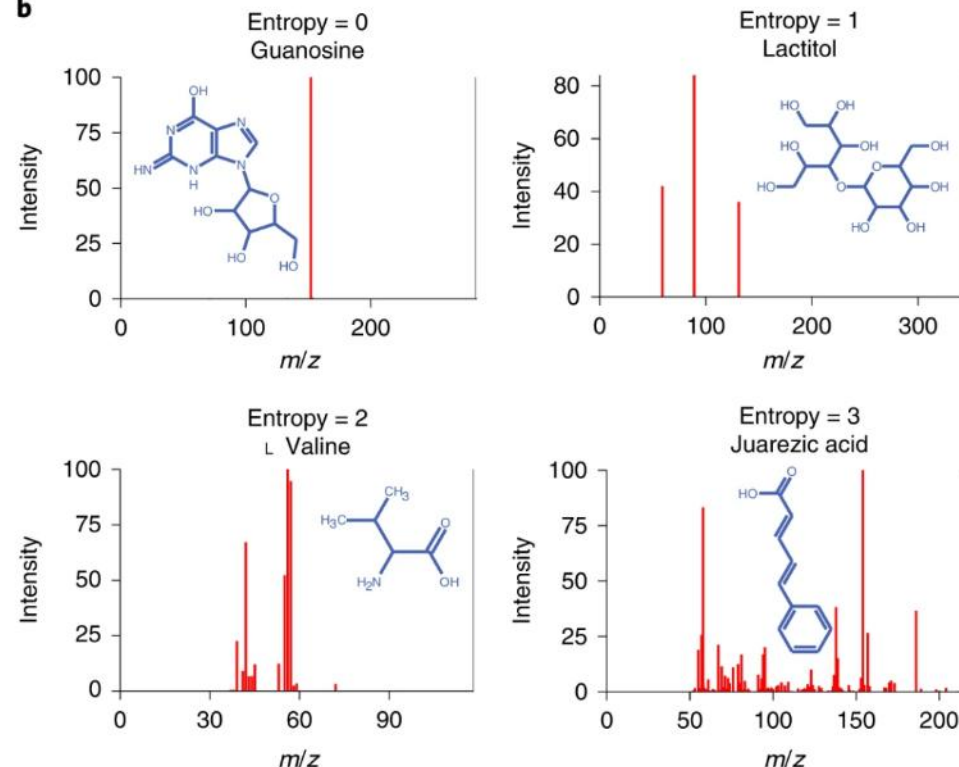
Spectral entropy

$$S = - \sum_p I_p \ln I_p$$



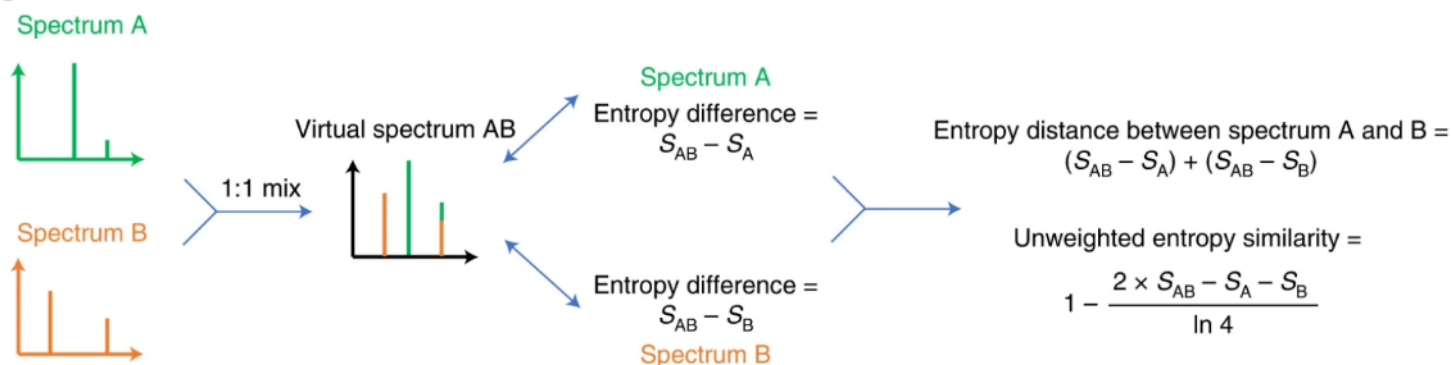
Peak number	3	3	3
Spectral entropy	0.11	0.35	1.09

b



Unweighted entropy

b



Options ✕

Title

Force Directed Visualization

Maximum Neighbor Number (topK)	10
Minimal Cosine Score Value	0.70
Max. Connected Component Size	1000

Force Atlas 2

Scale	30.0
Gravity	1.0

[Help](#)

ForceAtlas2, a Continuous Graph Layout Algorithm for Handy Network Visualization Designed for the Gephi Software



Mathieu Jacomy^{1,2,3*}, Tommaso Venturini¹, Sebastien Heymann^{3,4}, Mathieu Bastian³

1 Sciences Po, médialab, Paris, France, **2** Equipex DIME SHS, Paris, France, **3** Gephi Consortium, Paris, France, **4** LIP6 - CNRS - Université Pierre et Marie Curie, Paris, France

- **Repulsive Force:** Nodes experience a repulsive force that pushes them away from each other, reducing node overlaps.
- **Attractive Force:** Edges act like springs, pulling connected nodes together to maintain their relationship.
- **Equilibrium:** The layout reaches equilibrium when the forces are balanced, minimizing edge crossings and optimizing the graph layout.

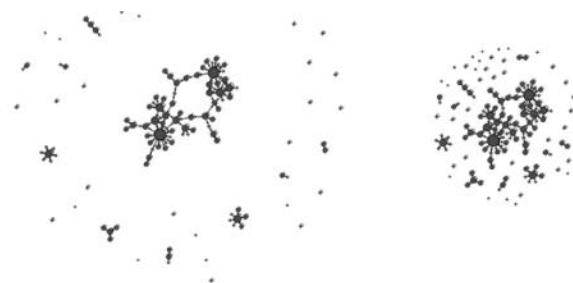


Figure 3. Effects of the gravity. ForceAtlas2 with gravity at 2 and 5. Gravity brings disconnected components closer to the center (and slightly affects the shape of the components as a side-effect).



ForceAtlas2, a Continuous Graph Layout Algorithm for Handy Network Visualization Designed for the Gephi Software

Mathieu Jacomy^{1,2,3*}, Tommaso Venturini¹, Sebastien Heymann^{3,4}, Mathieu Bastian³

¹ Sciences Po, médialab, Paris, France, ² Equipex DIME SHS, Paris, France, ³ Gephi Consortium, Paris, France, ⁴ LIP6 - CNRS - Université Pierre et Marie Curie, Paris, France

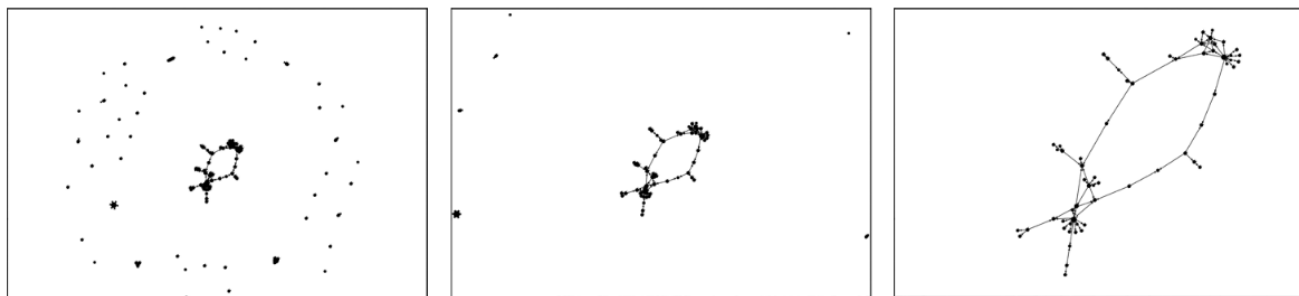
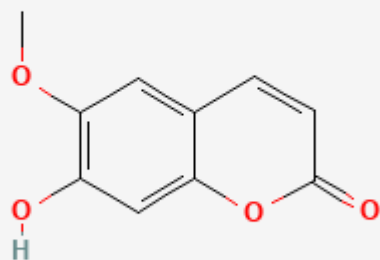
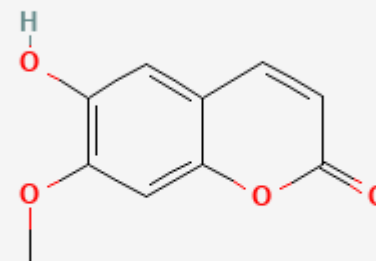


Figure 4. Effects of the scaling. ForceAtlas2 with scaling at 1, 2 and 10. The whole graph expands as scaling affects the distance between components as well as their size. Note that the size of the nodes remains the same; scaling is not zooming.

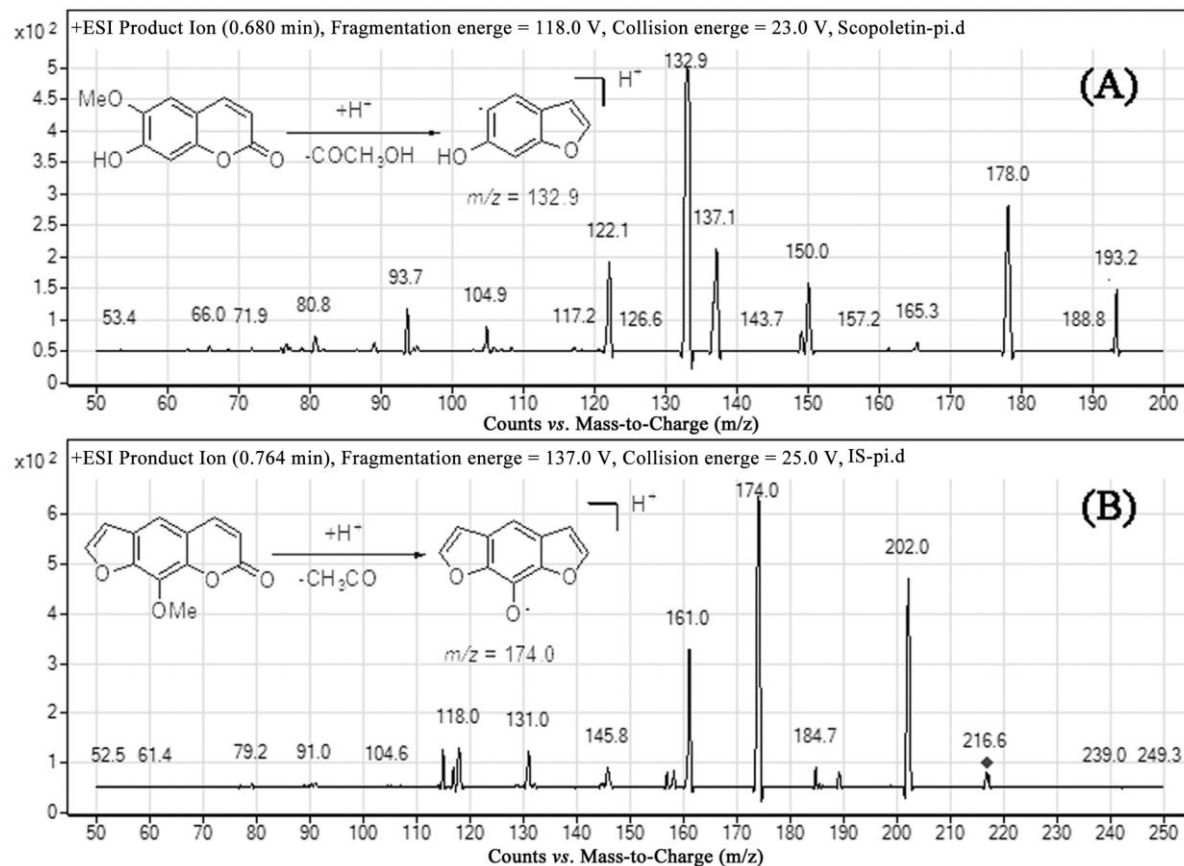
m/z 193.05

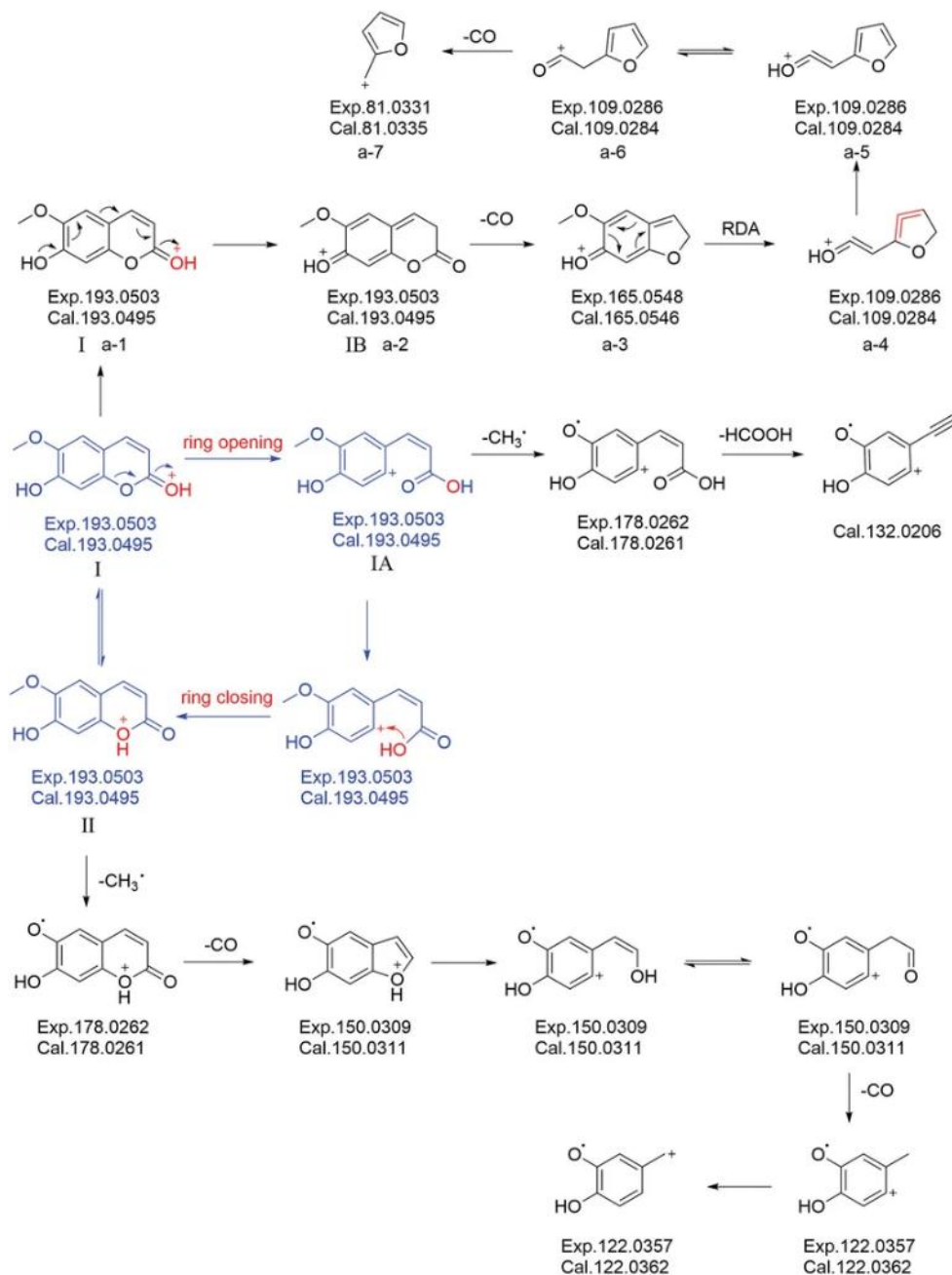


Scopoletin



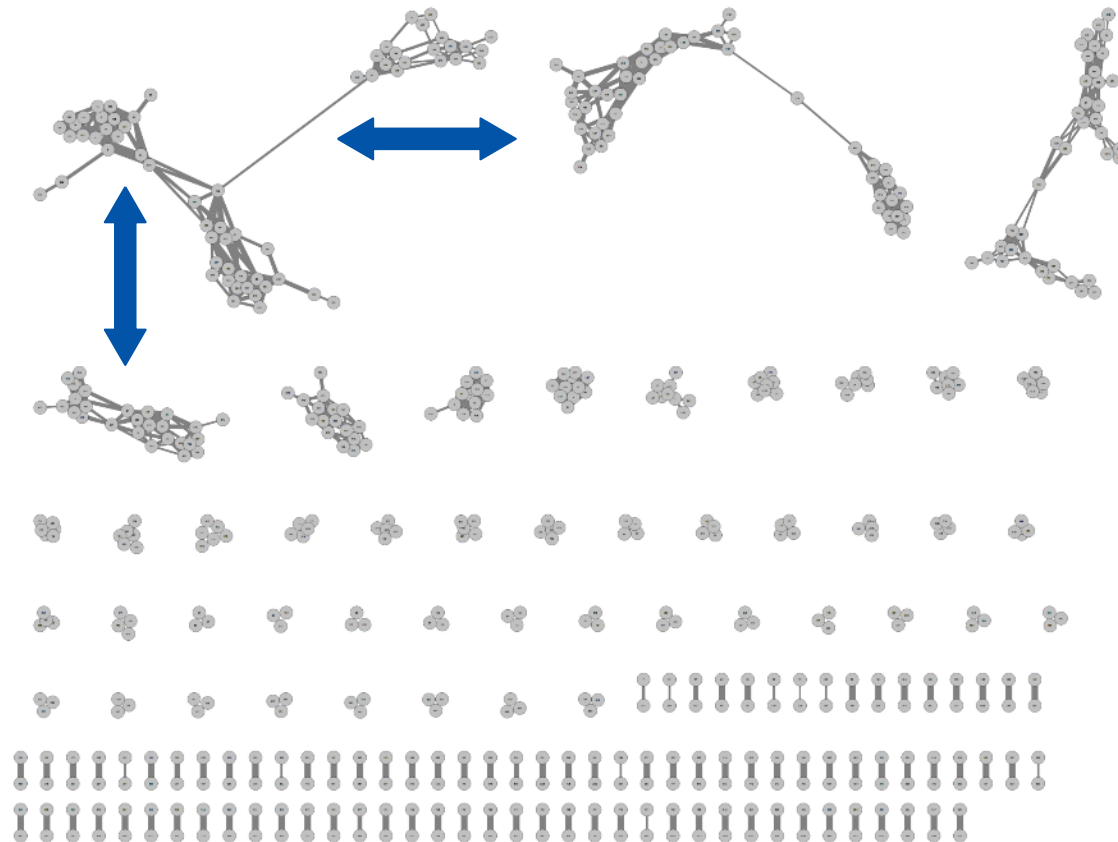
Isoscopoletin



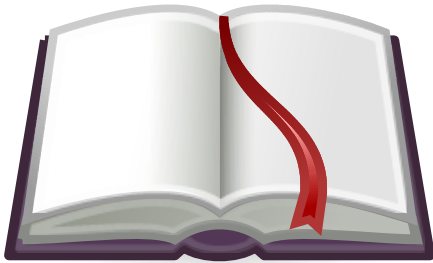


DOI: [10.1002/rcm.9481](https://doi.org/10.1002/rcm.9481)

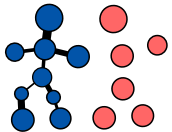
GNPS



t -distributed Stochastic Neighbor Embedding



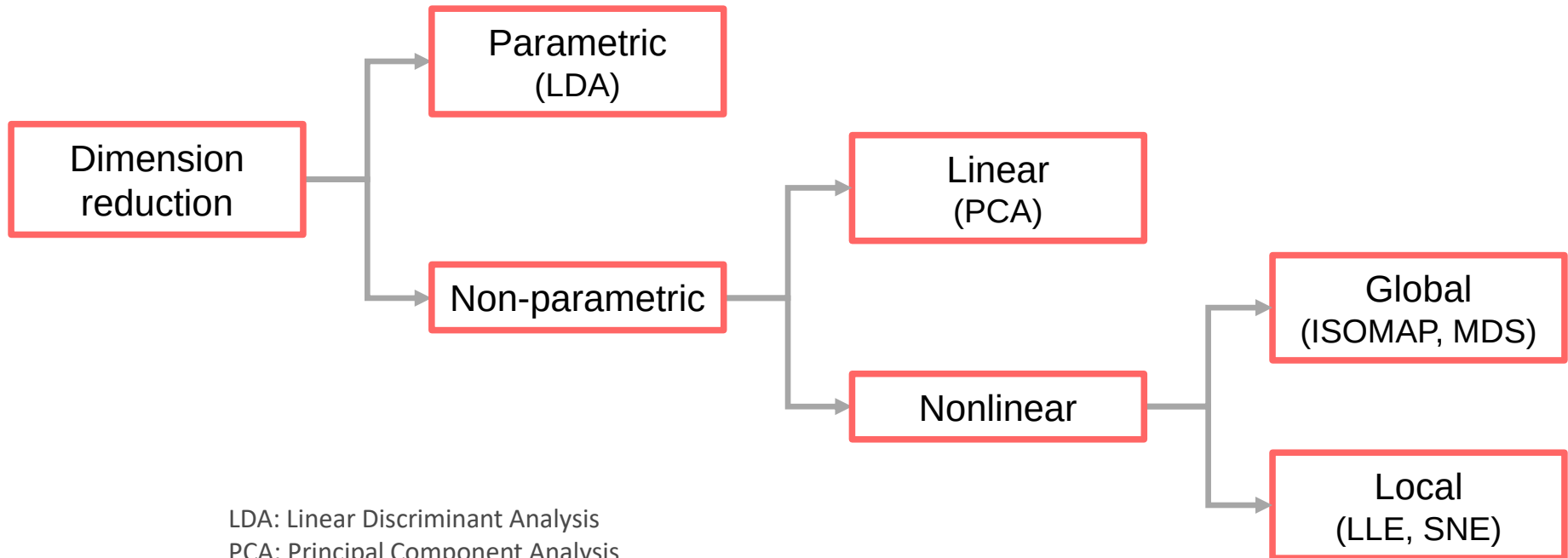
Non-linear projection of a n -dimensional space into a lower dimensional space with priority to local distances over global distances



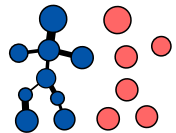
L. Van der Maaten *et al.*, *J. of M. Learning Research*, **2008**, 9, 2579-2605



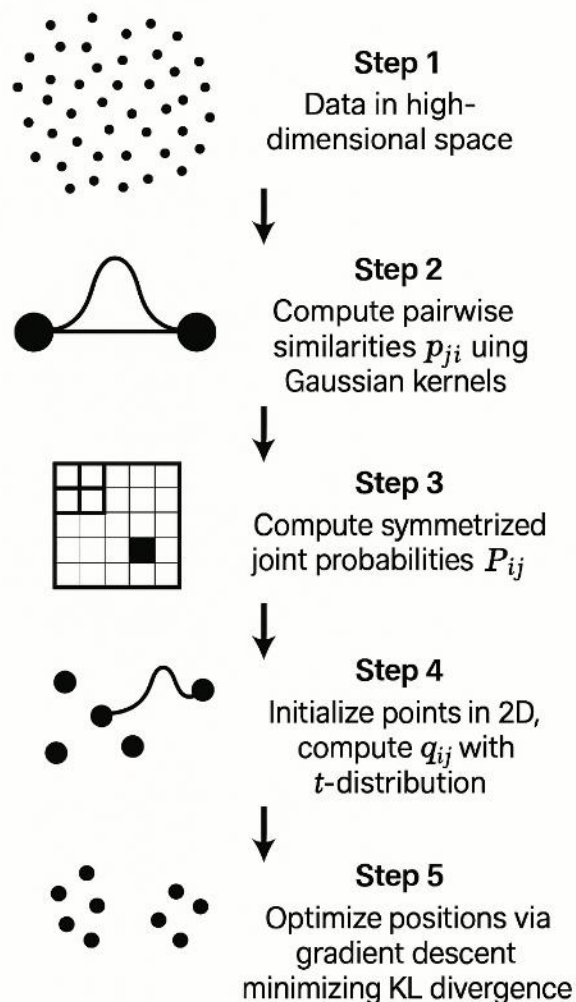
StatQuest: t-SNE, Clearly Explained: youtu.be/NEaUSP4YerM



LDA: Linear Discriminant Analysis
PCA: Principal Component Analysis
MDS: MultiDimensional Scaling
LLE: Locally-Linear Embedding
t-SNE: t-distributed stochastic neighbor embedding



t-SNE <https://distill.pub/2016/misread-tsne/>



1. High-dimensional data

Points distributed in a high-dimensional space (for example, 50D).

2. Gaussian similarity computation

For each pair of points, compute

$$p_{j|i}$$

using a Gaussian kernel centered on each point.

3. Symmetrization of probabilities

Construct the joint probabilities

$$P_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}.$$

4. 2D initialization + t-similarities

Randomly place the points in 2D, then compute

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

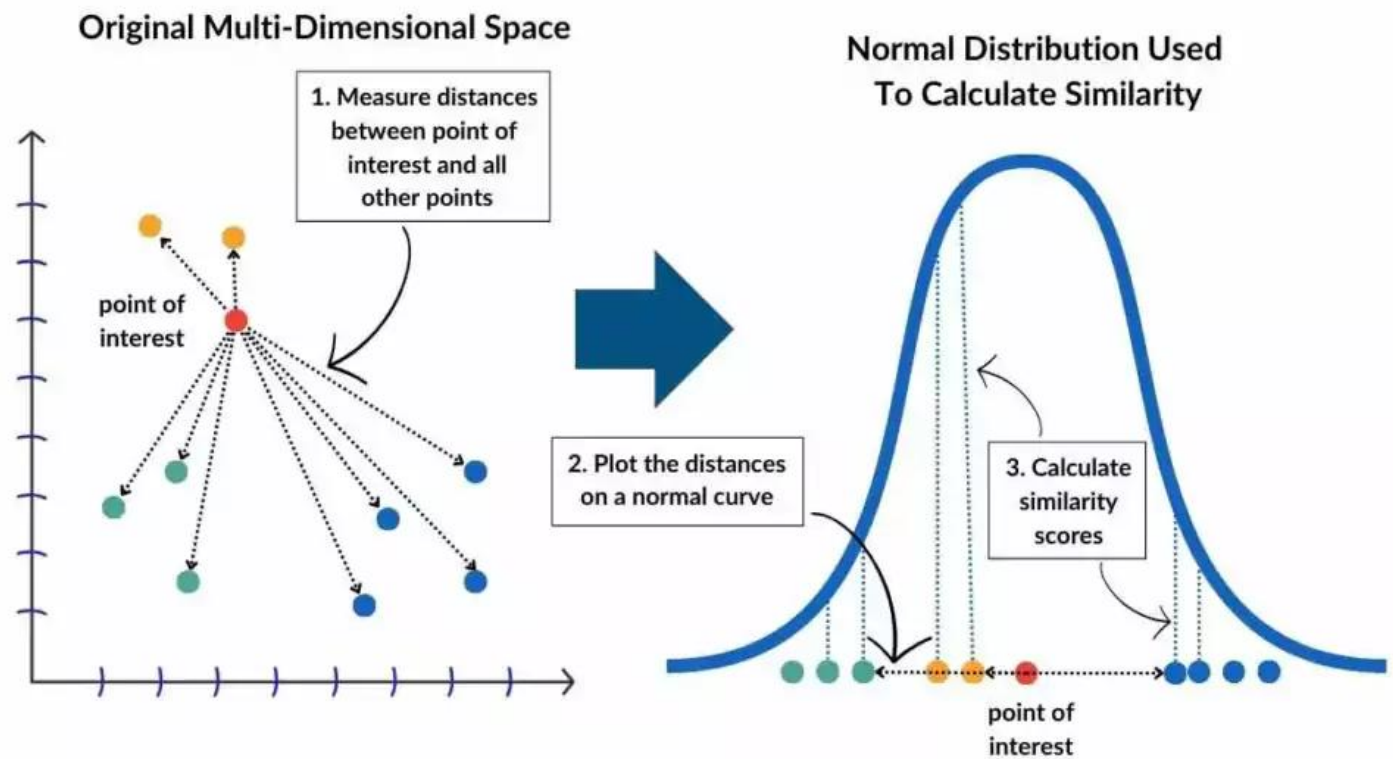
using a Student's t-distribution with one degree of freedom (long tails).

5. Optimization via gradient descent

Minimize the Kullback–Leibler divergence between P and Q :

$$\text{KL}(P \parallel Q) = \sum_{i \neq j} P_{ij} \log \frac{P_{ij}}{Q_{ij}}.$$

The 2D positions are then iteratively adjusted to preserve local neighborhoods.

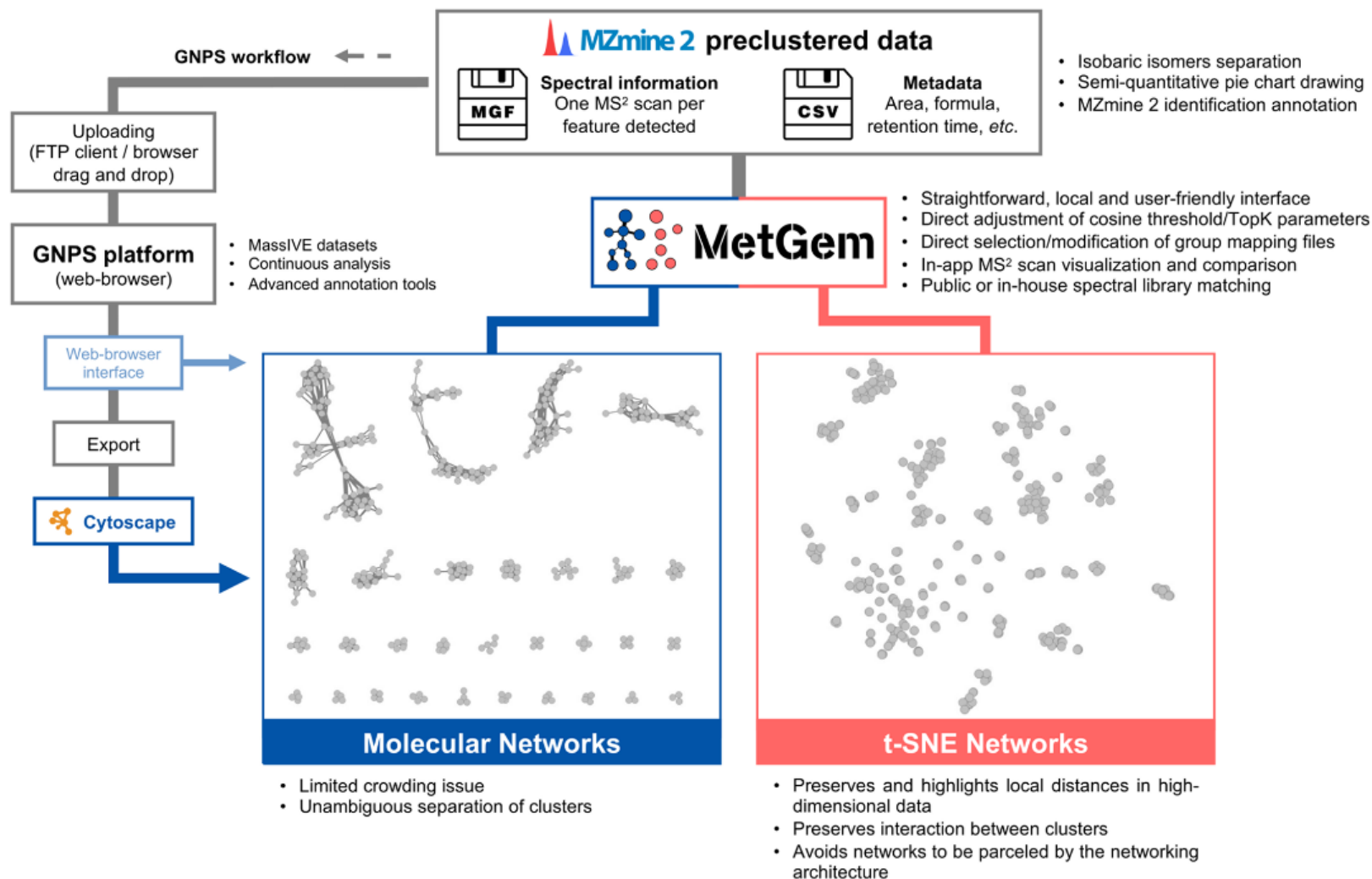


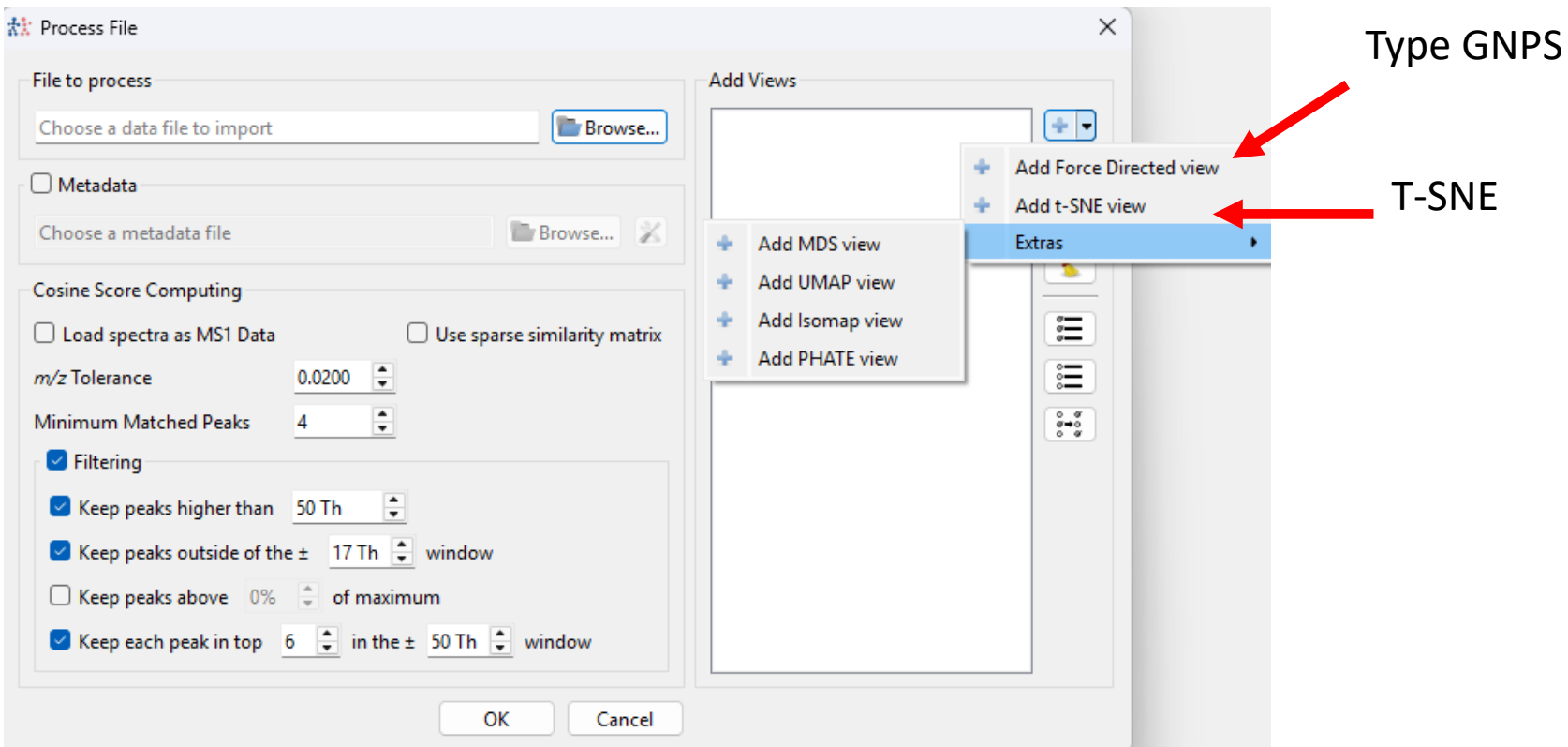
t-SNE <https://distill.pub/2016/misread-tsne/>

UMAP <https://pair-code.github.io/understanding-umap/>

Feature	PCA	t-SNE	UMAP
Principle	Linear orthogonal projection (SVD)	Non-linear embedding (KL divergence on pairwise similarities)	Non-linear manifold learning (fuzzy simplicial sets + cross-entropy)
Output	Ordered principal components (continuous coordinates)	Low-dimensional embedding (2D/3D, depends on perplexity & init)	Low-dimensional embedding (influenced by n_neighbors & min_dist)
Captures	Global variance structure	Local neighborhoods (very well)	Local & some global structure
Computational Complexity	$O(\min(ND^2, DN^2))$ via SVD	$O(N^2)$ (naïve) or $O(N \log N)$ with Barnes-Hut	$O(N \log N)$
Scalability	Handles very large N (millions) with optimized SVD	Moderate: ~100 k points (Barnes-Hut), several million with FFT-based optimizations, but slow	High: > 500 k points in minutes; near-linear scaling with approximate nearest neighbors
Reproducibility	Deterministic (no randomness)	Non-deterministic unless random_state and init are fixed	More stable (spectral init by default + random_state)
Key Hyperparameters	- Number of components	- perplexity- learning_rate- n_iter- init	- n_neighbors- min_dist- metric- init
Strengths	Fast & deterministic Linearly interpretable Good for noise reduction & downstream modeling	Excellent local separation Ideal for well-defined non-linear clusters	Balances local & global structure Faster than t-SNE on large data Preserves both clusters and continuous structures
Limitations	Only captures linear relationships May miss manifold structure	Computationally expensive for large N Preserves only local structure Absolute distances not interpretable	Sensitive to n_neighbors/min_dist settings May distort very local or very global scales

t-SNE <https://distill.pub/2016/misread-tsne/>





Options ✕

Title t-sne ✕

t-SNE Visualization

At least 1 cosine score(s) above 0.70

Number of iterations 10000 Perplexity 6

Learning Rate 10 Early exaggeration 12

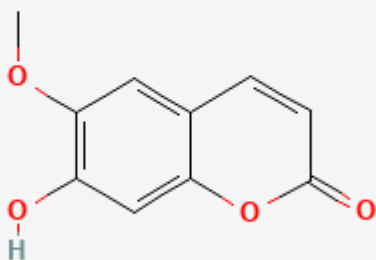
☐ Random Initial State

☒ Use Barnes-Hut approximation

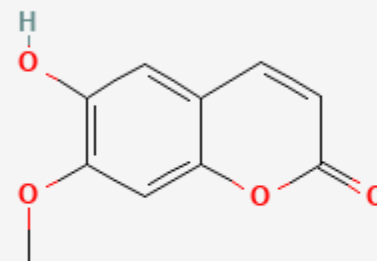
Angle 0.5

[Help](#)

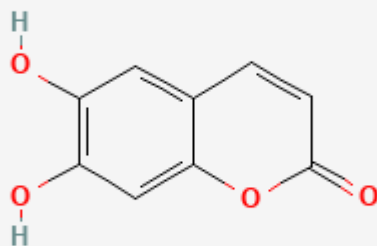
OK Cancel



Esculetin

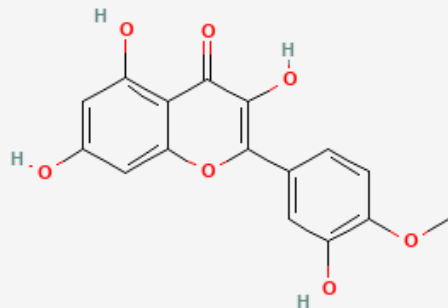


Scopoletin

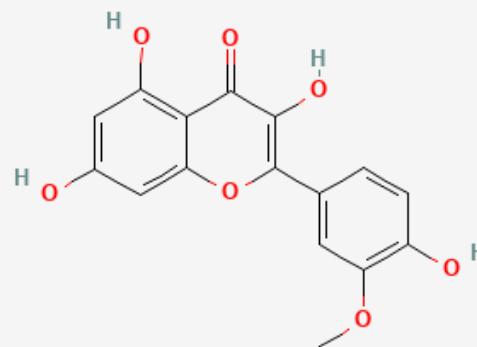


Isoscopoletin

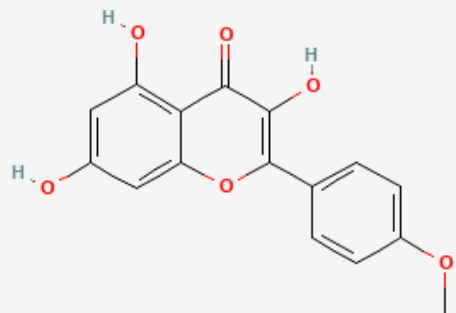
m/z 315.051 (negative ion mode)



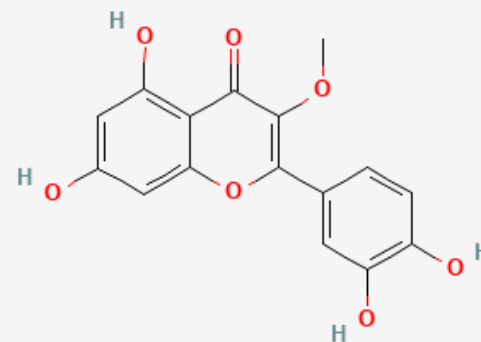
Tamarixetin



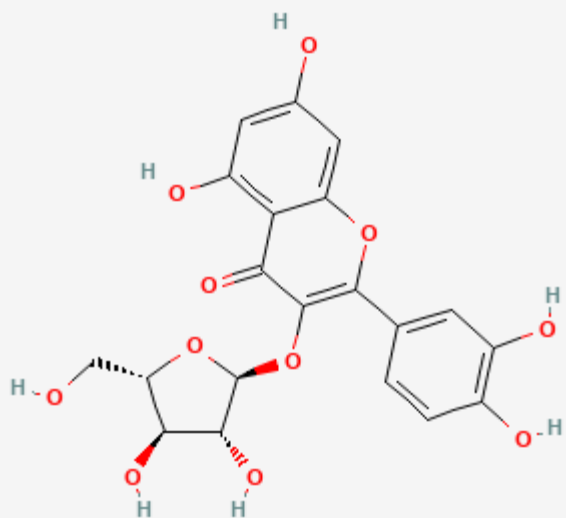
Isorhamnetin



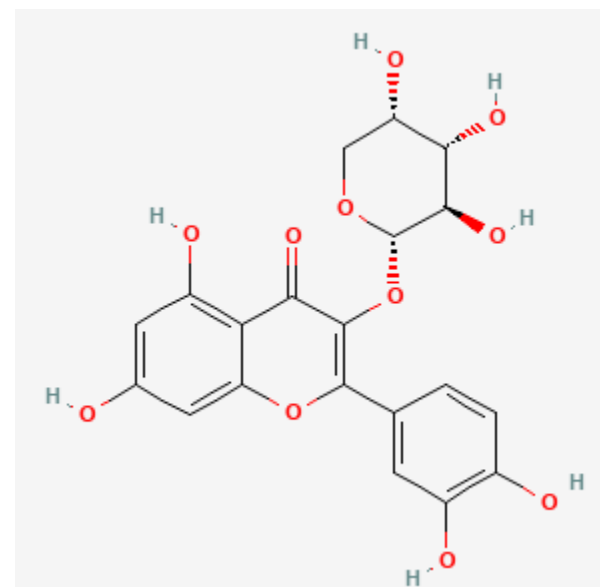
kaempferide



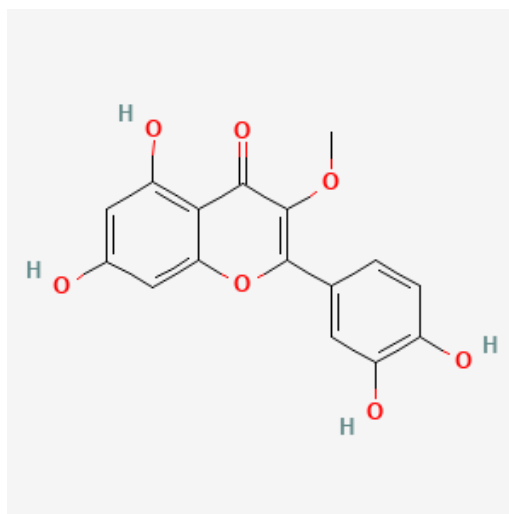
3-O-methylquercetin



Avicularin



Guaijaverin



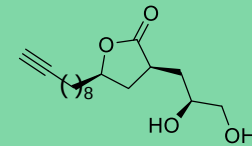
3-O-methylquercetin

Sextonia rubra: a holobiont model

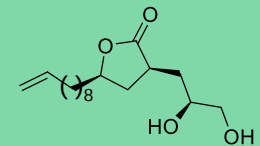
Sextonia rubra
→ Magnoliidae
→ Laurales
→ Lauraceae



Lactone derivatives



Rubrynolide



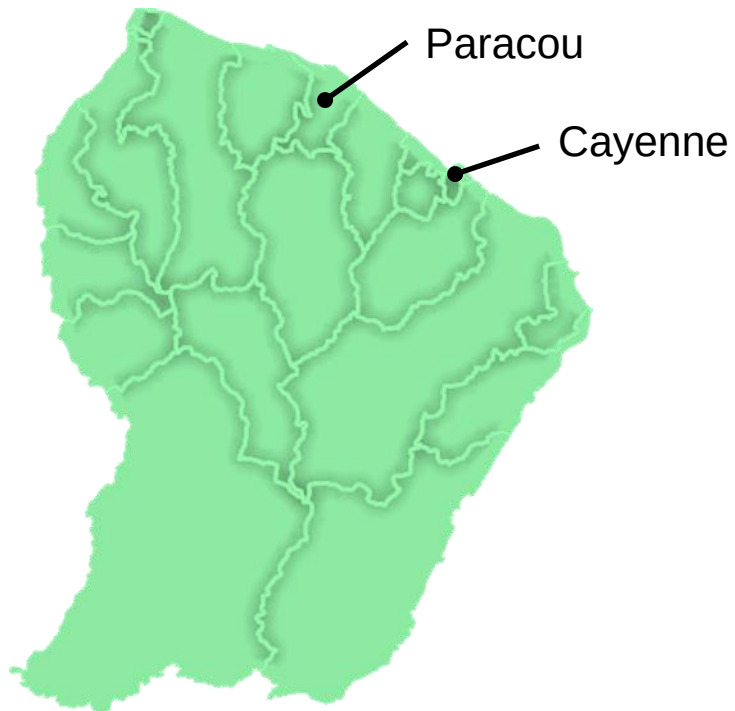
Rubrenolide

→ Fungicidal and termiticidal activity



→ 4th most exploited wood species
in French Guiana

Sextonia rubra sampling



Felling of a mature *S. rubra*

Sextonia rubra sampling

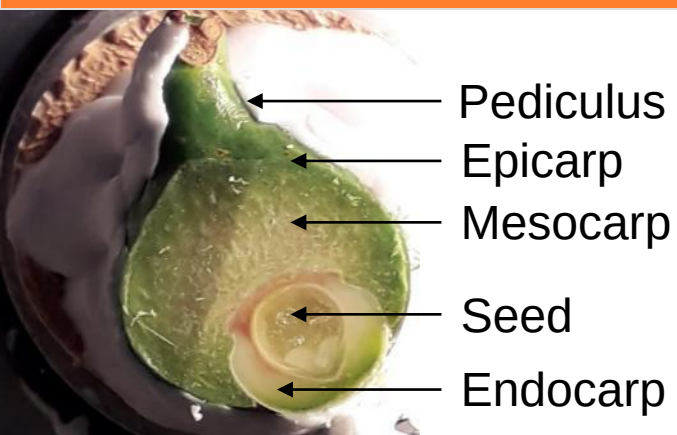


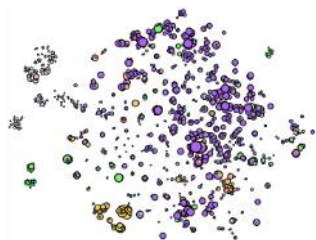
Felling of a mature *S. rubra*

Sextonia rubra sampling



Fruits chemical diversity – Volatile Organic Compounds

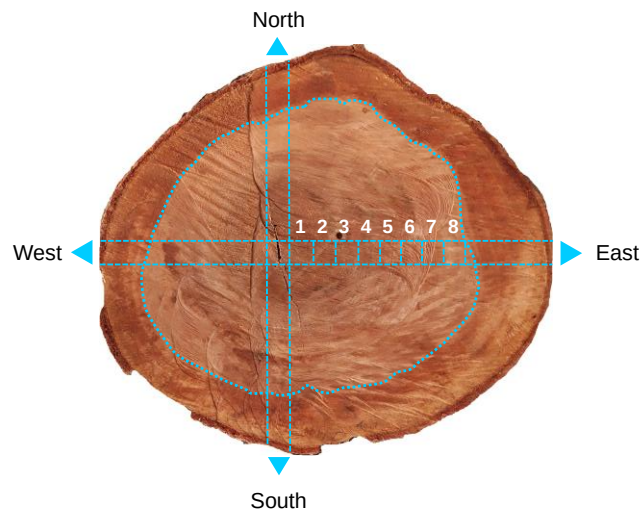


**EXTRACTION****DEREPLICATION****ANNOTATION**

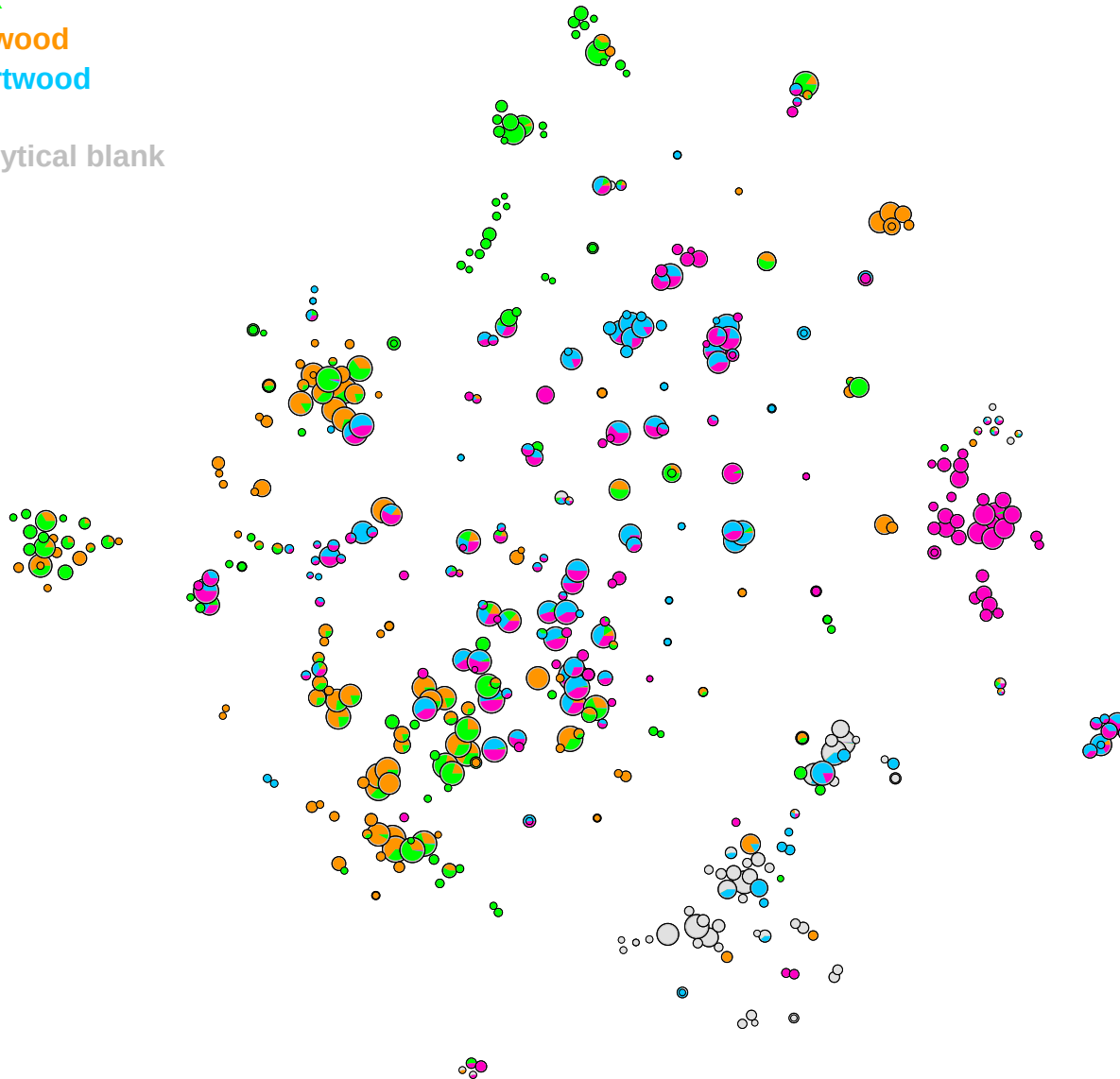
WILEY

NIST

LOTUS



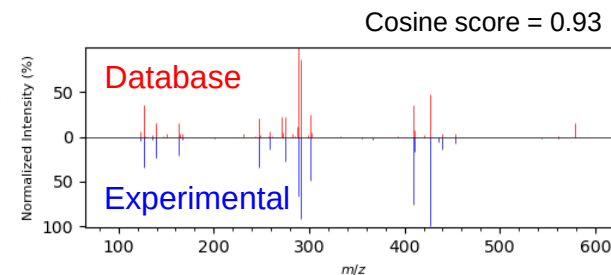
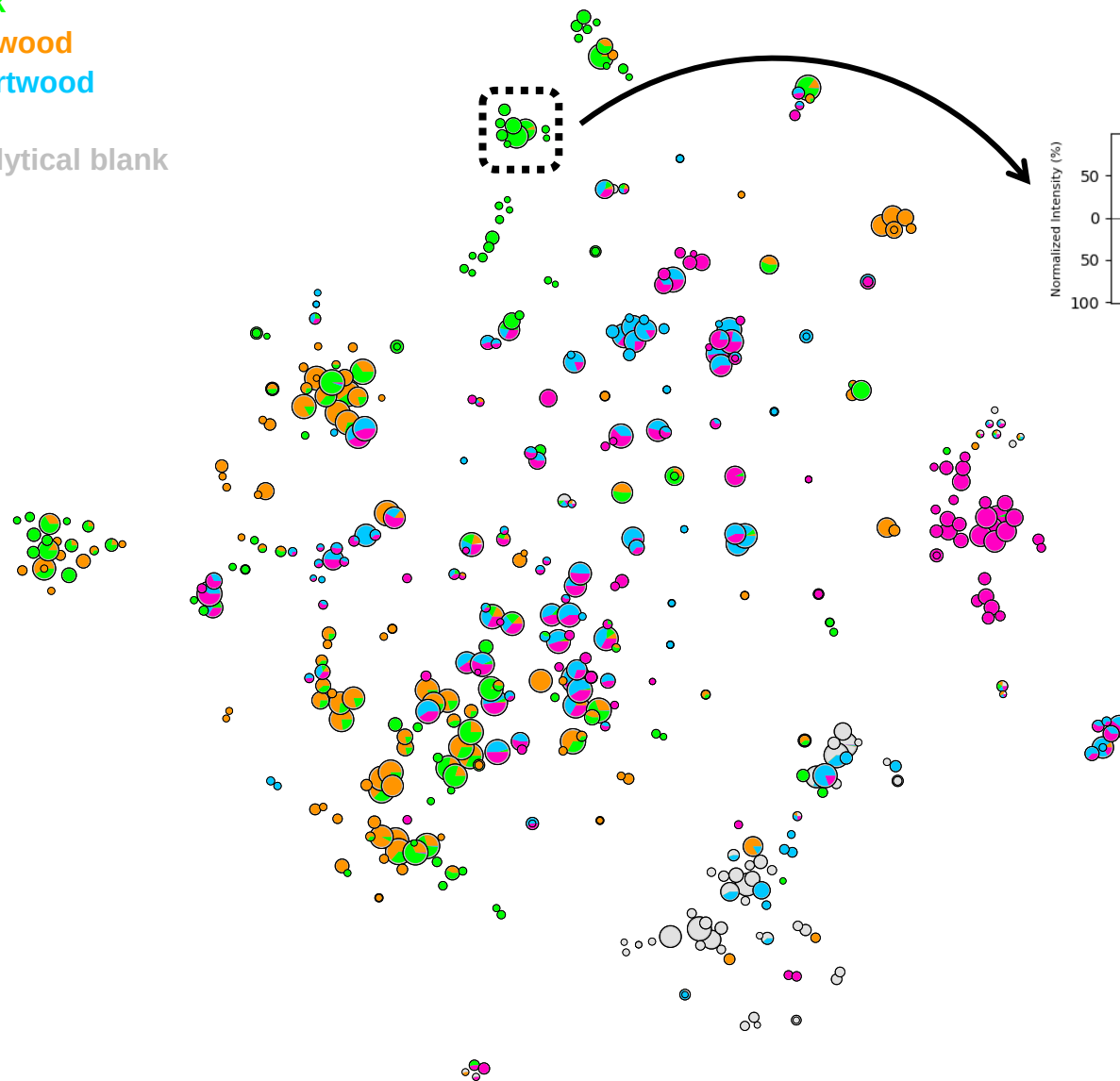
- Bark
- Sapwood
- Heartwood
- Pith
- Analytical blank



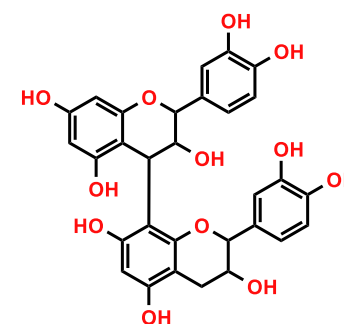
Exploration of ionizable chemical diversity by electrospray:

- Molecular networking
- Based on MS² data
- Dimensional reduction
- Spectral similarities
- Annotations with spectral banks

- Bark
- Sapwood
- Heartwood
- Pith
- Analytical blank

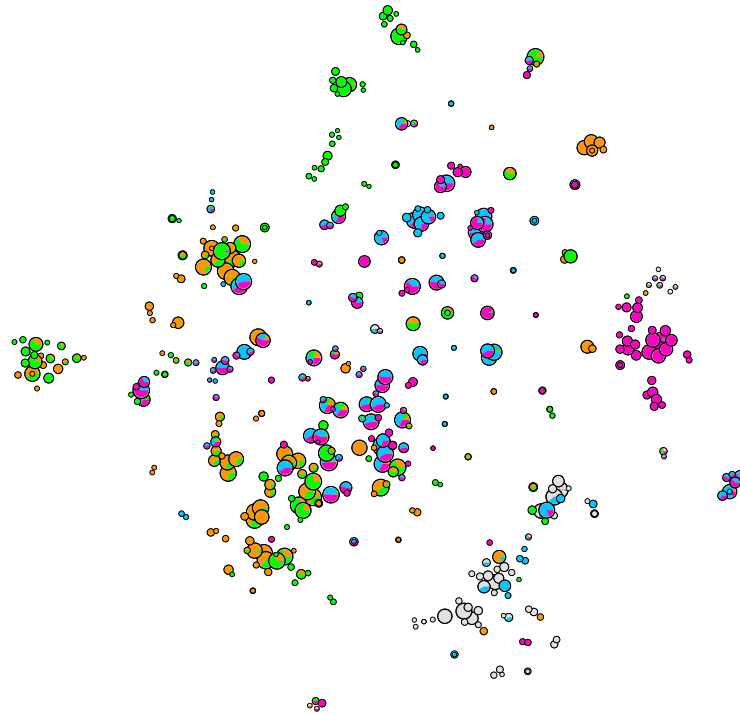


Procyanidin B2



$C_{30}H_{26}O_{12}$
 $[M + H]^+$
 $m/z_{\text{exp.}} = 579.1495$
 $\Delta_{\text{ppm}} = -0.3 \text{ ppm}$

- Bark
- Sapwood
- Heartwood
- Pith
- Analytical blank



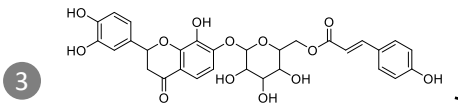
● **Bark**

- Sapwood

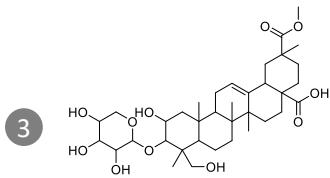
● Heartwood

- **Pith**

○ Analytical blank



Glycosylated flavanones



Glycosylated triterpenes

Adducts

Fatty acids

③ to ① annotation levels

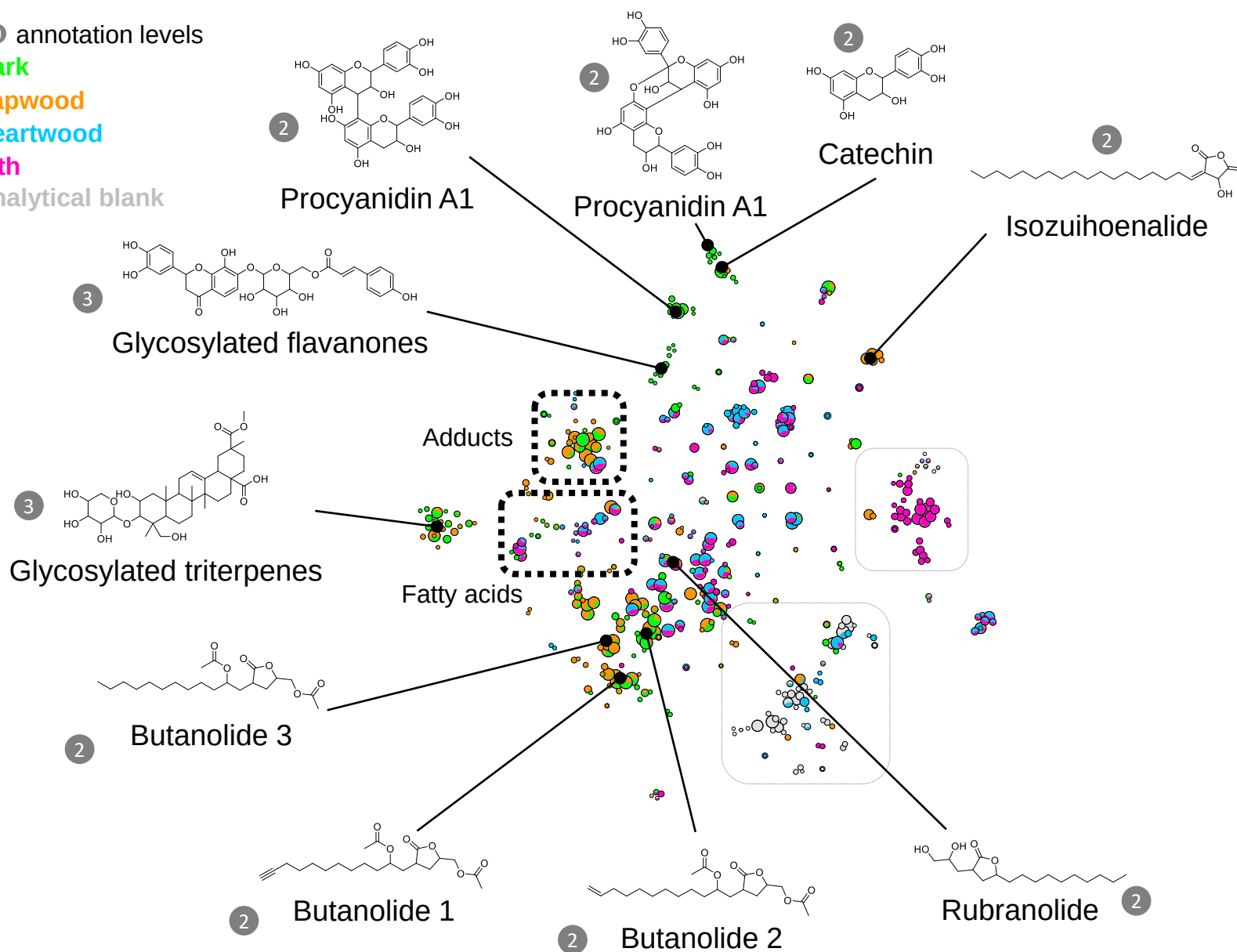
● **Bark**

● **Sapwood**

● **Heartwood**

● **Pith**

○ **Analytical blank**



③ to ① annotation levels

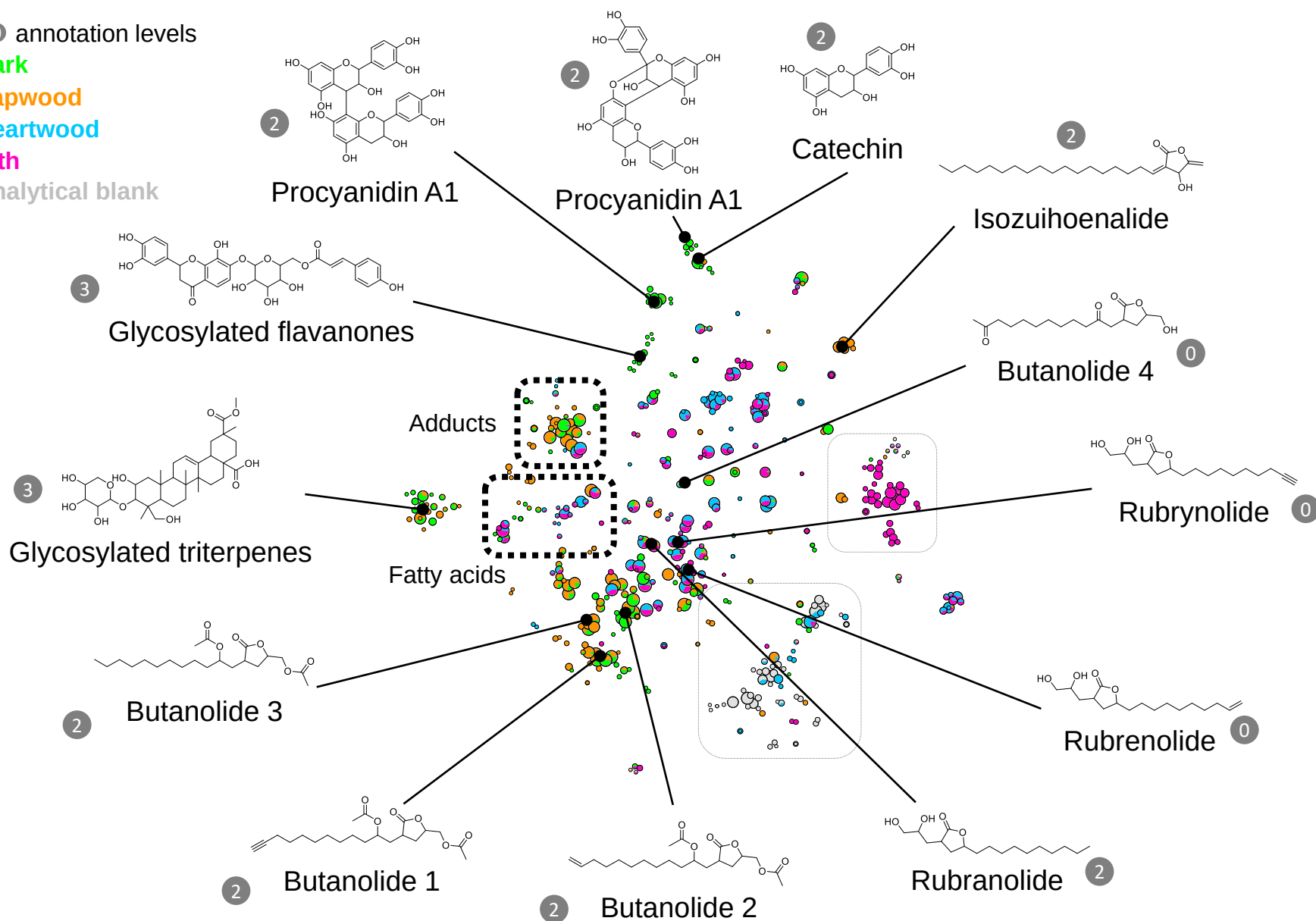
● **Bark**

● **Sapwood**

● **Heartwood**

● **Pith**

○ **Analytical blank**



③ to ① annotation levels

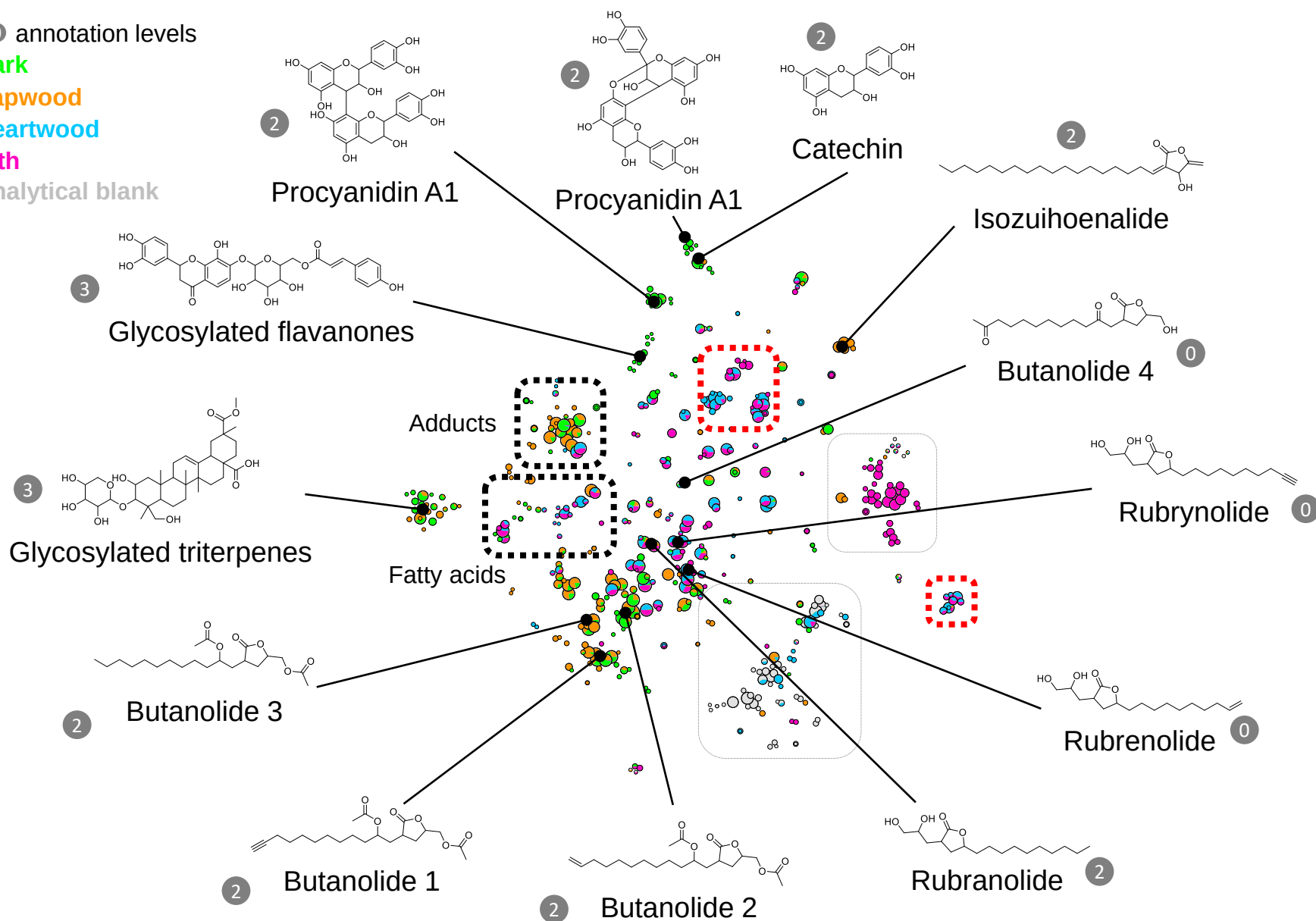
● **Bark**

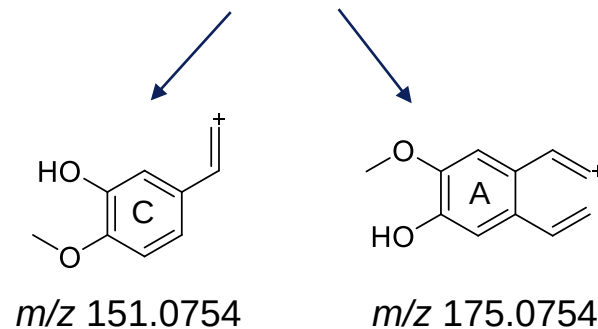
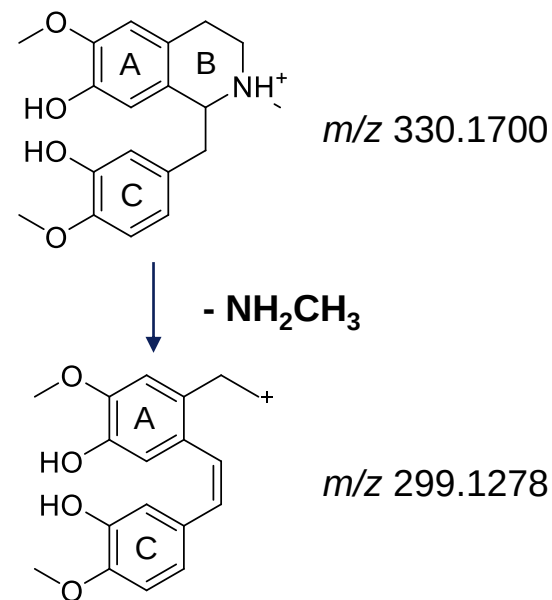
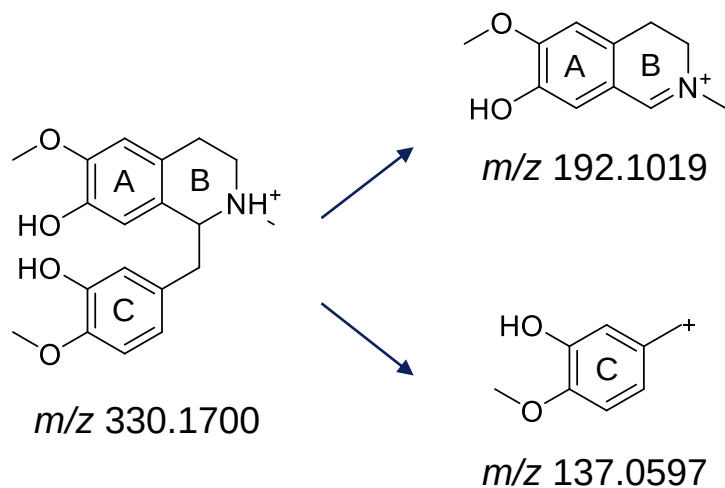
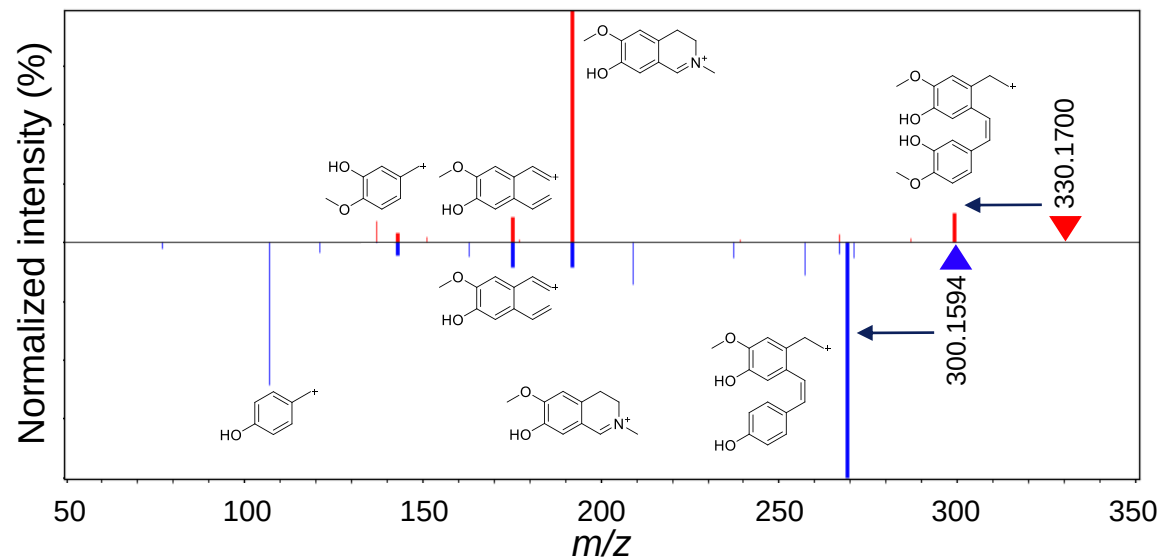
● **Sapwood**

● **Heartwood**

● **Pith**

○ **Analytical blank**





- Diagnostic fragments
- Diagnostic neutral loss

Level 1

Validated Identification
Confirmed structure
NMR or Standard comparison

Level 2

Putative ID
MS² match

Level 3

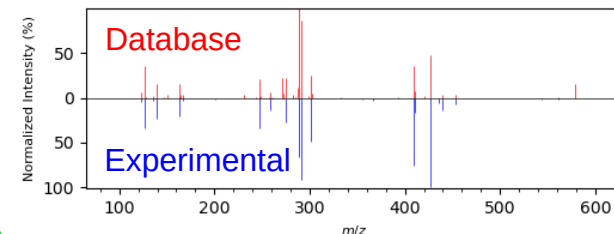
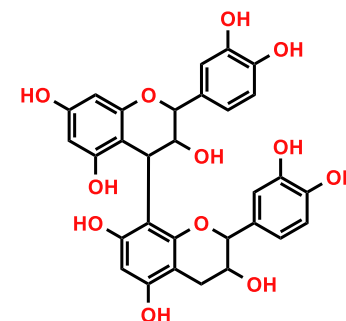
Tentative Structure
Molecular formula match

Level 4

Molecular Formula
Isotope distribution, charge state, adduct determination

Level 5

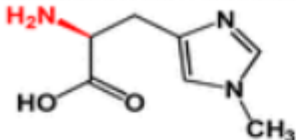
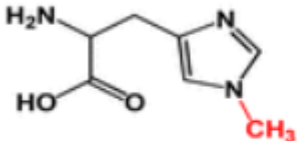
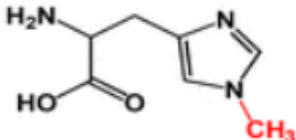
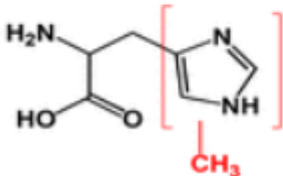
Unique Feature
Mass measurement only

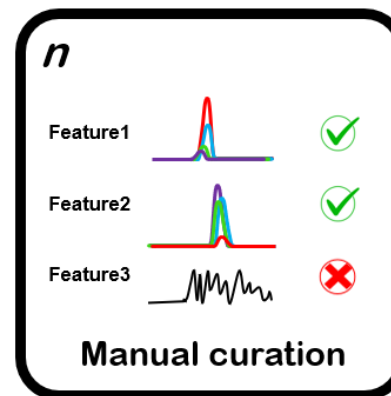
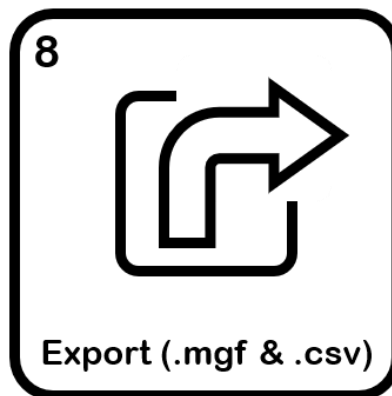
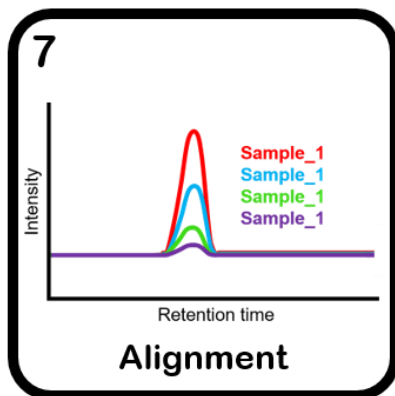
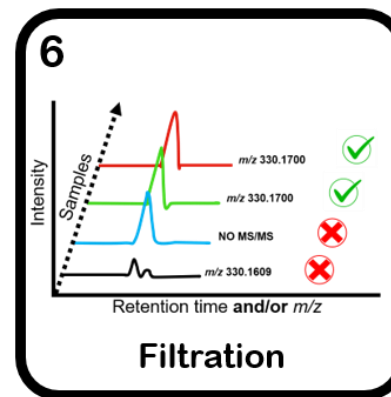
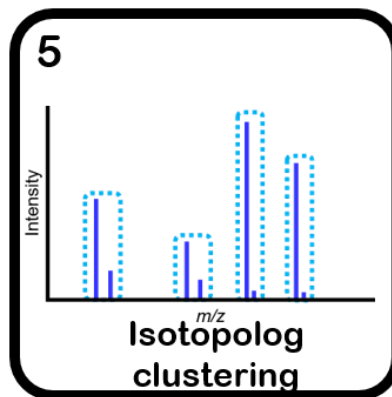
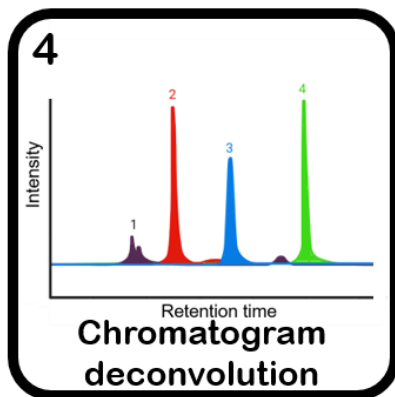
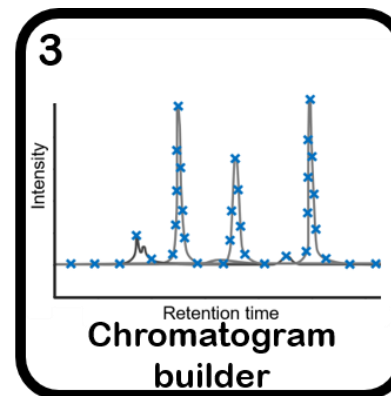
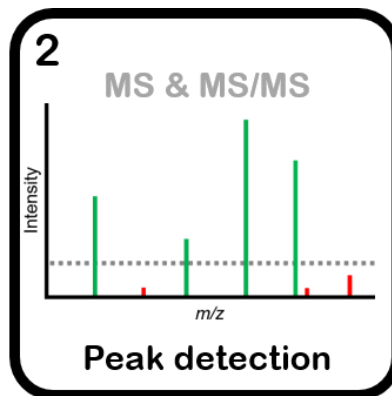
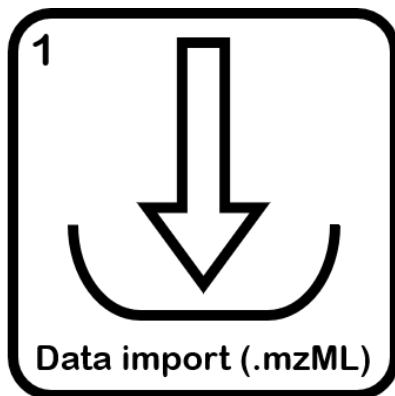


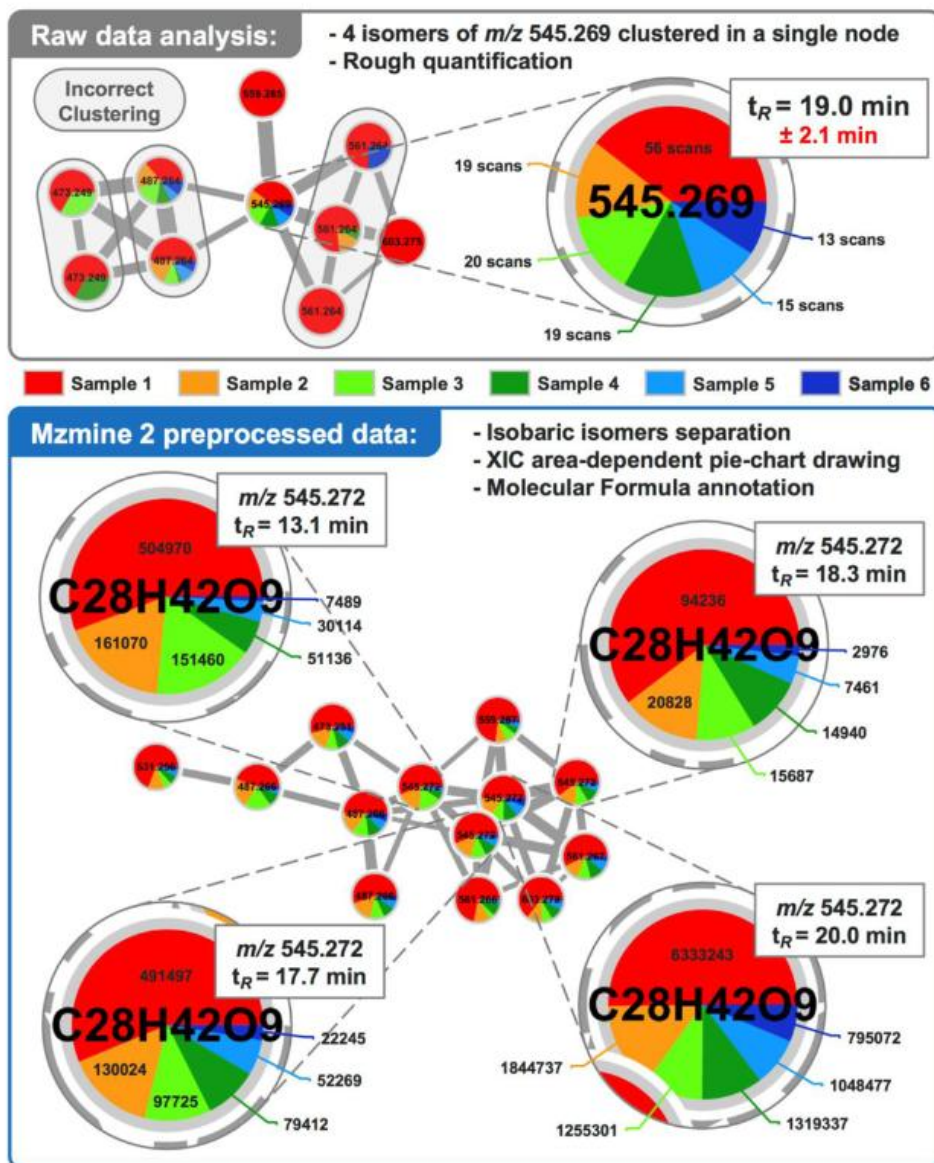
Procyanidin B2

$[M + H]^+$
 $C_{30}H_{26}O_{12}$

Feature ID: 1
 m/z 579.1497

	Level of confidence	Requirements
	Level 0: Unambiguous structure	Full structure and stereochemistry of isolated, pure compound
	Level 1: Confident structure	Reference standard match using at least two orthogonal techniques MS, MS/MS and RT matching authentic reference standards
	Level 2: Probable structure	Match to literature/database of at least two techniques e.g. MS, MS/MS, RT matching libraries
	Level 3: Probable structure/class Several candidates possible	Match of at least two techniques to support candidate structure/class e.g. MS, MS/MS, RT (indicative)
C7H11N3O2	Level 4: Unequivocal chemical formula, but no class identification	Match of Mass spectrum (mass/isotopes) with proposed formula
169.0851	Level 5: Unknown feature	No identification, exact mass only





MZmine 2

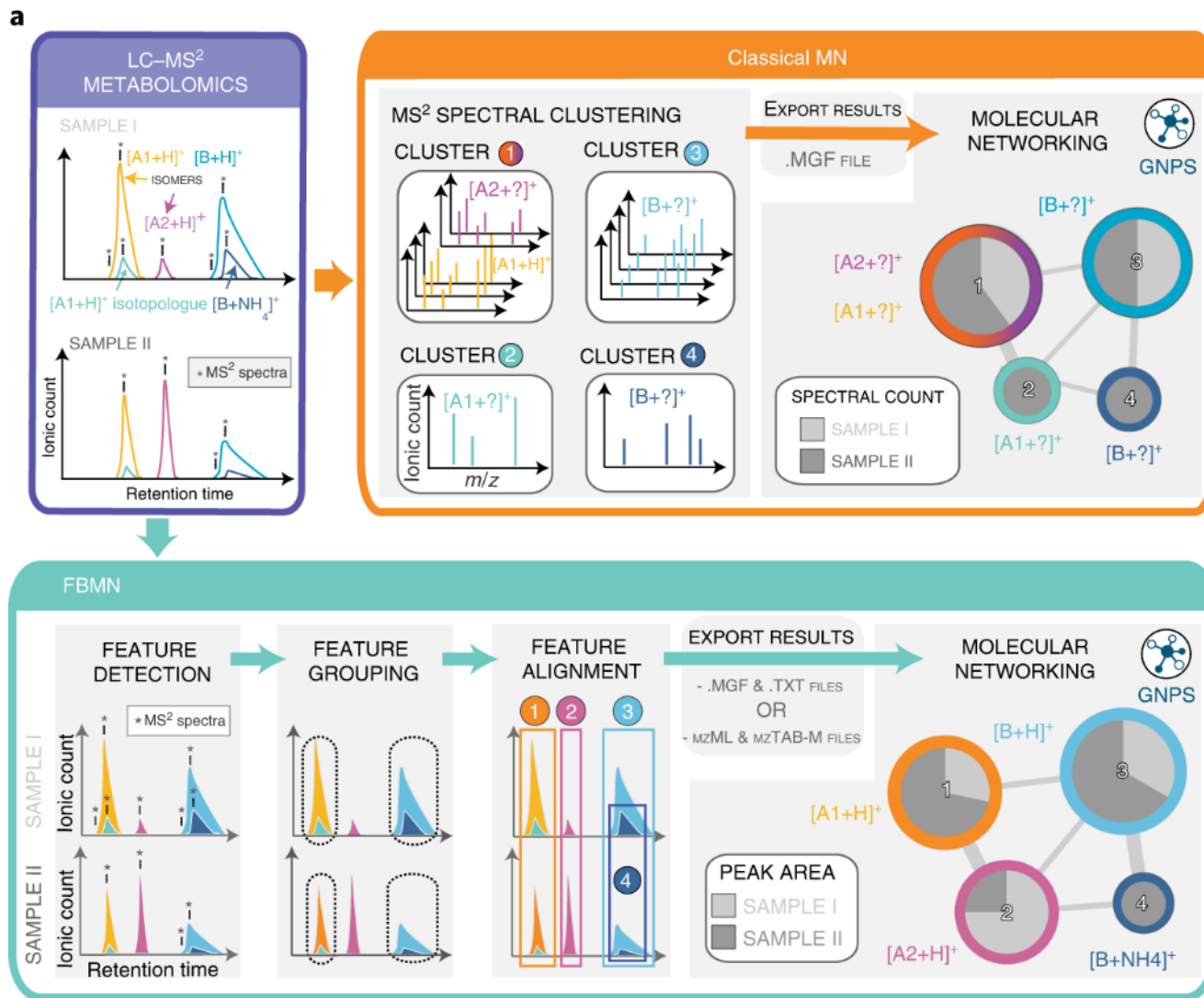
+



=

- Isobaric isomers separation
- XIC area-dependent pie-chart drawing
- Molecular Formula annotation

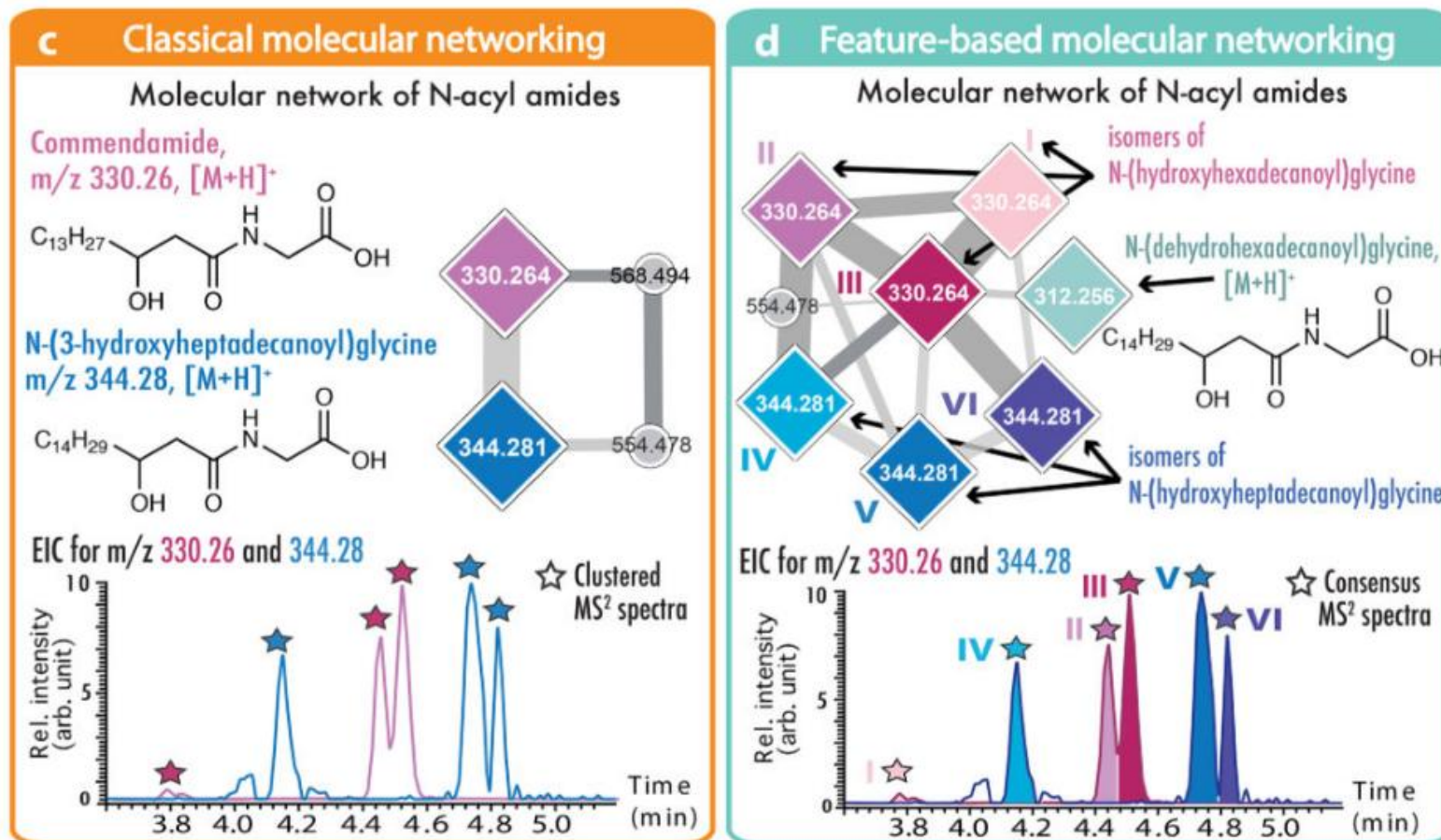
Received: April 28, 2017
 Accepted: June 23, 2017
 Published: June 23, 2017



Received
18 October 2019

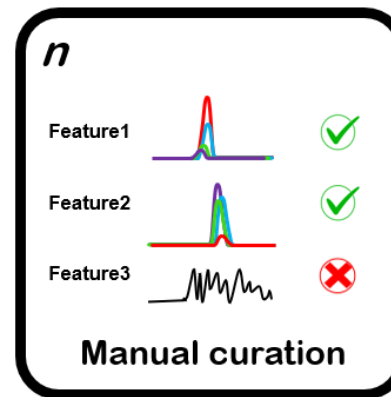
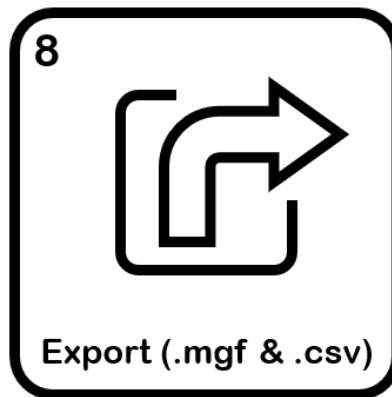
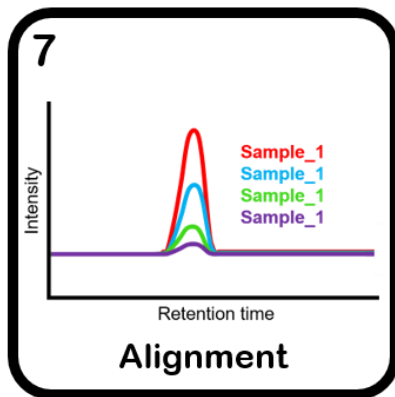
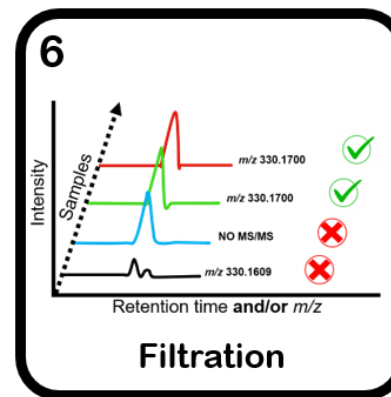
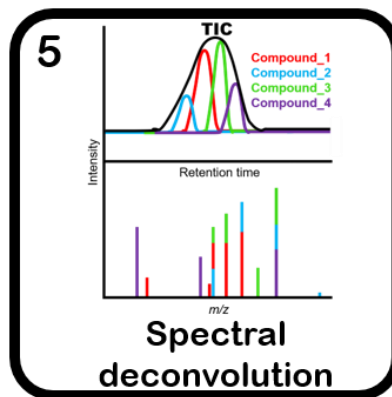
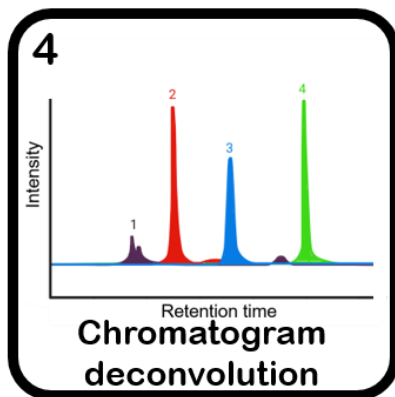
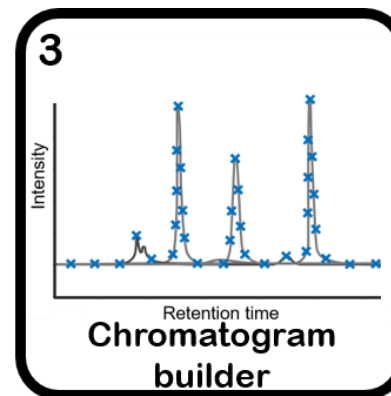
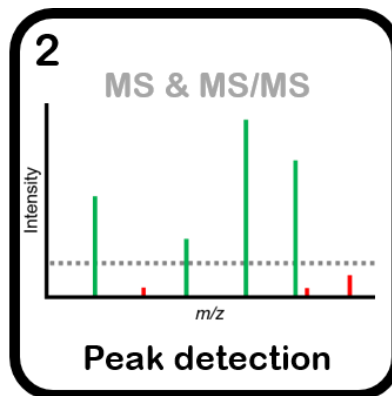
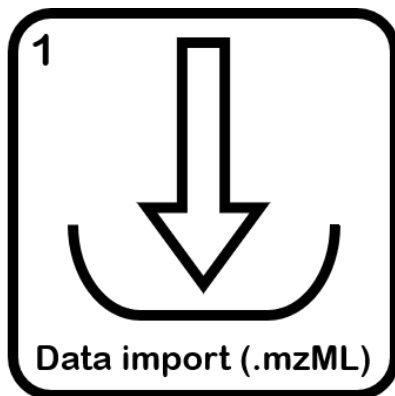
Accepted
22 July 2020

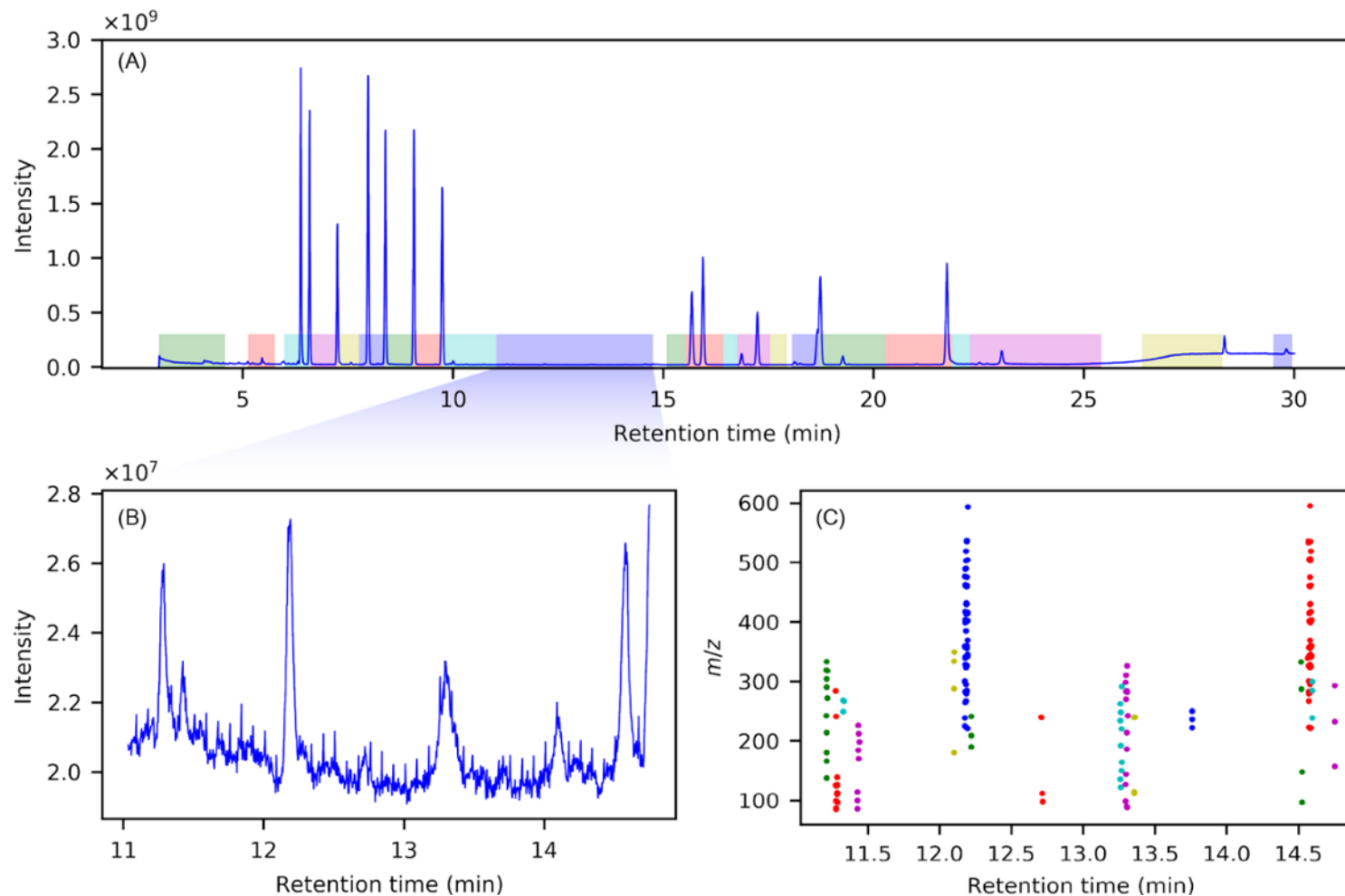
Published
24 August 2020



What about GC-MS ?







Smirnov, A., et al. (2018). *J. Proteome Res.* 17, 470–478. <https://doi.org/10.1021/acs.jproteome.7b00633>.

Alignment

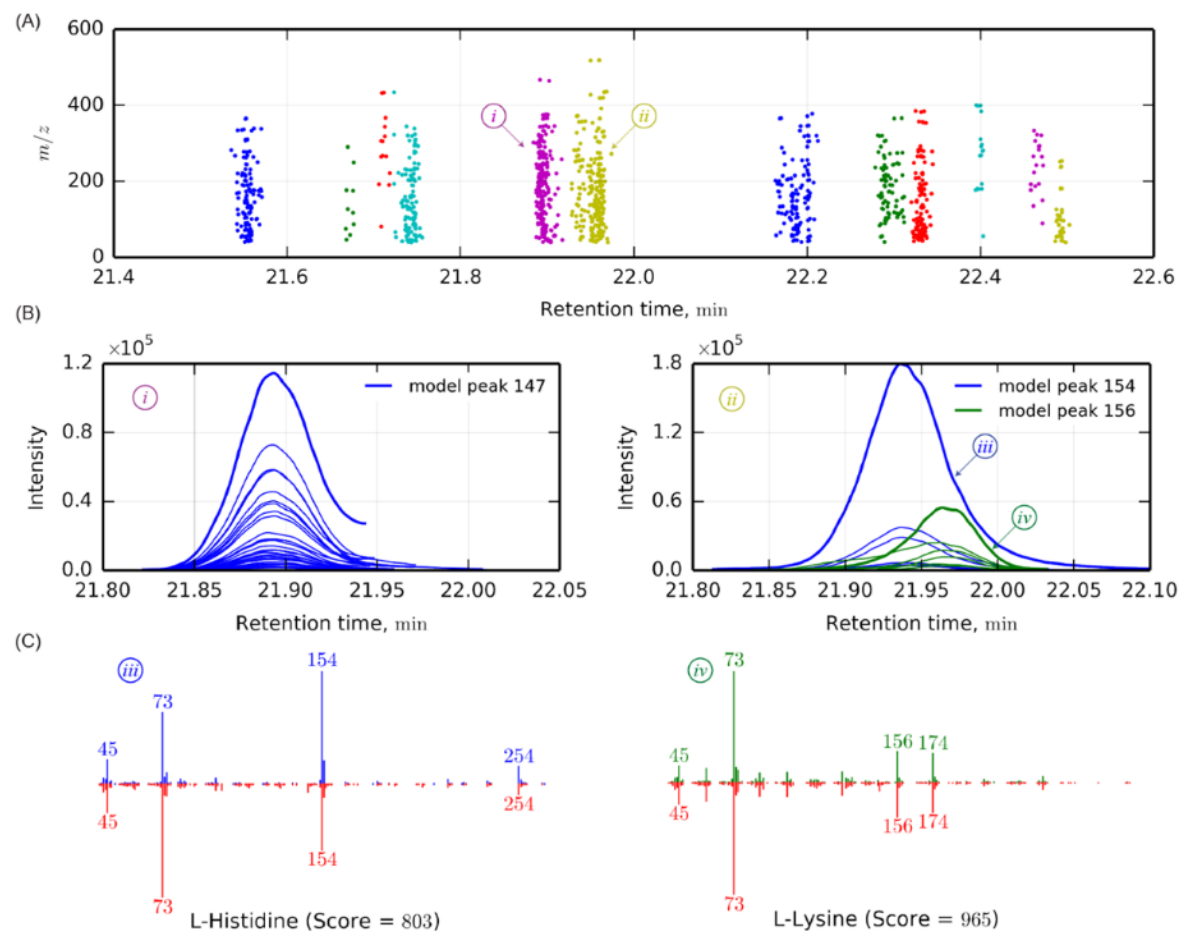
Export (.mgf & .csv)

Manual curation

1

2

3



Smirnov, A., et al. (2018). *J. Proteome Res.* 17, 470–478. <https://doi.org/10.1021/acs.jproteome.7b00633>.

Alignment

Export (.mgf & .csv)

Manual curation

File to process

☒ Metadata

Cosine Score Computing

☒ Treat as MS1 Data

m/z Tolerance

Minimum Matched Peaks

☒ Filtering

☒ Keep peaks outside of the $\pm$ window

☒ Keep peaks above of maximum

☒ Keep each peak in top in the $\pm$ window

t-SNE Visualization

At least cosine score(s) above

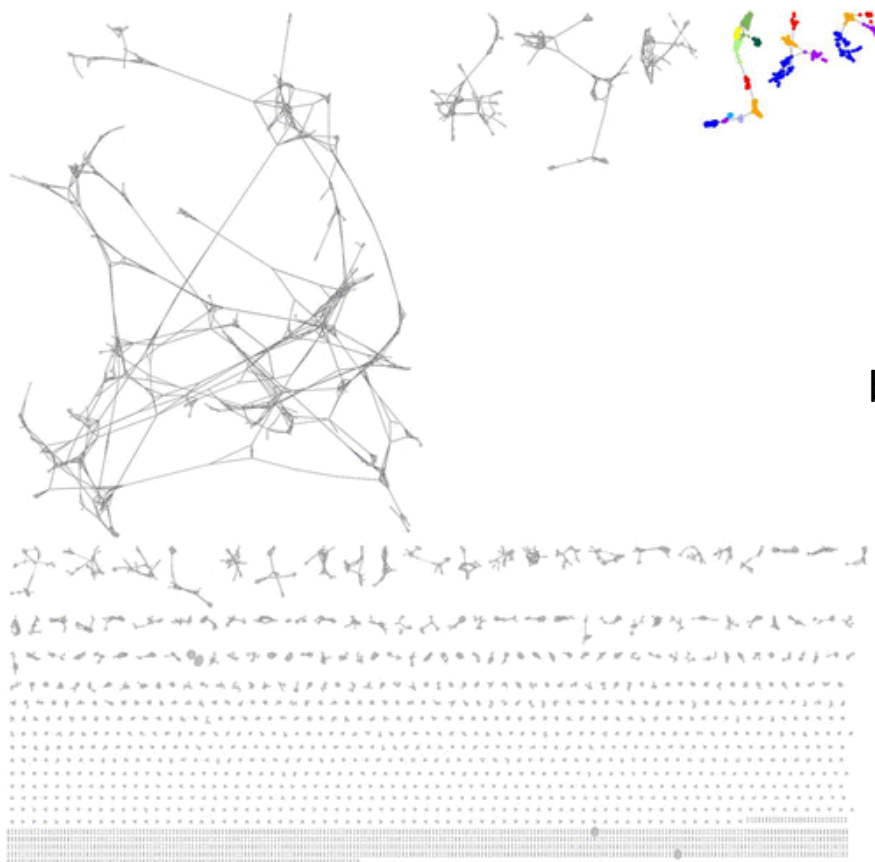
Number of iterations Perplexity

Learning Rate Early exaggeration

☐ Random Initial State

☒ Use Barnes-Hut approximation

Angle

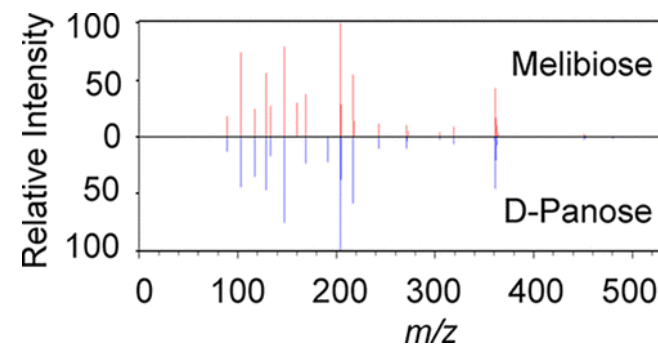
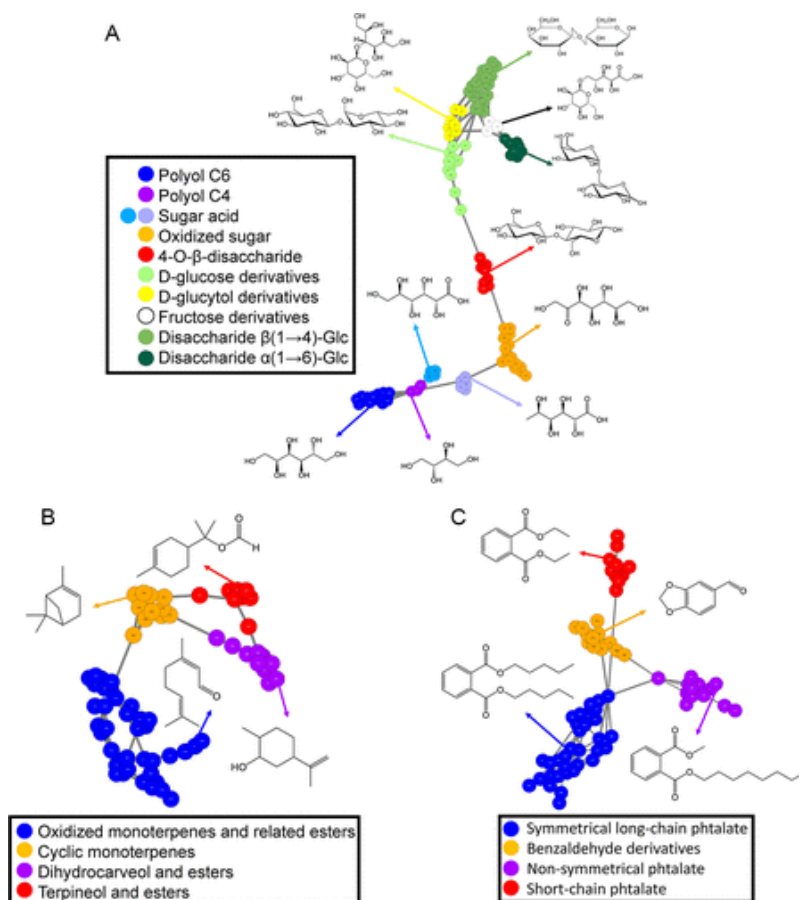


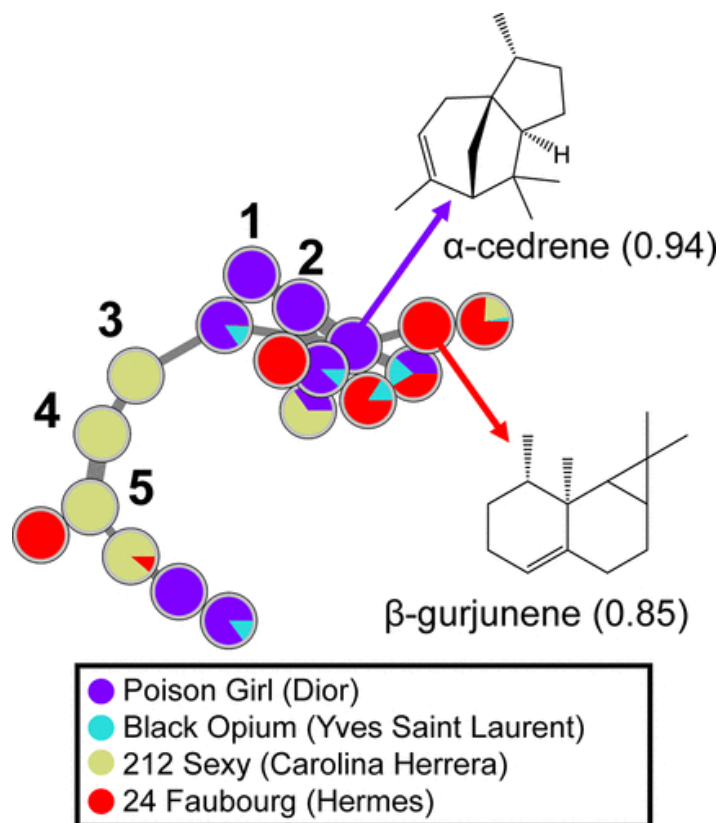
Global molecular network generated
by MetGem with 15.113 EI-MS spectra
INA

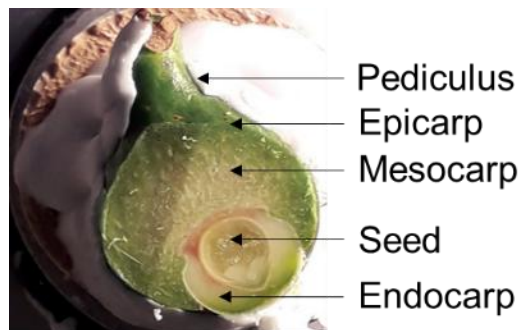
ISIPCA
From senses to expertise
- PARIS -

GRUPE
CLARINS

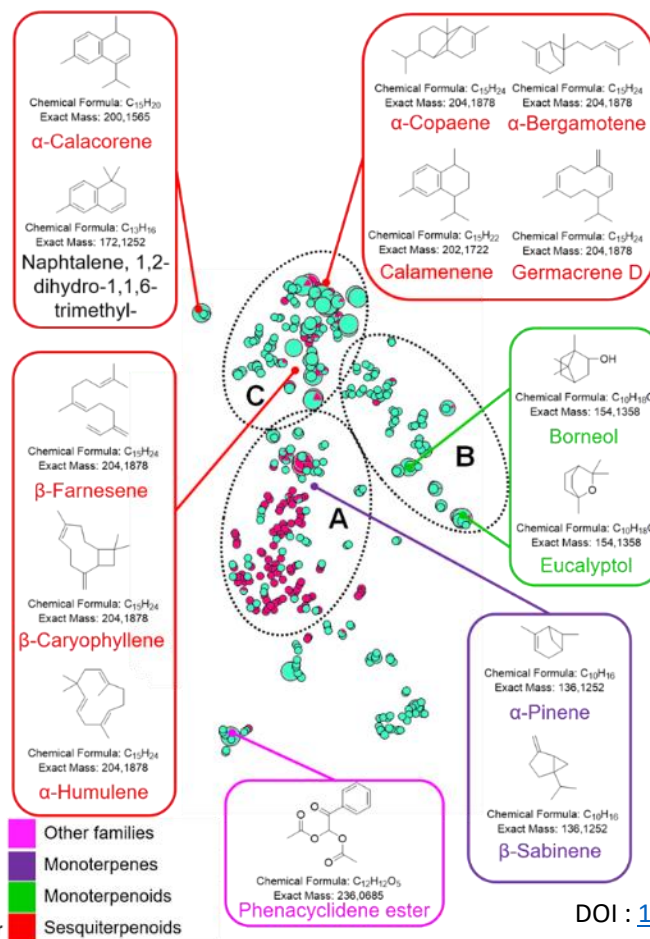
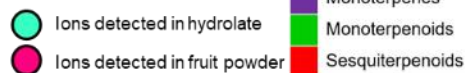
Anal. Chem. 2019, 91, 18, 11489–11492



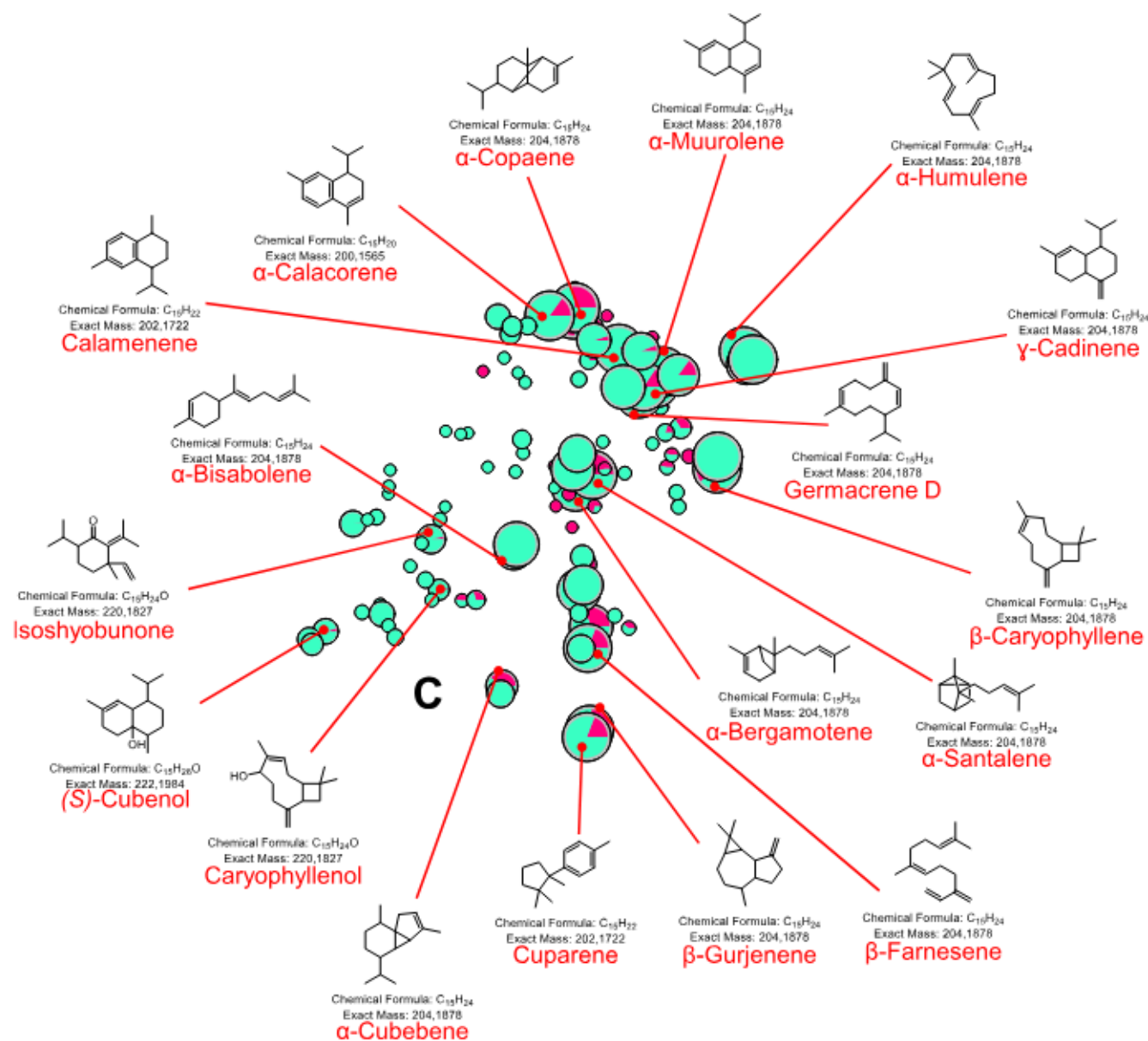




- MS¹ spectral similarities
- Accessing chemical diversity
- Query private IE databases (NIST/Wiley)
- Hydrolate → fungicidal activity ≤ 5% against *Tricophyton rubrum*

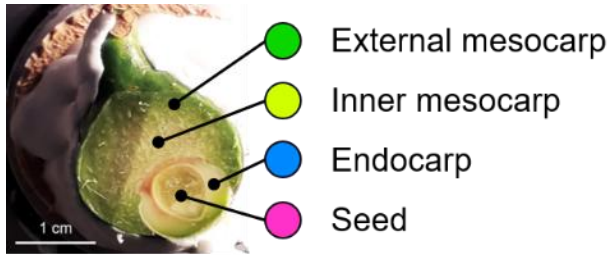


DOI : [10.5802/crchim.248](https://doi.org/10.5802/crchim.248)

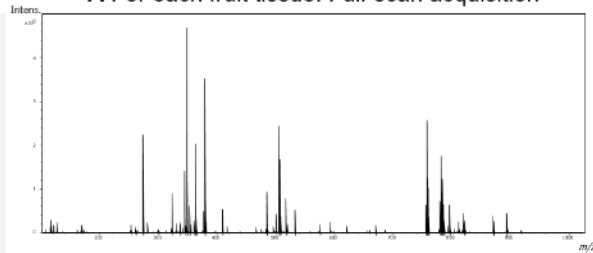


MALDI-MS/MS

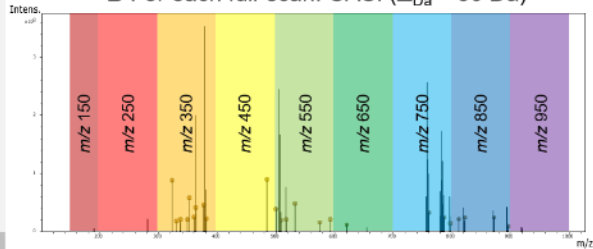
Very high-resolution



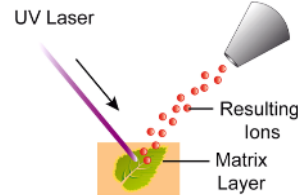
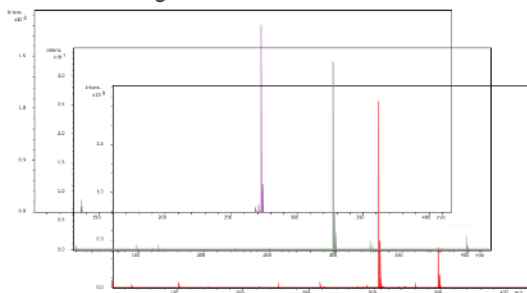
A For each fruit tissue: Full-scan acquisition



B For each full-scan: CASI ($\Delta_{Da} = 50$ Da)

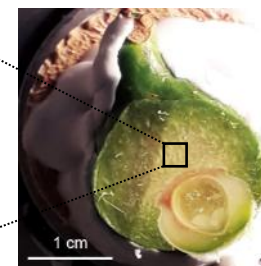
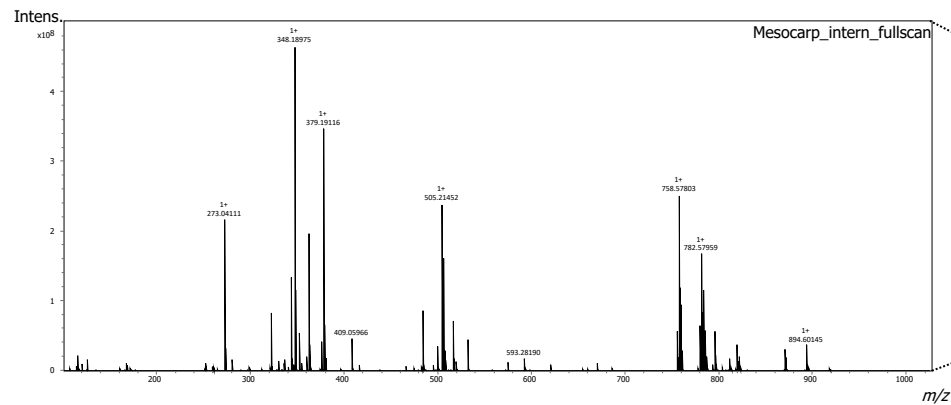


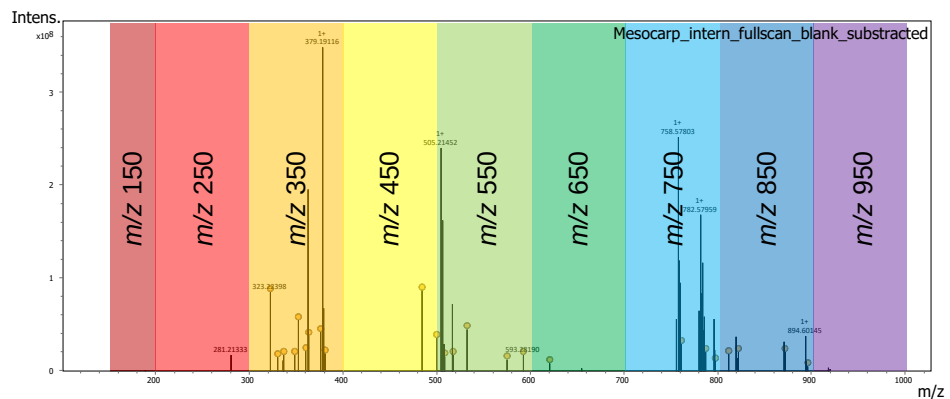
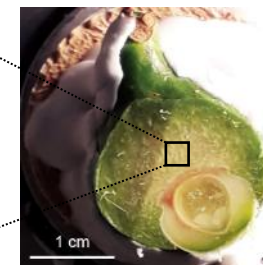
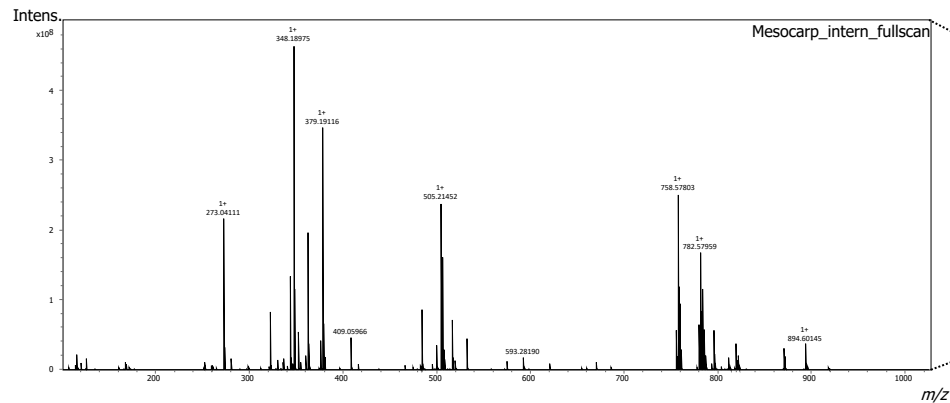
C For each m/z window:
Fragmentation of detected ions



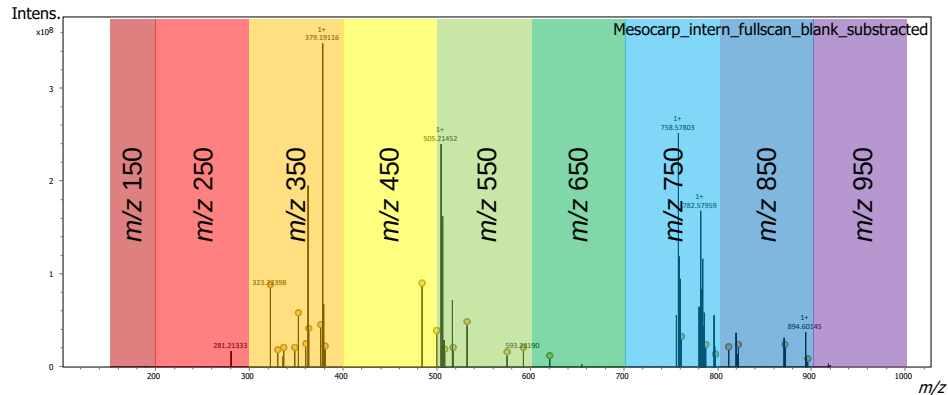
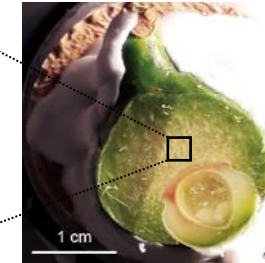
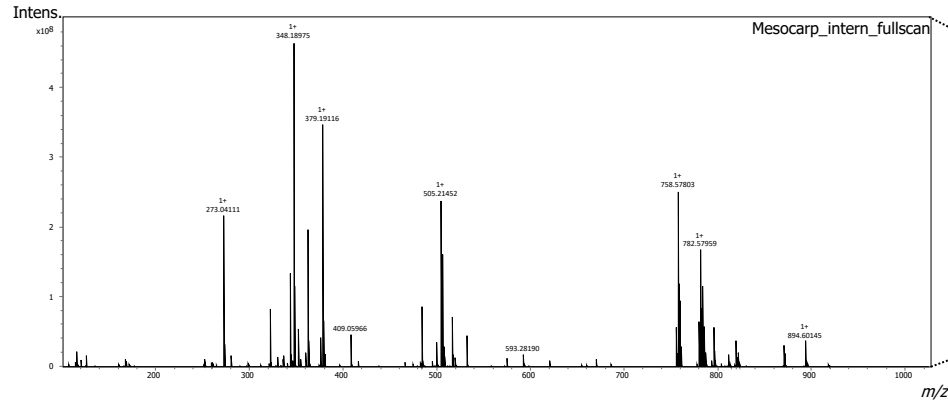
CONDITIONS

- Cuts – 20 μ m at -20°C
- Matrix – DHB in H₂O/MeOH (50/50)
- Ionization – MALDI (+)
- Laser intensity – 18%
- Spatial resolution – 20 μ m
- Full-Scan m/z 100 – 1200
- MALDI(+)-CID – 0 to 40 eV





CASI
Continuous Accumulation
Selected Ions
m/z window = 100 amu



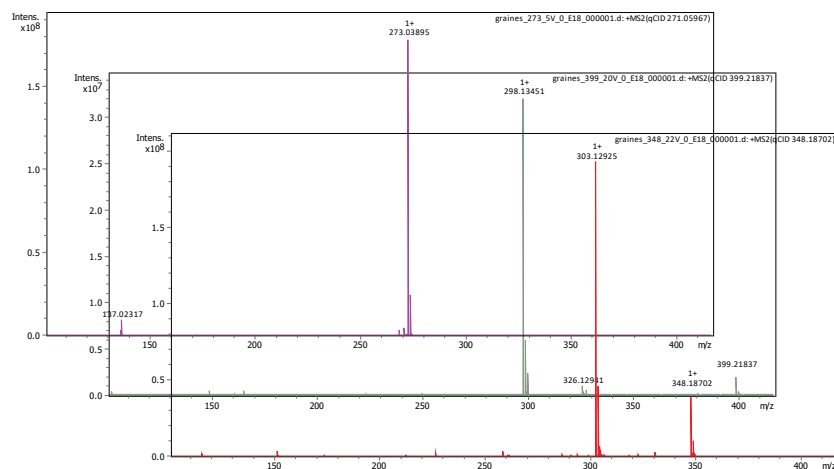
CASI
Continuous Accumulation
Selected Ions
 m/z window = 100 amu

Number	m/z	Adduct
1	m/z_1	$[M + H]^+$
2	m/z_1	$[M + NH_4]^+$
n	m/z_n	x

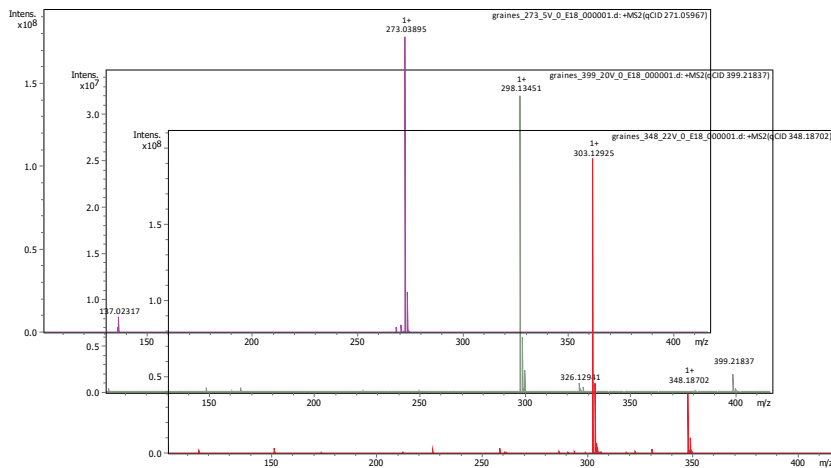
For each ion...

→ Ion fragmentation (CID with optimized 0-35 eV)

MS/MS spectra



MS/MS spectra

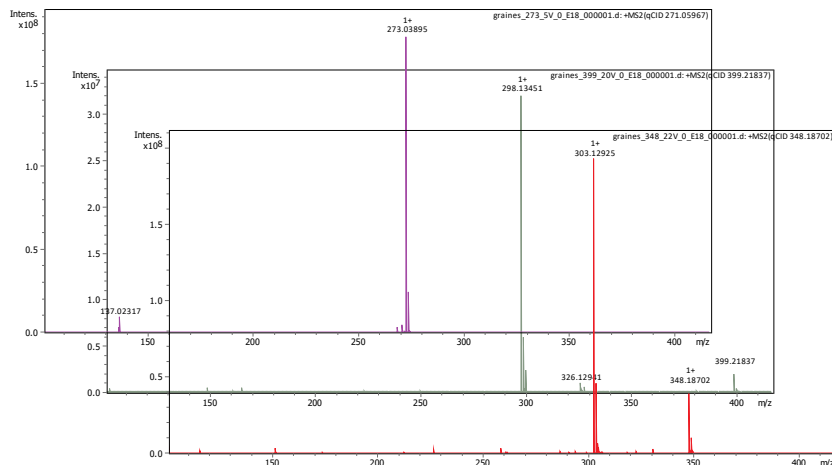


Smoothing
Savitzky-Golay

Baseline correction
SNIP

Peak detection
 $\text{SNR} > 3$

MS/MS spectra



Smoothing
Savitzky-Golay

Baseline correction
SNIP

Peak detection
SNR > 3

« Peak list »

.mgf

Parent ion: m/z

Fragment ions:

$m/z_1 + \text{Intensity}_1$

$m/z_2 + \text{Intensity}_2$

$m/z_n + \text{Intensity}_n$

...

Metadata

.CSV

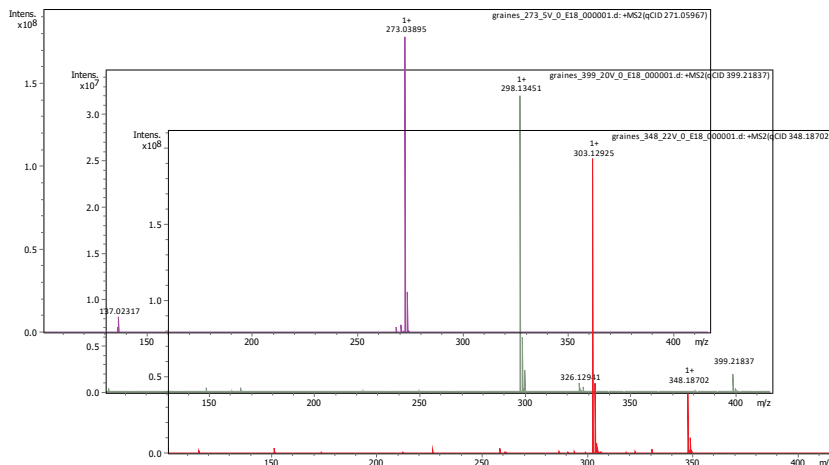
Parent ion: m/z

Intensity: x

Localisation: Tissu

...

MS/MS spectra



Smoothing
Savitzky-Golay

Baseline correction
SNIP

Peak detection
SNR > 3

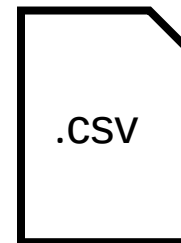
« Peak list »



Parent ion: m/z
Fragment ions:
 $m/z_1 + \text{Intensity}_1$
 $m/z_2 + \text{Intensity}_2$
 $m/z_n + \text{Intensity}_n$

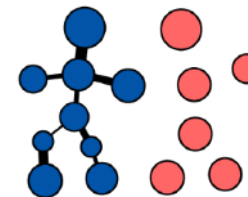
...

Metadata



Parent ion: m/z
Intensity: x
Localisation: Tissue

...

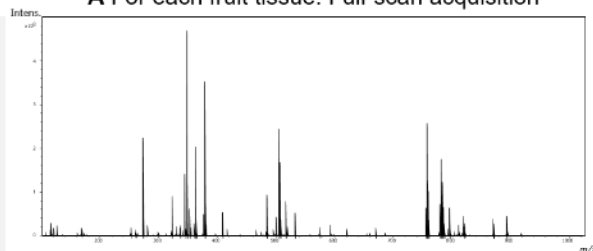


MetGem
v. 1.4.1

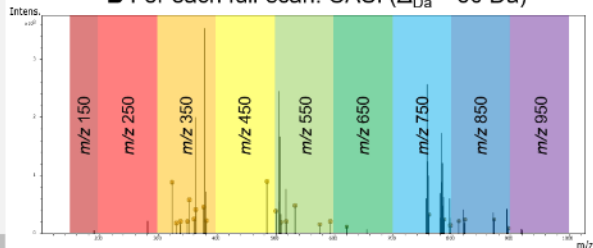
+ Spectral banks
GNPS
MassBank
MS-DIAL
...



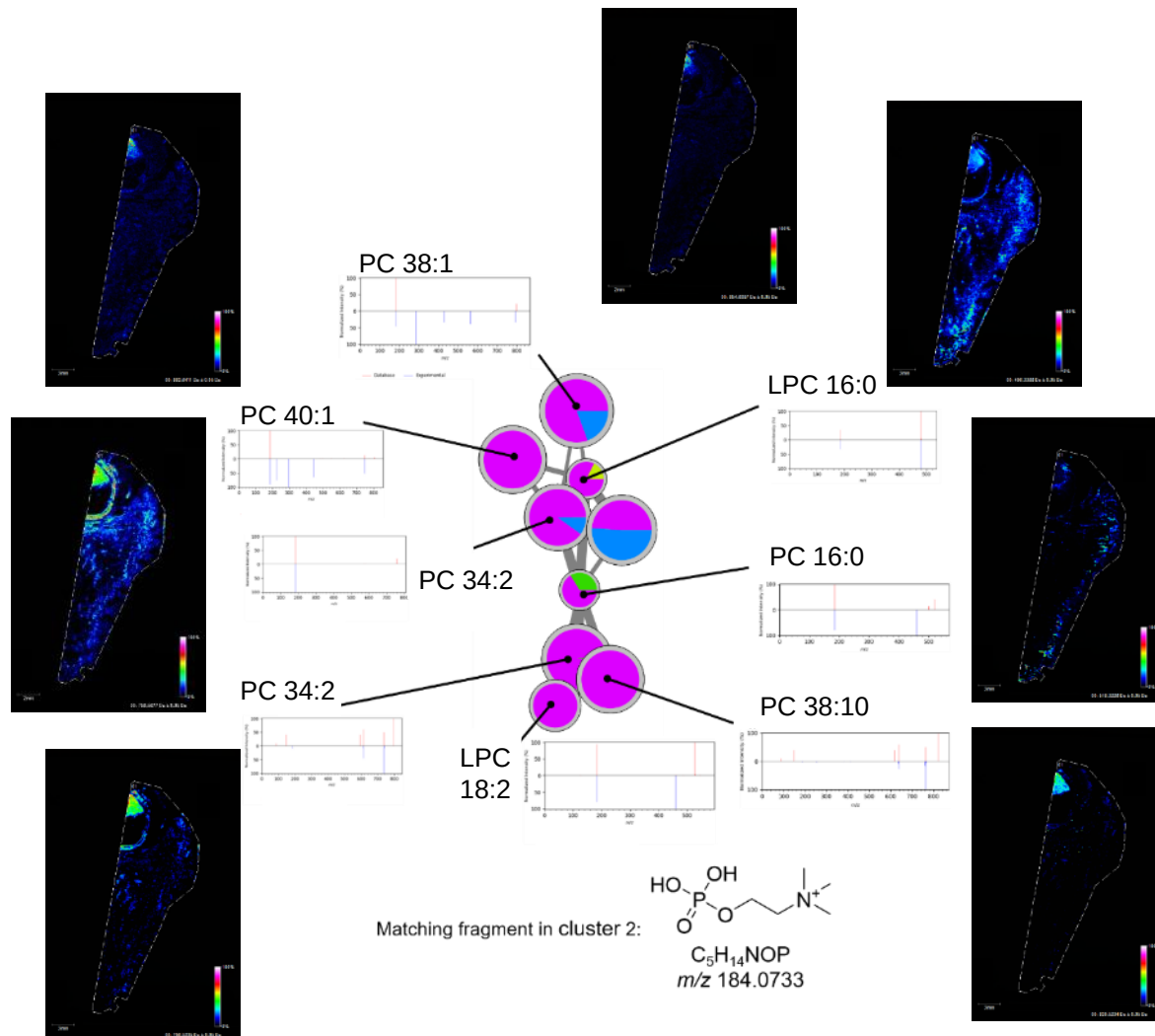
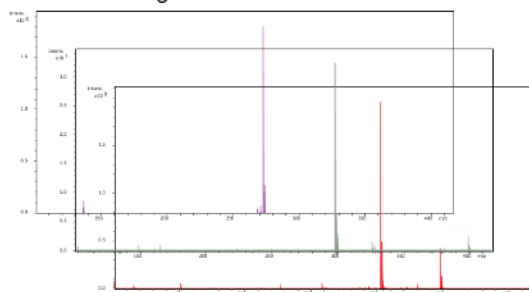
A For each fruit tissue: Full-scan acquisition



B For each full-scan: CASI ($\Delta_{Da} = 50$ Da)



C For each m/z window:
Fragmentation of detected ions

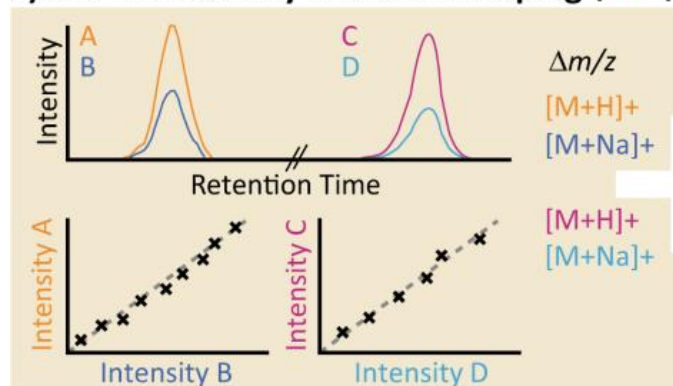


Enhancing metabolite annotation levels

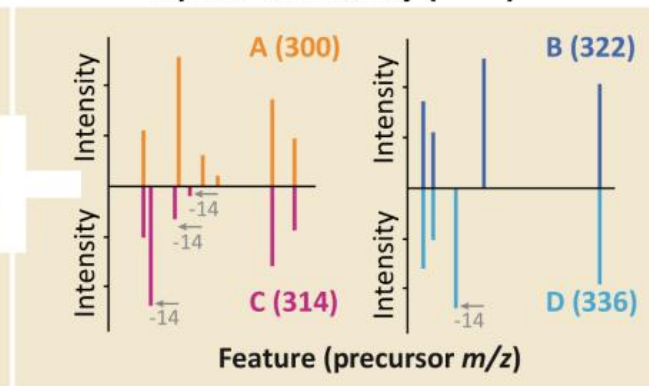
To go further ...



a) MS¹ Ion Identity Feature Grouping (.....)



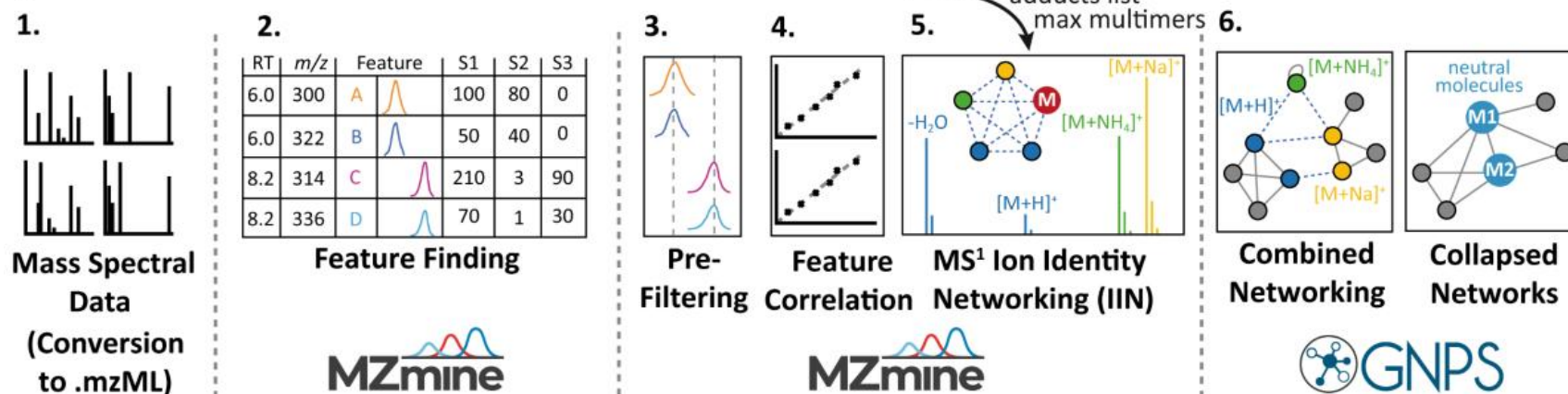
b) MS² Similarity (—)



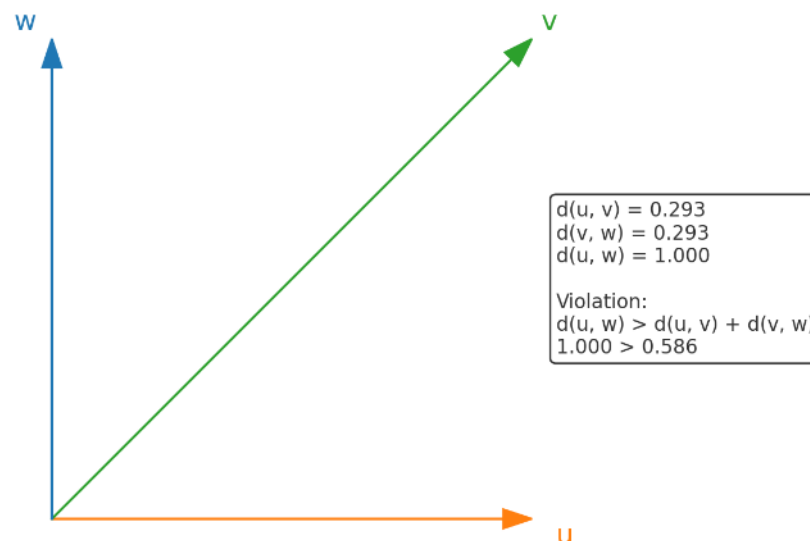
c) Ion Identity Molecular Networking



d) Workflow



Cosine Distance Triangle Inequality Violation



Since $1.000 > 0.586$, the triangle inequality $d(u, w) \leq d(u, v) + d(v, w)$ is violated. This visually confirms that cosine distance is not a true metric. [↗]

The spectral-entropy similarity $S_{\text{ent}}(A, B)$ you defined between two MS/MS spectra is **not** a true metric. If you turn it into a “distance” via

$$d_{\text{ent}}(A, B) = 1 - S_{\text{ent}}(A, B),$$

you get a nonnegative, symmetric function with $d_{\text{ent}}(A, A) = 0$, but it generally **violates the triangle inequality**. In other words, there exist triplets (A, B, C) for which

$$d_{\text{ent}}(A, C) > d_{\text{ent}}(A, B) + d_{\text{ent}}(B, C).$$

Workarounds & best practices

- Use a proper angular metric

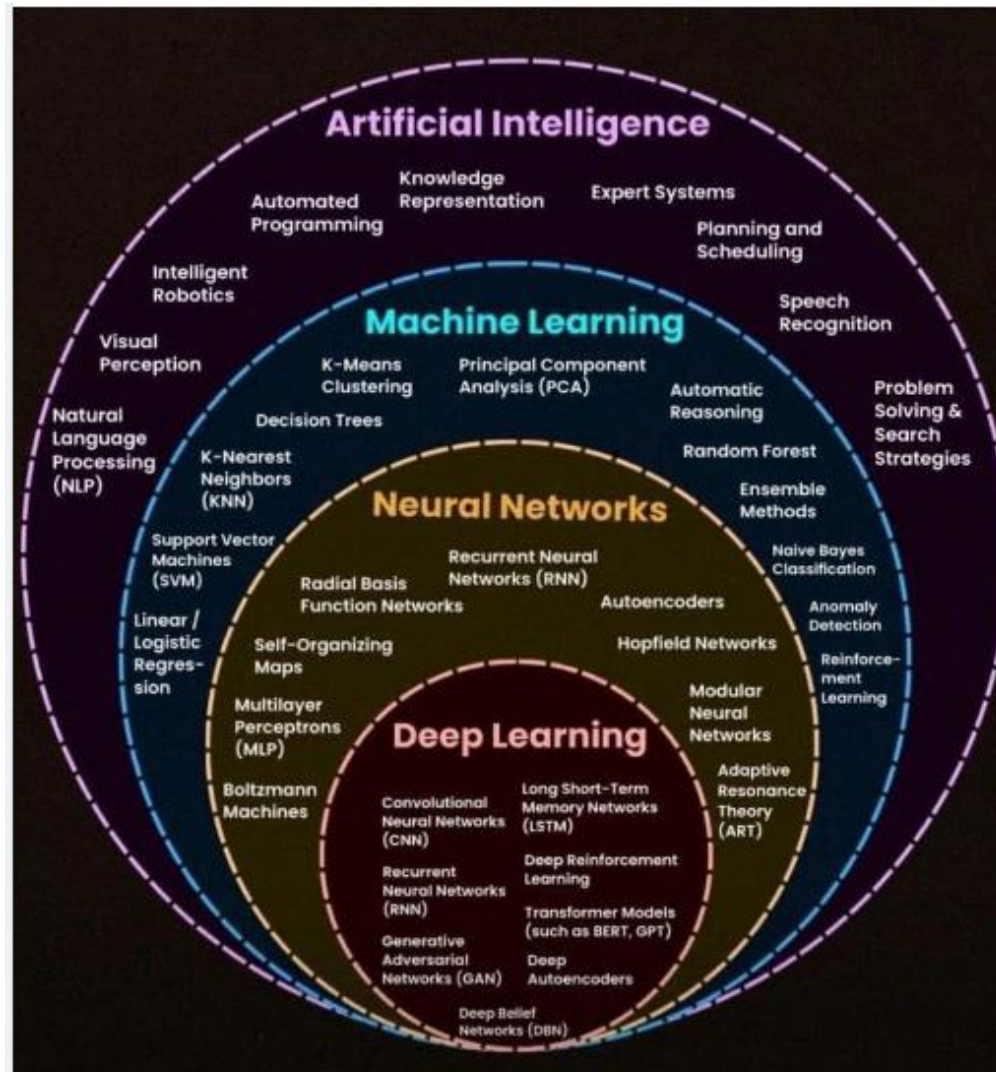
$$d_{\text{angle}}(u, v) = \frac{\arccos\left(\frac{u \cdot v}{\|u\| \|v\|}\right)}{\pi}$$

This *is* a true metric (it obeys the triangle inequality) and still reflects the same “angle” information.

- **Choose algorithms tolerant of non-metrics**
 - t-SNE itself is fine with any symmetric similarity.
 - For UMAP, use a brute-force or approximate-NN backend that supports cosine, or convert to the angular metric above so you can leverage metric-based trees.

Metric / Method	Principle	Computational Cost	Key Properties
Jensen-Shannon Distance	Symmetric divergence between distributions	$O(n)$	True metric; sensitive to all peaks
Hellinger Distance	Euclidean distance on sqrt intensities	$O(n)$	True metric; emphasizes moderate peaks
L_1 / L_2 on Binned Vectors	L_1/L_2 norm on binned spectra	$O(n)$	True metrics; loses sub-bin shifts
Bray-Curtis Distance	Normalized L_1 overlap in intensities	$O(n)$	True metric; ignores bin order
Earth Mover's Distance (EMD)	Minimum work to move mass along m/z	$O(n \log n)$	True metric; captures peak shifts

Metric / Method	Fragments (same m/z)	Neutral losses (m/z shifts)	Notes
Jensen–Shannon Distance	exact match	does <i>not</i> account for small shifts	Treats every bin separately; two spectra both missing or having a peak match well, but a shifted peak looks completely different.
Hellinger Distance	exact match	same issue as JS	Exactly like JS, but with a different weighting.
L₁ / L₂ on Binned Vectors	exact match	no notion of distance along m/z	Simple vector norms—cannot “see” a neutral-loss peak as related to its parent peak.
Bray–Curtis Distance	overlap of intensities	ignores m/z order	Only cares about total shared intensity, not <i>where</i> on the m/z axis it occurs.
Earth Mover’s Distance (EMD)	exact match	explicitly models shifts via “mass transport”	Moves intensity from one m/z to another at cost $\propto \Delta m/z$ —so a neutral-loss peak just costs its mass times the loss distance.





Chemometrics and Intelligent Laboratory Systems 23 (1994) 351-364

Chemometrics and
intelligent
laboratory systems

Pattern recognition / expert system for identification of toxic compounds from low resolution mass spectra *

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Research Triangle Park, NC 27711, USA

(Received 7 December 1993; accepted 8 February 1994)

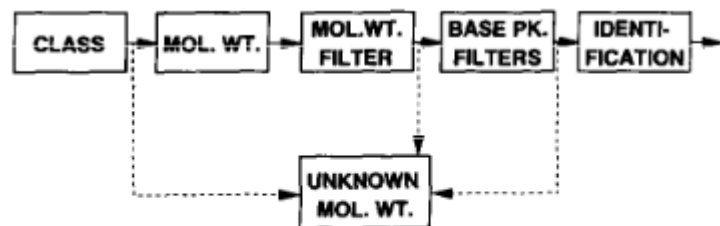


Fig. 1. Schematic diagram of classification and identification expert system.

Table 1
Training set composition and performance results

Class	Structure	Number in set
Nonhalobenzenes	Benzene	1
	Alkylbenzenes	10
	Alkenylbenzenes	1
	Phenylketones	1
	Phenylaldehydes	1
	Benzonitriles	1
Chlorobenzenes	Chlorobenzenes	4
	Alkylchlorobenzenes	3
	Alkenylchlorobenzenes	1
	Monochloroalkanes	4
Mono- and dichloroalkanes / alkenes	Monochloroalkenes	1
	Dichloroalkanes	8
	Dichloroalkenes	3
	Monochloroepoxyalkane	1
	Monochloroalkene	1
	Trichloroalkanes	5
Tri-, Tetra- and pentachloroalkanes / alkenes	Trichloroalkenes	1
	Tetrachloroalkanes	6
	Tetrachloroalkenes	1
	Pentachloroalkanes	1
	Monobromoalkanes	4
	Dibromoalkanes	7
Bromo- and bromochloroalkanes / alkenes	Tribromoalkanes	1
	Monobromoalkenes	2
	Monobromomonochloroalkanes	3
	Monobromodichloroalkanes	1
	Monobromotrichloroalkanes	1
	Dibromomonochloroalkanes	1
Unknown	Cyclic ethers	3
	Acyclic hydrocarbons	13
	Cyclic hydrocarbons	3
	Unsaturated hydrocarbons	3
	Aldehydes	3
	Ketones	3
Classification and identification accuracy:	Alcohols	2
	Acids	1
Molecular weight absolute deviations:		100 %
		median 0.0 Da
		average 0.17 Da

Table 2
Performance results with various test sets

Sets	Number	Molecular weight absolute deviations		Classification and identification accuracy (%)
		Median (Da)	Average (Da)	
Pollutant test	32	1.0	5.6	97
Random NIST	99	1.0	7.3	97
Pharmaceutical	400	2.0	10.0	97.5
Overall	531	2.0	9.2	97.4



Article <https://doi.org/10.1038/s41587-025-02663-3>

Self-supervised learning of molecular representations from millions of tandem mass spectra using DreaMS

Received: 22 April 2024

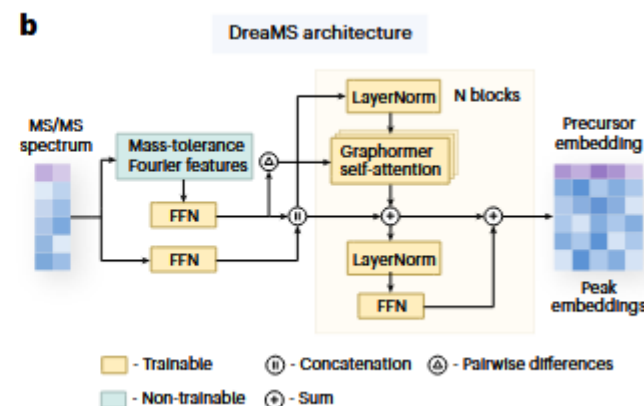
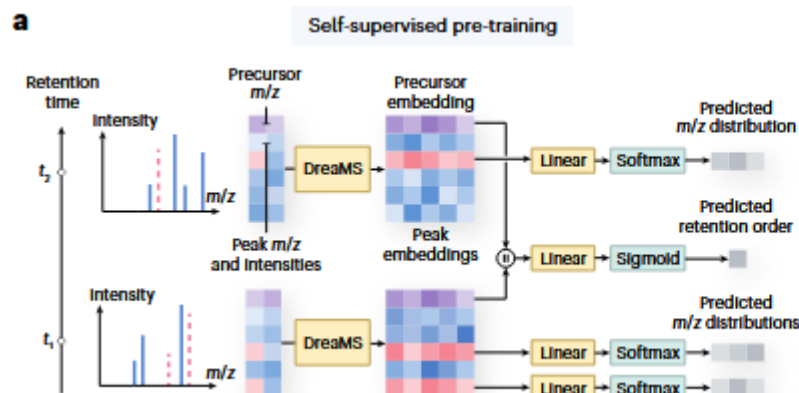
Accepted: 31 March 2025

Published online: 23 May 2025

Check for updates

Roman Bushulev^{1,2,3}, Anton Bushulev^{2,3}, Raman Samusevich^{1,2}, Corinna Brungs¹, Josef Sivic²✉ & Tomáš Pluskal¹✉

Characterizing biological and environmental samples at a molecular level primarily uses tandem mass spectroscopy (MS/MS), yet the interpretation of tandem mass spectra from untargeted metabolomics experiments remains a challenge. Existing computational methods for predictions from mass spectra rely on limited spectral libraries and on hard-coded human expertise. Here we introduce a transformer-based neural network pre-trained in a self-supervised way on millions of unannotated tandem mass spectra from our GNPS Experimental Mass Spectra (GeMS) dataset mined from the MassIVE GNPS repository. We show that pre-training our model to predict masked spectral peaks and chromatographic retention orders leads to the emergence of rich representations of molecular structures, which we named Deep Representations Empowering the Annotation of Mass Spectra (DreaMS). Further fine-tuning the neural network yields state-of-the-art performance across a variety of tasks. We make our new dataset and model available to the community and release the DreaMS Atlas—a molecular network of 201 million MS/MS spectra constructed using

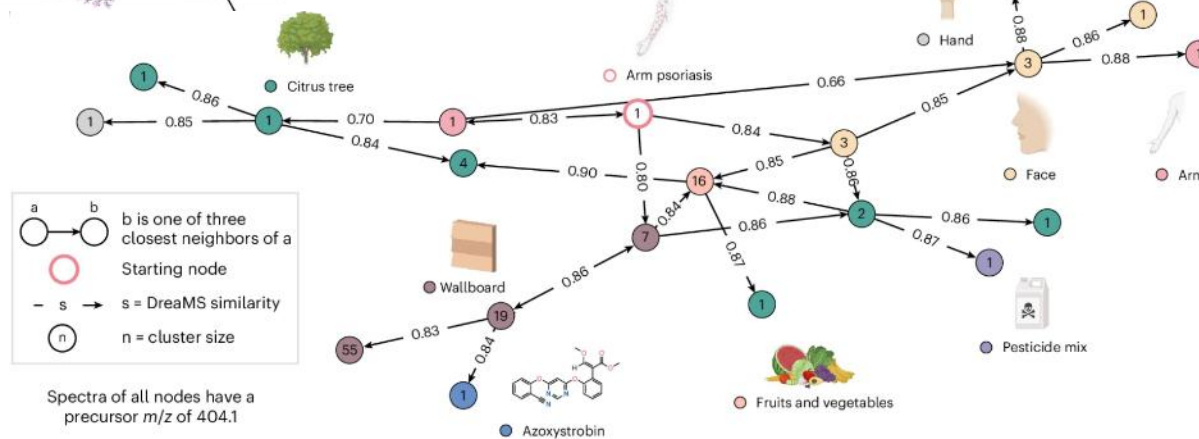
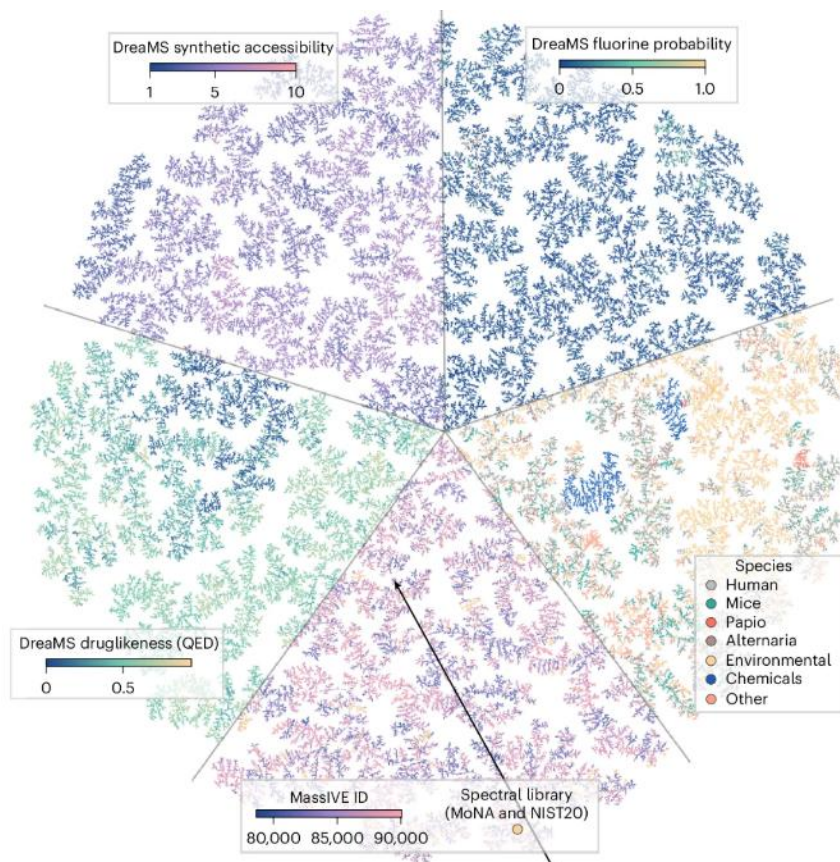


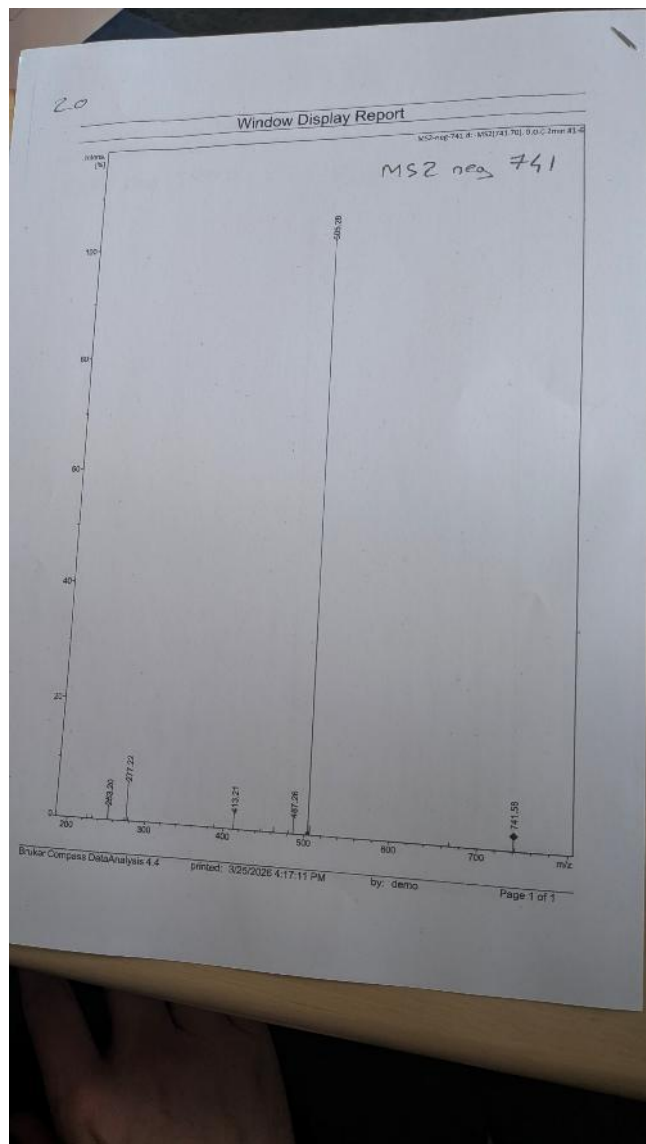
TMAP: (3-NN) DreaMS Atlas

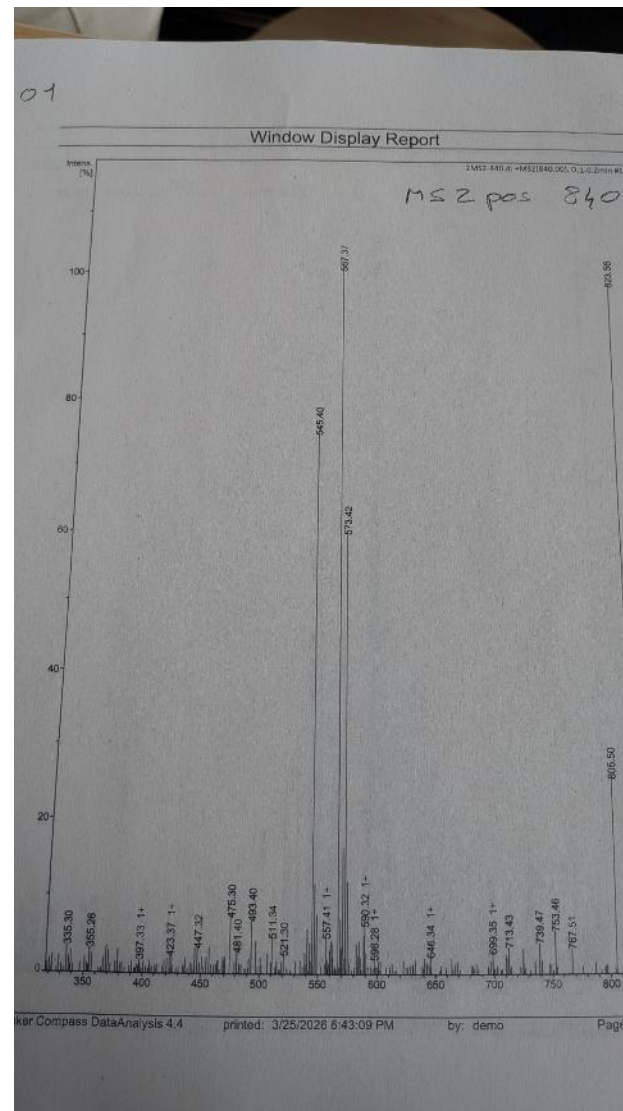
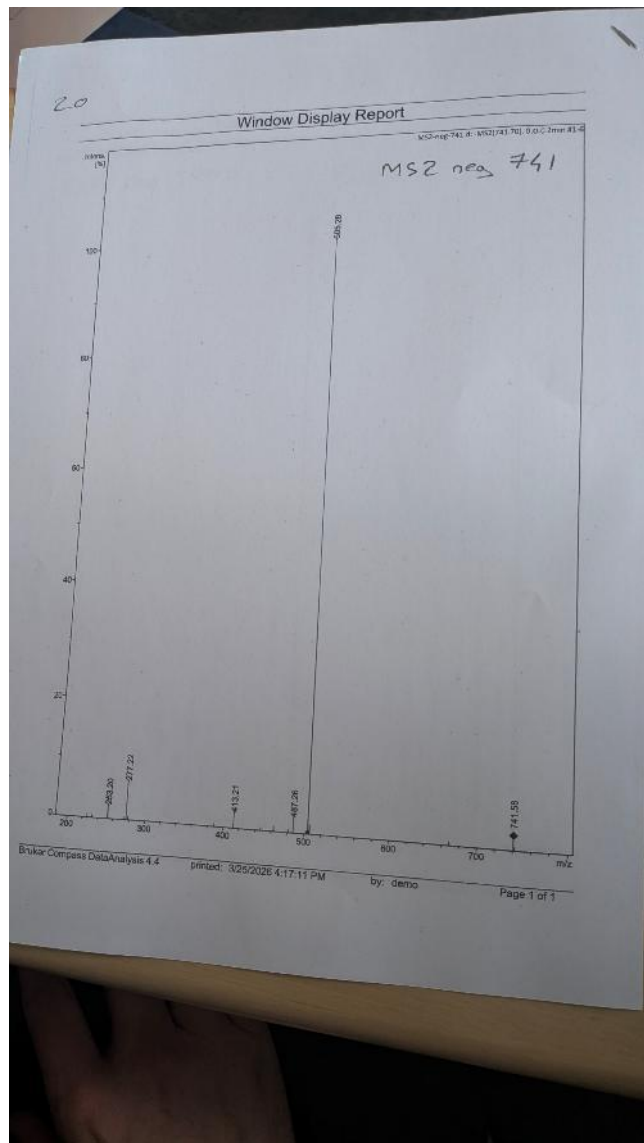
Neural network-based model
 Embedding similarities (MS/MS)
 1 M nodes showed (DreaMS ~ 201 M MS/MS)
 Massive metadata
 Processing time: 7 h with a laptop

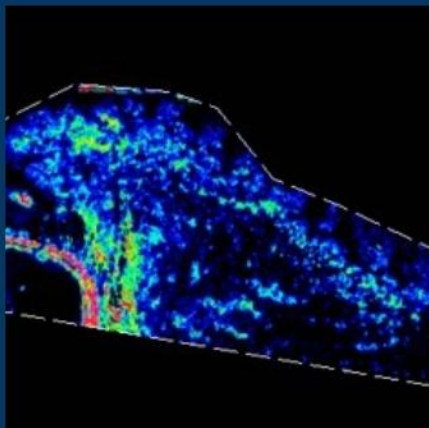
Directed three neighborhood of a selected node:

Research hypothesis generator









Mass Spectrometry Imaging

Spatial molecular analysis of tissues, organisms, plants and environmental samples.



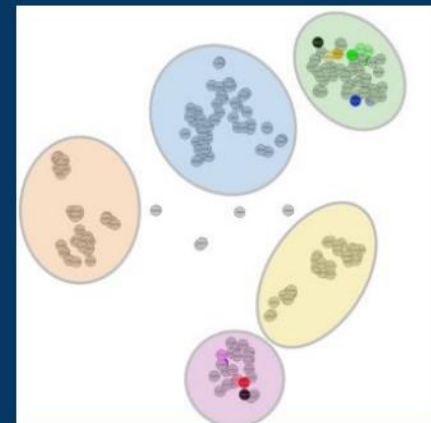
Natural Products Chemistry & Chemotaxonomy

LC-HRMS and MS/MS workflows for biodiversity, herbarium specimens and plant classification.



Metabolomics & Epitranscriptomics

Targeted and untargeted profiling of lipids, metabolites, specialized metabolites and modified nucleosides.



Computational Mass Spectrometry

Molecular networking, MS/MS annotation, AI-assisted curation and reproducible workflows.

<https://sites.google.com/view/david-touboul-research/home>

If interested to apply MS or NMR post-doctoral fellowship or CNRS research positions

david.touboul@cnrs.fr

covadonga.lucas-torres@polytechnique.edu

Linear sum assignment

- Classical combinatorial optimization problem
- Example 1: Three workers, three jobs to assign, find the best combination that will minimize the bill

			
	50€	25€	30€
	40€	36€	18€
	49€	24€	32€

Linear sum assignment

- Classical combinatorial optimization problem
- Example 1: Three workers, three jobs to assign, find the best combination that will minimize the bill

			
	50€	25€	30€
	40€	36€	18€
	49€	24€	32€

92€

Linear sum assignment

- Classical combinatorial optimization problem
- Example 2: Affinity matching. Find best combinations to maximize satisfaction.

			
	85%	5%	30%
	41%	47%	24%
	15%	60%	30%

Linear sum assignment

- Classical combinatorial optimization problem
- Example 2: Affinity matching. Find best combinations to maximize satisfaction.

			
	85%	5%	30%
	41%	47%	24%
	15%	60%	30%

Solutions

- General solutions :
 - Hungarian algorithm (aka Kuhn-Munkres algorithm)
 - Auction algorithm
- Work well but slow

